

THE BREAKING WAVE

Oceans reform in Aotearoa New Zealand



Greg Severinsen, Raewyn Peart, Bella Rollinson, Tracey Turner and Phoebe Parson

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List of terms

ACE	Annual catch entitlement	MSY	Maximum sustainable yield
AMA	Aquaculture management area	NBA	Natural and Built Environments Act
EDS	Environmental Defence Society	NES	National environmental standard
EPA	Environmental Protection Authority	NPS	National policy statement
EEZ	Exclusive economic zone	NZCPS	New Zealand Coastal Policy Statement
EEZ Act	Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012	QMA	Quota management area
		QMS	Quota management system
IMO	International Maritime Organization	RMA	Resource Management Act 1991
IPA	Indigenous protected area	TAC	Total allowable catch
ITQ	Individual transferable quota	TACC	Total allowable commercial catch
MACA Act	Marine and Coastal Area (Takutai Moana) Act 2011	UNCLOS	United Nations Convention on the Law of the Sea
MPA	Marine protected area	UNDRIP	United Nations Declaration on the Rights of Indigenous People

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We also note that while the Ministry for the Environment and Department of Conservation provided financial support for the project, the views expressed and any omissions or errors are solely those of the authors/EDS.

1 Introduction



West coast, Northland

Craig Potton

1.1 Background

Aotearoa New Zealand has jurisdiction over a very large marine domain. The state of that environment was specifically assessed in a joint report by the Ministry for the Environment and Statistics New Zealand (*Our Marine Environment 2019*).¹ It describes a space with many uses and much potential to benefit humanity, but also one which is suffering serious and concerning environmental degradation and imminent threats of species extinctions. Habitats are under threat and biodiversity is declining. Land-based activities are polluting our oceans and shorelines. Pest species are expanding in number and range. Climate change is affecting our seas, what can thrive in them, and how we can use them. And there are questions about how we, as New Zealanders, make the best use of what can be scarce and contested marine resources. All of this is tied up with deep questions about the role of Māori and the implications of te Tiriti o Waitangi for how we manage te moana.² It is a complex space. Yet while for many of us the sea may be out of sight in our day-to-day lives, we cannot afford for it to be out of mind. Many challenges like climate change, pollution and biodiversity loss are cross-cutting and interlinked. They need to be tackled head on, and together, rather than on an ad hoc, disjointed and piecemeal basis. In short, the system by which the country's marine areas are managed and protected is in need of significant change to make it fit for purpose in the 21st century.

The oceans management system is a very broad, and complex, thing. It encompasses many different human activities across multiple spaces: those in the marine area itself (estuaries, the territorial sea, the exclusive economic zone (EEZ) and the extended continental shelf) as well as those on land that impact those spaces and what people do in them. It includes norms, legislative frameworks, institutions and various tools. And an equally important aspect of the system is the *processes* through which decisions and interventions are made. These determine who is involved, how choices are made, and how different processes relate to each other. We define the oceans management system in more detail below and describe it in Chapter 3.

The need to provide a more integrated system for oceans management has long been recognised. The current legal framework has developed over more than 50 years into an uneven patchwork of provisions. There are multiple pieces of overlapping legislation and some significant gaps in coverage, including no marine protected area (MPA) legislation that applies outside the territorial sea. Ad hoc legislation for specific locations such as the Hauraki Gulf, Fiordland and Kaikōura is, at least partly, indicative of failings in the broader system. Some legislation is outdated and in need of radical revision, including, most notably, the Wildlife Act 1953, the Marine

Reserves Act 1971 and the Marine Mammals Protection Act 1978. Other legislation, such as the Resource Management Act 1991 (RMA), is being reformed but largely from a terrestrial rather than marine perspective. There is no overarching mechanism to help ensure that all legislation impacting on the marine environment is interacting coherently. While what we need is an oceans management *system*, arguably what we currently have is a mixed assortment of oceans management interventions.

The process for reforming the resource management system is now well underway, in the wake of the report of the government's independent panel chaired by Hon Tony Randerson QC (see Chapter 4).³ The marine space looks likely to be the next cab off the rank. The need for a conversation about fundamental oceans reform has been underscored by Cabinet papers referring to a “review of the marine system” following overhaul of the RMA.⁴ Since the 2020 election, and creation of a new ministerial portfolio for Oceans and Fisheries, all indications are that oceans reform is being seen by government as a priority. There has been the 2021 announcement of a marine “vision”, and a series of proposed reforms to the Fisheries Act 1996, as well as references to being open about deeper reforms to the oceans management system.⁵

In contemplating oceans reform we have an opportunity, not just to fix current problems in this system, but also to reflect on future risks and opportunities. There is potential to build something better, not just tinker with what we have. That requires deep thought about the nature of our relationship with our oceans, what they mean to us, and what we want to use them for. An oceans management system should reflect the values of modern society rather than the intellectual or ethical assumptions and inheritances of the past.

1.2 The EDS project

It is against this background that the Environmental Defence Society (EDS) is conducting a first principles policy project looking at the future of Aotearoa New Zealand's oceans management system. The project is intended to encourage and facilitate an open discussion about what the system could and should look like over the coming decades. It is doing this by (1) conceptualising the (currently fragmented) system as an integrated whole, (2) analysing both targeted and larger scale options for change to that system, and (3) exploring different starting points for what a future system as a whole *could* look like. This report is aimed at a wide audience including Ministers of the Crown, policy advisors, Māori, marine stakeholders and the interested public. A summary document is also available to highlight key points and options.

The analytical approach and structure of the report are explained further below. The approach is conceptually similar to that applied to EDS's multi-phase work on resource management system reform (which helped facilitate a discussion that led to the government's current package of reforms and overhaul of the RMA). In some important ways, however, the marine context is different and we have reflected this in the structure of this report.

As part of the project, EDS released a working paper in September 2021 which included a series of questions for readers to consider. We are grateful for the feedback received on the working paper, as well as from the workshops and interviews we have conducted around the country.

The timing of the project is deliberate. It is intended not just as a think piece or call for change, but as something that can support future marine reform efforts in a tangible way. The project's title is also deliberate: we feel that momentum for change in the marine space has built up over the past 20 years to the point where deeper systemic reform is now a wave waiting to break.

We should also note at the outset that the need for deeper oceans reform – which may take several years to progress – should not be used as an excuse to do nothing in the meantime. There are many elements of the system – legislative and otherwise – that require targeted attention. Many of these elements (such as MPAs, regenerating the Hauraki Gulf, supporting a sustainable and productive marine economy, reducing catchment-based pollution and minimising waste) are urgent and on the government's radar already. They need to proceed now. But careful thought needs to be given to how such short-term changes can be eventually knitted into longer-term reforms.

The project is not intended to be an exhaustive list of detailed problems with the current system. Nor is this report making firm recommendations or arriving at a single proposition for reform. We need an open conversation about options before choosing a way forward. We anticipate a Phase 2 of the work that will develop a single preferred model and a pathway to achieve it.



Hahei, Coromandel Peninsula

Raewyn Peart

1.3 The structure of the report

This report is divided into three parts. Part 1 looks at what we have now. This includes a description of Aotearoa New Zealand's marine environment, how we use it, and the problems/challenges this has caused (Chapter 2); a summary of the existing oceans management system and issues with that system (Chapter 3); and the context within which systemic reform would occur, including reform measures currently planned or underway (Chapter 4). This recognises that in undertaking oceans reform, policy makers will not be starting from a blank slate.

Against this background, Part 2 of the report looks at various options for reforming our oceans management system. The structure of this part largely reflects the way in which we are breaking up the system into manageable “chunks” for analysis. In short, this is according to cross-cutting “themes”: norms (worldviews, principles and objectives); tools (specific ways in which the system intervenes to shape people's behaviours); and structures (how legislation and institutions are split up and designed). Information and money are also important cross-cutting themes that flow through all of the above.

Exploring reform options in this way – through themes that cut across the whole system (rather than looking at a series of sectors, problems,

solutions or existing legislative frameworks in turn) – is useful to ensure the system is continually viewed as an integrated, connected whole. For example, one important element within the theme of “norms” is people's worldviews and ethics. It is more useful, at least as a starting point, to consider options for these across the whole system (what mix of worldviews should underpin human relationships with te moana?) rather than only for particular sectors (eg what are we aiming for in fisheries?), spaces (eg what should and should not be allowed in marine reserves?) or statutes (eg what should the purpose of EEZ legislation be?). It allows the bigger picture systemic questions to be tackled first.

We include **blue boxes** that summarise key points and discussion in the text. Where concrete options for change appear, we highlight these as building blocks (🧱 – noting that these particular blocks are non-plastic and fully biodegradable). Some of the building blocks represent alternative options, while others complement existing mechanisms. Building blocks are summarised at the end of each chapter to give a sense of some of the reforms that could be mixed and matched.

Part 3 of the report is more exploratory and seeks to draw some of the theme-based threads and building blocks of Part 2 together. This recognises that the choices between the myriad options presented for each theme (eg whether the RMA should be integrated with the Fisheries

Raewyn Peart



Point Chevalier Beach, Auckland

Act or not) are not just to be made on their own merits. They must also provide the pieces for a carefully constructed, internally coherent system, with strong connections between its parts and with a powerful overall narrative. This means that contemplating the merger of legislation, for example, would require close consideration of the institutional settings that support it (would there be multiple government ministries or just one?, and what role would the courts play?), the norms that underpin it (would a combined statute have different purposes and principles depending on the type of decision being made?) and the types of tools used (would there continue to be a need for both RMA-style national direction and Fisheries Act-style sustainability measures?). And even this element would only be one part of a much larger system.

Part 3 therefore culminates in the sketching out of several possible approaches for what a reformed system could look like. This is not intended to be a set of comprehensive models to be adopted or rejected. Instead, it is designed to show which granular options described in Part 2 could work together, and to challenge readers to think about different ways of approaching systemic reform – what a starting point could look like. Each will have pros and cons.

Chapter 1	Introduction
Part 1	What we have now
Chapter 2	The marine environment
Chapter 3	The current oceans management system
Chapter 4	The context of reform
Part 2	Options for the future
Chapter 5	Conceptualising a future oceans management system
Chapter 6	What is the rationale for having a system?
Chapter 7	Ethics, principles and objectives
Chapter 8	Reconsidering the toolkit
Chapter 9	Spatial protections in the toolkit
Chapter 10	Strategic and integrative tools
Chapter 11	Legislative design
Chapter 12	Institutional design
Part 3	Drawing the threads together
Chapter 13	Visions for the future

Figure 1.1: Structure of the report

Te ao Māori/tikanga Māori and te Tiriti o Waitangi need to underpin thinking about a new system. These are not just “subjects” of a system that is otherwise assumed to be “Western” in its foundations (things to be contained *within* it or protected *by* it).⁶ Tikanga and te Tiriti are also living and evolving things that exist *outside* the oceans management system. Indeed, they are guiding factors (among others) in designing what the system should look like in the first place (including how problems are identified and articulated and how tools are deployed).

Te ao Māori and tikanga Māori are not just “subjects” of the oceans management system to be provided for or protected within that system. They are factors that should guide what the system looks like and the design choices made across all themes.

1.4 What is the oceans management system?

Although it is a convenient label, an oceans management system constructed by humans does not actually manage the oceans per se. The oceans have managed *themselves* since time immemorial. What the system does is manage *people* and their *interactions* with the oceans. It would be more accurate – albeit probably also more confusing – to call it the marine *people* management system.

But what actually is it? Above, we have referred to the “oceans management system” in a fairly imprecise manner. That is partly because it describes an artificial idea with fuzzy edges, not a concrete thing or universally agreed set of rules and processes. There is no statutory or dictionary definition of it. Indeed, the concept is seldom talked about at all, with most preferring to think in terms of specific sectors (eg the fisheries system), statutes (eg the RMA), issues (eg marine pollution), solutions (eg MPAs) or disciplines (eg law or philosophy). The oceans management system arguably includes all such things and more, and there is no single right or wrong way to set its boundaries. Yet because it is the subject of this report, it behoves us to offer some more specific thoughts on how the system can be defined.

The oceans management system can be thought of as the collection of laws, institutions, incentives and norms that collectively manage or influence outcomes in marine areas under Aotearoa New Zealand’s jurisdiction. This is a broad definition. But we need to be careful not to define the system so widely that it loses its usefulness. For example, a meaningful analysis of the system cannot include a deep dive into

everything (eg labour laws for marine workers, trade rules for goods imported by sea, criminal law for those committing offences in the EEZ, or security and defence frameworks). That quickly becomes an overwhelming prospect requiring links to be made into many other complex systems. Instead, the focus needs to be narrowed somewhat, onto managing human interactions *with* the marine “environment” and the natural and physical “resources” within it.⁷

But the system should not be so narrowly defined that a reform package misses key components, thereby undermining the point of the exercise. To be useful, it needs to be a broad concept. We are therefore conceiving of the oceans management system as a subset of the broader resource management system (which is much wider than just the RMA).⁸ For example, recurring efforts to reform the RMA – including the most recent through the Randerson Panel process⁹ – have covered some elements of oceans reform but have consistently excluded consideration of fisheries, conservation laws, and related aspects like environmental education and local government settings. It is not therefore really oceans “system” reform. Nor should marine reform fail to address land-based stressors on the marine environment such as sediment, plastics and chemical pollution. Too narrow a view of the system also sits uneasily with *te ao Māori*, where *te taiao* (the interconnected and holistic relationship between all living things in respect of our natural world)¹⁰ and *te moana* (the ocean) have no clear boundaries within them or between them and people.¹¹

As a starting point, the oceans management system can be seen as the collection of laws, institutions, incentives and norms that collectively manage or influence human interactions with the marine environment and the natural and physical resources within it under Aotearoa New Zealand jurisdiction.

The best way to approach system-wide reform may be to focus more on its core, rather than defining a sharp boundary or dismissing peripheral subjects as irrelevant. This reflects the idea that the system is about components that are tightly interlinked, and our focus should diminish as those connections become weaker. For example, while the entirety of marine criminal law (eg penalties for customs breaches) is intuitively beyond the scope of the system we are concerned with, aspects of it are still relevant (eg when it comes to enforcement and penalties for unauthorised pollution or extractive activities). The Education Act 1989 (which guides the curriculum that establishes the knowledge and ethics of future decision-makers) is relevant to the oceans management system as



Raewyn Peart

Container ship, Auckland harbour

well as other areas of public policy. And even higher-level, non-legislated, concepts like capitalism and democracy are important elements to consider in a future system, even though they have much wider resonance across society. In short, there is a lot of overlap between different “systems” and sharp boundaries cannot realistically be drawn. Any flow-on effects between systems need to be carefully considered when placing one system – like oceans management – at the forefront of analysis.¹²

For that reason, we have chosen not to ring-fence the oceans management system by reference to particular pieces of existing legislation like the RMA, Fisheries Act or Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012 (EEZ Act). Doing so could prevent consideration of expanding the scope of existing statutes, or redrawing their boundaries in fundamentally different ways, which are legitimate options for reform. It could also dismiss other frameworks, which are not specifically “marine”, as unimportant (eg the Environmental Reporting Act 2015 or Companies Act 1993), whereas they (or parts of them) can play important roles in marine outcomes in the future. Furthermore, some reform measures may even be so systemic as to not comfortably fit within a specific statutory framework at all. For example, many public authorities operate within powers of general competence, including government departments and councils. That means their interventions can be extremely diverse (eg funding various initiatives, deploying new technology, or undertaking projects), and not obvious from looking at the statute book.

Using existing statutory boundaries would be a sub-optimal way to define the oceans management system.

With this in mind, in Figure 1.2 we provide a more detailed definition of the system for the purposes of this project.

"Aotearoa New Zealand's oceans"	The oceans management system is about Aotearoa New Zealand's oceans, which include areas where the country has sovereignty or sovereign rights. The landward boundary of the ocean the mean high water springs level, and the seaward boundary is either the edge of the EEZ or (where it is further out) the edge of the continental shelf. For the purposes of this project, it does not include areas within Aotearoa New Zealand's Antarctic marine jurisdiction¹³ or areas of the high seas where the country has jurisdiction over its flagships. Oceans include all natural and physical resources in those areas, whether they are animate or inanimate; public, private or neither; or natural or built.¹⁴ To the extent that seeing the oceans in terms of "resources" is considered unacceptably instrumentalist – a fair point – this term can be replaced with the marine "environment" or "moana".¹⁵ The point is that the system includes everything in the oceans. We use the terms seas, oceans, moana and the marine environment interchangeably.
"Management"	Management includes the following <i>public</i> interventions in relation to the marine resources/environment described above: • Regulation: requiring or restricting human action (you must/must not). • Behavioural incentive: influencing human action (you should). • Resourcing/funding: enabling human action (you can). • Strategy: making a plan for how the above three interventions will happen over time and space. In order to generate the following kinds of actions: • Limiting or preventing human activities to manage adverse impacts on the environment or other users (whether resources are used, and how). • Influencing the use of resources for environmental, social and economic benefit (economic and social planning, and environmental enhancement – how and why resources are used). • Shaping the spatial distribution of resource use, protection and enhancement (spatial planning – what happens where) . • Shaping the temporal distribution of resource use, protection and enhancement (strategic planning – when things happen). • Distributing resources to different parties or communities of interest (allocation – who gets what). The concept of "management" therefore includes all human activities that use or protect natural and physical marine resources (eg conservation, recreation, shipping, fisheries, mining etc).¹⁶
"System"	The system is the framework of norms, structures (eg legislation and institutions), tools and networks within which all of these public interventions are supported and interact.

Figure 1.2: A definition of Aotearoa New Zealand's oceans management system for the purposes of this project

A few points about the definition above should be explained. First, the main purpose of this project is to contribute to *government*-led reform. As such, it deliberately limits the system to questions of “management” – that is, public interventions that are imposed (or deliberately not imposed) to influence public and private action (such as regulations, funding and other incentives). That is not to demean the importance of behaviour change at the level of individuals or companies, which can be affected by other drivers (eg markets, ethics, and public and consumer opinion). Nor is it to lose sight of how much technological innovation could help. It simply reflects what government – and therefore this project – can influence. While a focus on public interventions means that purely private actions are not conceived of as forming part of the “system”, public actions that *influence* private behaviour (eg tax settings, education campaigns, psychological nudges, government financial

support for research and development or community groups, regulation and compliance measures) are.¹⁷

Similarly, an “oceans management system” could be defined to include tikanga Māori (rules, protocols, practices and norms). This is a culturally important system through which tangata whenua¹⁸ manage the relationships between people and te moana, although those relationships are couched in very different terms to how a system built on a Western foundation frames them. In other words, it is a system of oceans management in its own right. However, this project is not about *reforming* tikanga, but is considering its interface with, and influence over, the system we have defined above (one defined by *public* intervention).¹⁹

Craig Potton



Rawene, Northland

Secondly, the fact that activities occurring outside the marine environment (eg on land, in water catchments and in the air) can impact on the oceans and what is in them, means that “management” includes interventions outside the marine space itself. In other words, the geographical constraints of the system (“oceans”) is about where outcomes manifest rather than where interventions occur.

Thirdly, we have defined “management” to include interventions that generate various “actions”. This word has been deliberately chosen instead of “outcomes”. Our view is that, in a first principles rethink, the boundaries of the system should not be arbitrarily defined according to the outcomes sought, because these – the objectives of the system – are one of many key design choices that need to be debated. While some objectives or principles, such as ecosystem-based management

and sustainability, might be obviously desirable,²⁰ what we want the system to achieve ultimately depends on our worldviews and ethics (see Chapter 7).

Finally, although they are important matters (especially with respect to migratory species and their connectivity with the Pacific and Antarctica), this project is not directly concerned with management of the high seas or the seabed beyond national jurisdiction.²¹ These have a more complex international framing and warrant separate attention. Nor are we directly concerned with Antarctic waters themselves. We are also not exploring in any depth land-based “coastal” issues like coastal erosion, managed retreat or coastal landscapes. Such things are about how the sea impacts the land. Our eyes are firmly focused on the sea itself and what affects it. With this in mind, we can turn to describing the moana, and the challenges we face within it.



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Cocks Creek, Auckland

Endnotes

- 1 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019). Similar conclusions, although delivered in a different framework based on te ao Māori and the stars of Matariki, have been presented in the government's most recent environmental reporting synthesis report *Environment Aotearoa 2022* (at 46-53).
- 2 Te Tiriti o Waitangi refers to the Treaty of Waitangi as set out in English and in Māori in Schedule 1 of the Treaty of Waitangi Act 1975.
- 3 Resource Management Review Panel *New Directions for Resource Management in New Zealand*, June 2020).
- 4 Office of the Minister for the Environment "Comprehensive review of the resource management system: scope and process" (21 June 2019) at 10.
- 5 Minister for Oceans and Fisheries "Oceans and Fisheries portfolio – ensuring healthy ecosystems" (2 July 2021). See also Minister for Oceans and Fisheries "Fisheries system reform agenda" (2 July 2021); Minister for Ocean and Fisheries "Fisheries Amendment Bill: Strengthening fishing rules and policies: landings and discards" (2 July 2021); Minister for Oceans and Fisheries "Fisheries Amendment Bill: Strengthening fishing rules and policies: offences and penalties and agile decision-making" (2 July 2021); Minister for Oceans and Fisheries "Revitalising the Hauraki Gulf – Government Sea Change Strategy" (2 July 2021); Minister for Oceans and Fisheries "Initial response to Prime Minister's Chief Science Advisor's report on commercial fishing" (2 July 2021); and Minister for Oceans and Fisheries "On-board cameras across the inshore fishing fleet" (2 July 2021).
- 6 Despite being often treated like that in practice in the current system (for example, in requirements to consider the Treaty of Waitangi, have particular regard to kaitiakitanga, or protect sites of cultural significance).
- 7 It is important to resist the temptation to define the scope of the system with reference to its objectives (eg everything relevant to achieving a "sustainable blue economy"), because those objectives are features of the system. In other words, the scope of the system needs to be defined before one considers what that system should be aiming to achieve, because there are many options for the latter that require discussion rather than arbitrary definition.
- 8 See Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation* (Environmental Defence Society, Auckland, 2018). Other important legislative frameworks include the Land Transport Management Act 2003, Climate Change Response Act 2002, Local Government Act 2002, Conservation Act 1987.
- 9 Resource Management Review Panel *New Directions for Resource Management in New Zealand* (June 2020).
- 10 Department of Conservation *Te Mana o Te Taiao – Aotearoa New Zealand Biodiversity Strategy 2020* (Department of Conservation, Wellington, August 2020) at 6.
- 11 See Robert Joseph and others *Stemming the Colonial Tide: Shared Māori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake, Waikato, 2020) at 53-55.
- 12 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation* (Environmental Defence Society, Auckland, 2018) at 31.
- 13 These areas have a further overlay of specific international legal requirements.
- 14 This is excluding human resources. The oceans management system as defined here does not include, for example, labour laws for workers on ships, maritime insurance law, or health and safety laws. It is about human interaction with the marine environment. However, it does include non-natural resources. This is important because a key part of managing oceans is mediating between different human activities that can impact on *each other* over and above any impact on the natural elements of the marine environment. For example, the system might need to choose whether a particular space is used for aquaculture, recreation, a port or something else. However, often activities that impact on other users will also impact on the natural environment (eg sediment can impact on aquaculture and fishing activities as well as ecosystems and habitats). There is often not a bright line between such impacts, making it important to include them all. Most human uses of the marine area rely on ecosystem services to support them, so an impact on a user is often an impact on the environment on which that user depends (eg the destruction of fish spawning habitat impacts both the natural environment and the fishing industry).
- 15 See the extremely broad definition of "environment" in the RMA, s 2.
- 16 Conservation is as much a "use" as sectors like fishing or mining. Using a resource does not always mean a consumptive use.
- 17 Some grey areas exist between public and private intervention, partly because it can be unclear whether an entity taking an intervention (such as a council-controlled organisation, a state-owned enterprise, or a heavily regulated private institution) is itself public or private in nature.
- 18 *Tangata whenua*, in relation to a particular area, means the iwi, or hapū, that holds mana whenua over that area; and *mana whenua* means customary authority exercised by an iwi or hapū in an identified area. See RMA, s 2.
- 19 Which can include public institutions where Māori are decision makers and operate under legislation.
- 20 Ecosystems based management is the framing under which the Sustainable Seas Science Challenge is being undertaken. See Sustainable Seas Science Challenges Ko ngā moana whakauka <www.sustainableseascchallenge.co.nz>.
- 21 The "Area" is defined under international law as the common heritage of mankind and is managed by the International Seabed Authority. One important issue of domestic law with respect to the high seas is the regulation of New Zealand flagships.

2 The marine environment



What strikes many as ironic is that we have long called our planet the Earth, when – and this is of course especially noticeable when our blue and green spheroid is seen from outer space – it manifestly should more properly be called the Ocean.¹

2.1 The importance of context

Before looking at options for reform, Part 1 of this report considers what we have now. In it, we look at three things:

1. Aotearoa New Zealand's marine environment and problems/challenges being faced in it (Chapter 2);
2. the existing system by which that environment is managed and problems with it (Chapter 3); and
3. the socio-political and historical context within which that system is operating and evolving (including reform measures that are planned or underway) (Chapter 4).

It is necessary to outline these matters to provide general context and to explain why a conversation about whole-of-system reform is urgently required. For the following reasons, they are also important considerations when thinking about reform options.

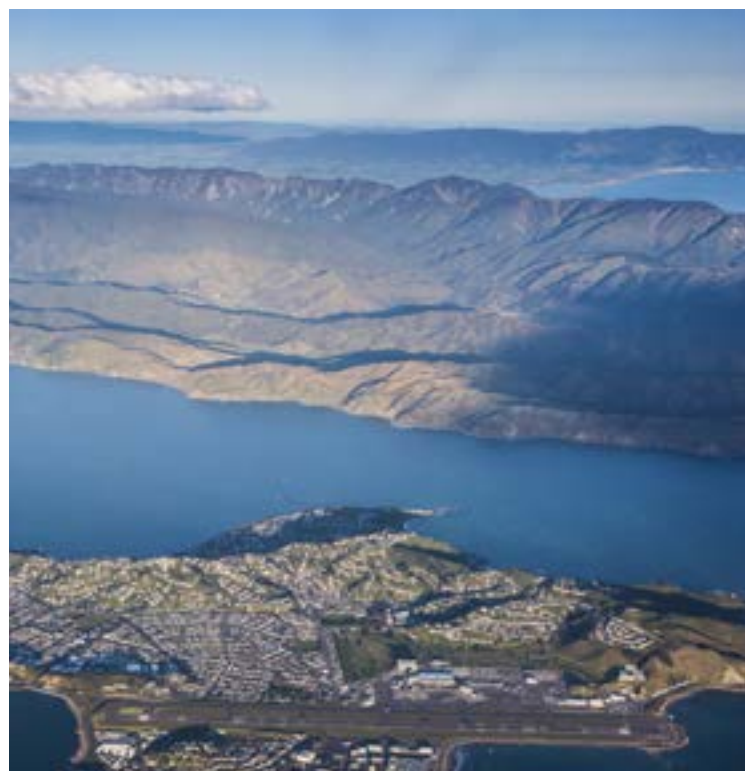
1. The way problems and challenges are articulated (and why they are considered to be problematic at all) can have implications for what the objectives of a future system should (and should not) be.²
2. Reform choices need to be informed by practical matters such as political and historical context, the perspectives of stakeholders, the degree of disruption that change could cause, and the amount of resourcing that may be required. The evolving relationship between the Crown and Māori is particularly important here.
3. A future system needs to be cognisant of reforms already underway, to ensure they are not replicated and to potentially mould them rather than reinvent them entirely.
4. A future system must comply with Aotearoa New Zealand's international legal obligations.

There are important links between Part 1 of the report (describing the marine environment, the existing system and the context of reform), and Part 2 (options for reform).

2.2 What is the marine environment?

The obvious place to start is by describing the marine environment itself. Aotearoa New Zealand is an island nation. The sea is never far away; it connects us to the rest of the world. Most of the country's population is not far from the sound of the waves,³ and is shaped both directly and indirectly by its presence. But what do we mean when we speak of the marine environment? The answer may seem obvious, but there are two ways in which the question of definition can be approached.

The first is the technocratic answer we might find in a geography textbook. This sees the oceans as a spatially delineated area, mapped in two or three dimensions; it is something on a map (see Figure 2.1). That is useful in conveying the geographical scope of what we are looking at: Aotearoa New Zealand's territorial sea (out to 12 nautical miles from land),⁴ its EEZ (which extends out to 200 nautical miles from land),⁵ and its extended continental shelf (which extends out further than the EEZ and was formally claimed in 2008 through an international legal process).⁶ These collectively span a vast area from sub-tropical to sub-Antarctic waters. From a "vertical" point of view, the marine environment is much more than just the water column and things within it – it includes the seabed below and the air above.



Wellington harbour

Raewyn Peart

The outer boundaries of the territorial sea and EEZ are legalistically (and somewhat arbitrarily), rather than ecologically, defined. In contrast, the boundary of the extended continental shelf is defined geologically. In practice, this means that the country has significant areas of “deep seabed” – beyond the geological continental shelf – within its EEZ jurisdiction. Aotearoa New Zealand has either sovereignty or a reasonably wide range of “sovereign rights” in all its waters and extended continental shelf. This affords sufficient jurisdiction to cover most things “resource management” (see Chapter 3 on international marine law).⁷

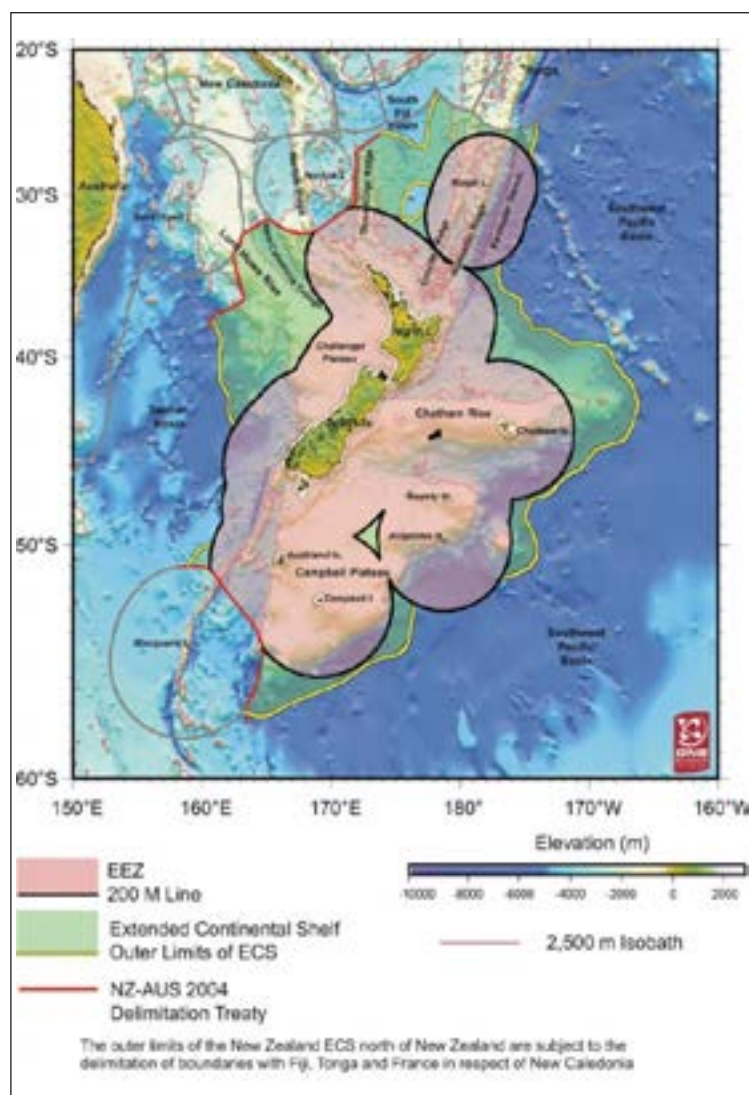


Figure 2.1: Aotearoa New Zealand's maritime zones
(Source: Ministry for the Environment, 2013)

The most striking thing about Figure 2.1 is how extensive the country's oceans are. Indeed, Aotearoa New Zealand is a “continent” in its own right (often described as Te Riu-a-Māui/Zealandia), which is many times larger than what peeks above the waves at this point in geological history. The territorial sea extends over almost 170,000 square kilometres. That is significant, as is the country's more than 15,000 kilometres of coastline. If the EEZ is included, this figure jumps to over 5.8 *million* square kilometres. It is not exaggerating to say that Aotearoa New Zealand has jurisdiction over a marine empire – one that is many times larger than our land mass – albeit one whose main inhabitants are non-human.

Aotearoa New Zealand has responsibility for a huge marine area. This includes the territorial sea/coastal marine area within 12 nautical miles from land, the EEZ out to 200 nautical miles and the extended continental shelf.

The marine environment can also be described in a more values-based manner. Just as we can define a human being with reference to a physical body, we can define the oceans as something that takes up a particular space on a map. But just as a person is not only a bundle of DNA (we have a concept of *humanity*, which is much more), so too we can ask: what *are* our oceans?

That question is closely linked to our worldviews and ethics (see Chapter 7) and we do not resolve it here. But it is far from an academic question, because if we stray beyond a neoliberal mindset of the oceans as a supermarket shelf of resources to exploit, we might think differently about how we interact with our watery backyard as well as the kinds of “management” tools we use. For example, are the inhabitants of the sea commodities to be harvested and traded, or non-human persons with rights and interests to be defended? Is te moana itself a space to be managed in order to maximise human welfare, or the embodiment of ngā atua (deities like Tangaroa or Hinemoana) to be respected?

People's sense of what the oceans are is reflected in the stories they tell and the emotions they feel when they think about, look at, or experience the sea. In particular, the oceans are deeply embedded in te ao Māori, and they have been an integral part of Māori life and culture for centuries. Te moana is an ancestor, linked through oral tradition and whakapapa (kinship) to the people that use and protect it. Te moana evokes images of Māori deities, and a spiritual connection not only with the water (waitai), but also with the life within it. There is a rich genealogy to the moana in Māori culture:

Traditionally, Māori divided the natural world into realms ruled by various gods. These gods, the children of Ranginui (sky) and Papatūānuku (earth), were seen as the original kaitiaki (guardians) of their realms. Kaitiakitanga was based around these ...⁸

Water was considered to be an energy possessing myriad characteristics, shapes and natures. It upheld life, yet was also able to bring terrible destruction. This energy with all its forms, moods and expressions is called Tangaroa. The common translation, 'god of the sea', does not adequately convey its meaning.⁹

To Māori, the answer to the question *what are the oceans?* is therefore one with deep spiritual significance. It follows that harm to the oceans has significant spiritual, emotional and personal impact on mana whenua¹⁰ and kaitiaki.¹¹

Species depletion and imposition of harvesting bans have prevented harvesting practice and thereby caused loss of traditional knowledge, such as understanding life cycles, species management and food harvesting methods. Locally specific knowledge and skills are no longer used, and therefore are not able to be passed on to subsequent generations. [This affects] the passing on of stories and knowledge that was part of the communal experience of collecting, preparing and eating local foods. ... Younger generations now have less familiarity with the foods that are part of tribal tradition, and how to prepare them, and lack broader knowledge about their ecology. ... Ultimately resource depletion affects iwi and hapū identity.

On the flipside, a healthy marine environment has a positive impact on the mauri of the people. Many non-Māori New Zealanders also have a deep spiritual and emotional connection with the sea that surrounds them. This is reflected in the myriad ways we use the marine space (see further below), but also in the stories we tell. For example, one fisher has said:¹²

I like seeing fish come up out of the water. I love the fact that you don't have to be that good at reading and writing but you can use your skill at sea to benefit yourself. Every day's different. You can be frightened at sea, which will get blood rushing around your body. With fishing, there are a thousand different things – currents, moons, time of year, depth, baits, time of day, habitat, whether sharks are in the area and whether small fish are getting the bait. When you line it all up it's a good feeling, very satisfying. Then other days you can do your 18 hours a day for \$100 or a \$6 an hour wage. On the calm days it's beautiful. We also get to go surfing and diving. So it's the job for me.

People who live or work in or near the sea may express similar sentiments and, for many, jobs out on the water or pastimes like surfing, fishing or diving are ways of life. Just as a home is more than a house, so too can the oceans for many people be more than a space on a map.

The oceans can be defined in a technocratic way, but to many this may miss what the oceans truly *are*. In other words, a definition of the marine environment is inextricably bound up with how we, as people, perceive and value it.

2.3 The natural environment

There are many ways to describe the natural marine environment surrounding Aotearoa New Zealand. It is useful to start in generalities, recognising that all parts of the marine environment are connected.¹³

Most obviously, the thing that defines the marine environment is salt water. This enormous body of water is constantly moving. Powerful currents shift seawater around the globe. From a big picture perspective, surface water reaches the country from the Southern Ocean and the subtropical Pacific (via Australia), with these two currents meeting at a latitude roughly intersecting with Fiordland and the Chatham Rise. Where this warm water meets cold, there is an efflorescence of marine life. There are, of course, many more localised currents that shift seawater around the coast too.

Although it is harder to see, water moves vertically as well. Wind blowing across the sea can create surface currents, which move seawater towards or away from the coast. Deeper, more nutrient (nitrate and phosphate) rich water rises to replace surface water moving away from land. This "fertilises" the water and promotes phytoplankton growth – creating the basis for a rich food chain and marine life. This phenomenon occurs in many places, but is particularly noticeable along the northeast coast of Te Ika-a-Māui/the North Island, and the west coast of Te Wai Pounamu/the South Island.

Just like the land, the seabed is far from uniform geologically. Some parts are much deeper than others. Generally, the further from land one goes the deeper the sea becomes. While 75 percent of the country's total marine area has depths of more than one kilometre, some parts of the EEZ are relatively shallow. The Kermadec Trench, to the northeast of Te Ika-a-Māui/the North Island, is up to 10 kilometres deep. That is deeper than Mount Everest is tall.

Differences in latitude, the physical and chemical composition of water and seabed, the climate and temperature, and interaction with land, has created a wide variety of environments for marine life. We described some of these in the project's working paper.¹⁴ The living world of the oceans can be conceptualised as a series of food webs, which cut across these myriad environments. The basis of marine food webs is the sun, which plants use for photosynthesis. As the sun does not reach very deep into the water column, shallow coastal waters host the highest diversity and abundance of flora and other photosynthetic marine species.

There is a huge variety of marine plants in our seas, including seagrasses and seaweeds, but the largest group of oceanic organisms belong to a group called plankton. Plankton are not *just* plants; it is a generic name for tiny, often microscopic, organisms that rely on the currents for movement, and includes zooplankton (tiny animals such as worms and the tiny larval stage of creatures like crayfish). But plant plankton in its many varieties – called phytoplankton – forms the most significant pillar of the marine food chain.¹⁵ As we described in *Sustainable Seas*:¹⁶

Plankton are believed to be the most important life form on Earth. Through photosynthesis, phytoplankton convert water and carbon dioxide into organic material, or food and oxygen. They are at the base of the oceanic food chain, and without them, few other life forms in the ocean could exist.¹⁷ Plankton are also responsible for manufacturing a significant amount of the Earth's oxygen.

Phytoplankton are fed on by zooplankton, which are then preyed upon by fish and other marine creatures. Feeding relationships are extremely complex. For example, some fish, including parore, silver drummer and butterfish, graze on seaweed. Others, including maomao, trevally and demoiselle sift plankton from the seawater. Still others, including snapper, blue cod, kahawai, John Dory and moki, prey on smaller fish and invertebrates living in the rocky reef environment or on adjacent areas of sediments.¹⁸ Bottom-feeding fish such as flounder, tarakihi, snapper, red cod and warehou prey on common residents of the coastal soft sediment seafloor including worms, crabs, shellfish, starfish, sea urchins and sea cucumber.

Small changes to marine food webs, whether through the removal of species, the addition of species or the introduction of stressors (eg human-induced climate change or pollution), can have significant impacts on their structure. In particular, plankton might look uniform and resilient, but these tiny species have evolved to be tolerant of specific environmental conditions (eg temperature, salinity, light). They can be a useful early indicator of environmental change, but damage to them is not necessarily

easy to reverse. The fortunes of entire marine ecosystems are dependent on their tiniest members.

Our marine environment is extensive and diverse. The deep seabed looks nothing like the coastal environment, and powerful currents carry water around our shores, forming a constantly moving water column that is the home of many forms of marine life. At the foundation of complex marine food webs is phytoplankton.

As on land, marine habitats (and the relationships between organisms within them) differ enormously, from soft sediments to reefs to underwater volcanoes. We can define habitats with reference to their geology (eg rocky reefs), the biota that live in them (eg seagrasses), or their geographical location (eg estuaries). Habitats provide shelter and food for the marine life that inhabit them. Biogenic habitats – those created by plants and animals, rather than just non-living landforms – are particularly important because they contribute to maintaining the health of the marine environment (eg shellfish beds filter pollutants from the water column).

Habitats support highly localised populations of marine life. The ecosystems they are part of can be complex, dynamic and often unique. Fish can use the seabed and its flora as a spawning/nursery ground as well as a food source and place to hide from predators. Marine habitats also support species that pass through them on much more ambitious journeys, often assisted by the currents. Some fish and marine mammals travel vast distances within and beyond the country's jurisdiction, and some have life cycles that take them on incredible migrations, not just across marine habitats but also into terrestrial and freshwater environments:¹⁹

Some species will spend all their lives in one habitat whereas others will use a range of different habitats during their lifecycle. For example, adult snapper generally spawn their eggs in highly productive areas close to estuaries and harbours, such as the Hauraki Gulf. The juvenile snapper move into sheltered coastal areas to mature, and then once they have grown to adult size, they often move further out to sea.

Studies have shown that the larvae of longfin eels actually hatch far away from New Zealand, possibly near Tonga, and one tagged female longfin eel took 161 days to swim from Canterbury's Lake Ellesmere to a point 160 kilometres north-east of New Caledonia.²⁰

New Zealand waters are visited by numerous migratory fish and they provide a critical habitat for many of them. Species such as the southern bluefin tuna and striped marlin migrate seasonally across the vast ocean basins.

The diversity of the marine environment is astonishing. An estimated 30 percent of Aotearoa New Zealand's biodiversity is in the sea.²¹ Over 17,000 species have been identified in the EEZ.²² Endemic species include around 95 percent of all known sponge species, over 80 percent of bivalves and gastropods, and three quarters of sea squirts.²³ It has also been pointed out that:²⁴

the animal kingdom Animalia²⁵ ... is by far the richest in species numbers with 13,415 marine species, followed by Chromista (a eukaryotic²⁶ supergroup) with 2,644 species, Plantae (mostly red and green seaweeds) with 702 species, Fungi with 89 and Protozoa²⁷ with 43 species.²⁸

Furthermore, the 412 species of marine invertebrates that have been assessed are thought to represent only five percent of the total number of existing species. That is a humbling thought to those who think we can “manage” the marine environment. How can humans possibly make such a claim when they don't even know what is there?

There is an astonishing diversity of marine life in Aotearoa New Zealand. We still have much to learn about ecosystems and how they operate. Thousands of marine species are yet to be identified. Species exist in complex relationships with each other and make their homes in environments as diverse as reefs, estuaries, seamounts and hydrothermal vents.

The marine environment contains species that are seen as particularly important beyond their role in maintaining a stable and healthy ecosystem or food web. Some are rare or threatened (and people's focus tends to come into sharp relief when there is the prospect of an entire species dying out). Others underpin valuable seafood export industries, such as crayfish, hoki and squid. But for others, like whales and dolphins, their value is based on more complex moral considerations (for example, they are seen as iconic, intelligent and altruistic, and more like people). Therefore is it “wrong” to harm them, even if they are not threatened and no significant damage is done to the ecosystems to which they belong?

The marine environment is a vibrant and diverse place filled with a variety of a variety of marine life including fish, marine mammals like whales, dolphins and sealions, seabirds, invertebrates, and flora like seagrass and kelp. Some species are particularly highly valued because they are rare or threatened, are economically valuable, or for other more complex ethical reasons.

There is a lot of regional and local variation in the sea. One reef is not like another reef, nor is one estuary the same as another estuary, whether because of their physical characteristics, the balance of life forms that live there, or the activities that people undertake within them. It is also not always easy to differentiate between what an environment “looks like” and the cultural or historical meaning it has to people, including from a mātauranga Māori (Māori knowledge) perspective. Many places around Aotearoa New Zealand have a special significance for Māori based on stories and histories of events that happened there or because of their place in the broader cosmogony of te ao Māori. For example:²⁹

Also known as Te Rerenga Wairua or Te Rēinga, [Cape Reinga] is one of the most sacred Māori places in New Zealand. Tradition says that the spirits of the dead travel along two pathways to Cape Rēinga, at the northernmost tip of the country. One path begins in the south and runs along Te Oneroa-a-Tōhē (Ninety Mile Beach), and the other starts at Kapowairua (Spirits Bay). The spirits congregate at Cape Rēinga before leaping into the water; they surface after crossing the ocean to Manawatāwhi (Three Kings Islands). There they sing a last lament for the loved ones they have left behind before proceeding to their spiritual home in Hawaiiki.

The social and cultural context of a particular marine environment means it can be described quite differently by different people. Thus, to some, the space between Cape Rēinga and Ohaua is not the “sea” or a series of ecosystems – it is the path of the spirits.

There is much about the marine environment that people do not know or understand. Much remains hidden. There are vast areas where habitats are not mapped. Thousands of species have yet to be studied in detail.³⁰ New species are being discovered all the time. Their interactions with each other, the physical environment, and with the atmosphere and land are not well known. Most significantly, the interactions between the natural environment and human activities are still poorly understood.

Our oceans are also changing over time. Some of this change – especially over long time periods – is natural. Many of our inshore coastal environments, and the species in them, have evolved slowly as geological, climatic and biological pressures have played out over thousands of years. They will continue to do so. However, much rapid change in natural structures and processes over the past decades and centuries, has been induced or accelerated by people.

Every part of our marine environment is unique. Many areas are defined not just by their topography and ecosystems, but by their significance to people. Particular places hold deep emotional and spiritual connections in te ao Māori as well as for other New Zealanders. But our oceans are changing, and a lot of that is down to human activities and pressures.

Raewyn Peart



Te Rerenga Wairua

2.4 Humans in the marine environment

People in Aotearoa New Zealand are highly active in the marine environment. We are a maritime nation. Māori have a long-standing and deep relationship with te moana, going back centuries. The sea does not just exist – we use it and rely on it. How we do so says a lot about why we value it.

The idea of the sea – even the deep sea – being a wilderness may also be slowly changing. Some nearshore areas would no longer be recognisable to those who lived a hundred years ago, let alone Māori who first used them. People are no longer just boating on it or harvesting and landing the sea's bounty. They are increasingly making their presence felt by farming it, reclaiming it, mining it and harnessing its energy. Humanity is going deeper and wider. We are staying there longer. There are increasing conflicts over who gets to use things that we are realising are finite. In some places, particularly those that are close to large centres of human population (eg the Hauraki Gulf), the sea is becoming increasingly congested and subject to contention.³¹

The marine space is of significant and growing value to the people of Aotearoa New Zealand. Environmental reporting points out that the marine economy – comprising many things – was worth \$7 billion in 2017.³² The total sector employs around 70,000 people.³³ Commercial uses are varied, and increasingly diverse. Some particular parts of the marine environment are especially valuable. It has been suggested that the Hauraki Gulf, for instance, “supports the livelihoods of around one third of New Zealand’s population”.³⁴

Some uses of the sea are extractive. Fishing is the main extractive use, occurring around all of the country. The seafood sector contributes over \$4 billion per year to the economy, \$1.4 billion in export earnings, and \$1.1 billion in GDP.³⁵ In 2017, the financial contributions of fishing and aquaculture accounted for 29 percent of the marine economy, and these sectors employed over 13,000 people.³⁶ Wild commercial catch has remained stable at about 450,000 tonnes per year over the last decade. It comprises over 100 species caught using a variety of methods. As of March 2019, there were 37 large deep-water trawl vessels conducting around 25,000 tows per year, with 140 smaller trawl vessels conducting over 50,000 trawls annually.³⁷ The total commercial fleet (including inshore) comprises around 860 vessels.³⁸ Māori are heavily involved in the fishing industry, and around a third of fishing quota is owned by iwi interests.³⁹ The primary use of fish is for food, and it is exported as well as sold through supermarkets and local businesses. Fishing supports a wide range of businesses and livelihoods, from processors to high end restaurants to fish and chip shops.

Aquaculture is a large and developing industry. Total revenue from the sector in 2018 was over \$600 million, with the majority of revenue generated by mussel farming operations.⁴⁰ Recreational fishing is also significant commercially, in that it supports many economic activities (eg boat building, equipment, charter boat businesses).

Marine mineral mining is another extractive use of the sea. In particular, there has been increasing interest in recent years in deep seabed mining for phosphate nodules (which are ground up for fertiliser), massive sulphides (from hydrothermal vents, containing deposits of copper, zinc, lead and gold), manganese nodules (containing various metals) cobalt, and iron sands. Although proposals have been put forward for iron sand and phosphate mining, these have met significant opposition and litigation, and no applications have yet been approved.⁴¹ It remains an emerging industry. However, sand has been mined in shallower coastal marine environments for many years.⁴² Oil and gas – notably off the Taranaki coast – has been a significant activity for decades and still forms one of the mainstays of the region's economy.⁴³ However, the future of the industry remains uncertain, with government policy being not to allow new offshore oil and gas permits. Overall, marine minerals comprised 27 percent of the marine economy in 2017 – a significant figure.⁴⁴

Other commercial uses of the marine space are non-extractive. Shipping now provides the biggest contribution to our marine economy, including port operations, boat building and maintenance, and freight and passenger transport.⁴⁵ Busy passenger ferry services exist in the Hauraki Gulf, Wellington harbour, and across the Cook Strait. Around 99 percent of all exports are transported by ship.⁴⁶ The size of ships, and traffic volume, have increased in recent times. Most major coastal towns have a port, and many have deepened their access channels to accommodate larger ships. The sea is used in the communications and electricity sectors too – notably through the fibre optic and electricity cables that span the Cook Strait and the Hauraki Gulf.

Marine tourism – including sightseeing, whale watching, dolphin swimming, shark diving and (until recently, due to Covid-19) cruise ships – is also significant. It made up over 40 percent of the marine economy prior to Covid-19, employing over 43,000 people.⁴⁷ Marine species viewing is a significant element of this sector. Kaikōura, where whale-watching is a predominant part of the economy, hosted around 100,000 passengers on viewing trips prior to Covid-19.⁴⁸

Not all marine tourism is provided by commercial operators; it can be hosted by non-profits, community groups, sports groups or governmental bodies. Māori providers of marine tourism activities and experiences are a key part of the industry in Aotearoa New Zealand, not merely as economic players, but also as shapers of tourism values and practices. The places where tourism occurs are inseparable from the relationships and obligations iwi and hapū have with the land and sea.⁴⁹

How people use the marine environment might look quite different in the future. Aside from the potential growth of some existing industries such as aquaculture (especially into deeper, offshore waters, and new species like seaweed for stock feed, human consumption and nutraceuticals), the slowdown of oil and gas operations, and the emergence of new markets for existing resources,⁵⁰ there is the prospect of entirely new activities being established. There has been some interest, for example, in offshore wind energy (which is deployed extensively overseas) and tidal energy.⁵¹ Although it is by no means devoid of environmental impacts and has a number of technical and cost-related challenges,⁵² offshore wind may have a number of benefits (such as the ability to deploy farms at greater scale, potentially greater social license due to landscape concerns on land, and the prospect of stronger and more reliable wind offshore). A large-scale offshore wind farm off the Taranaki coast is currently being explored, which could power over 650,000 homes and represent over 11 percent of current demand capacity.⁵³

As mentioned, deep sea mining is still a prospect, and efforts continue to establish operations (with an iron sands mining proposal before the courts at the time of writing).⁵⁴ The development of incentives for sequestering “blue carbon” could also see new operations (eg seaweed farming) alongside potential for marine carbon geo-sequestration (whereby carbon dioxide from point source emissions is compressed and injected deep below the seabed and stored in perpetuity).⁵⁵ The latter has been a live prospect for over a decade, with technical studies looking at the geological potential of sub-surface offshore formations to store carbon dioxide securely, and a detailed exploration of the legal and other barriers to deployment commissioned by the government in 2013.⁵⁶ Technically, geo-sequestration is entirely possible (with Norwegian operations in the North Sea having been undertaken since the mid-1990s). But the main impediments here appear to be the cost of the technology (combined with an insufficient economic incentive through carbon pricing), a lack of legal clarity on a number of fronts (including under the RMA), and uncertainty about social license.

A spotlight on desalination

With Auckland's water shortages causing concern, a growing population, the prospect of a changing climate and increasing difficulties getting consent to take water from the Waikato River, there is the possibility of another use of the marine environment in the Hauraki Gulf, Manukau Harbour and potentially elsewhere: desalination for drinking water. Seawater desalination is an increasingly popular technology in drought-prone communities. The first plants were built during the 1960s and there are now some 20,000 desalination plants operating worldwide. In Melbourne, a desalination plant provides a third of the city's drinking water supply. The technology has improved markedly over the years, and during the last three decades, the cost of desalination has dropped by more than 50 percent.⁵⁷ It has potential to provide a secure supply of safe drinking water from a source that is much more abundant than freshwater catchments.

Seawater desalination involves the removal of salt and impurities from seawater to render it fit for human consumption. There are two types of desalination processes; thermal where the water is heated and condensation captured, and reverse osmosis where fine membranes act as microscopic strainers to remove the salt. Earlier plants used thermal processes but more recent ones have mainly adopted reverse osmosis. Both require a large amount of energy. This means an Auckland plant could potentially put additional pressure on the marine environment through the development of local infrastructure for electricity generation (eg tidal or offshore wind). As energy is a major cost of operating these plants, much research effort is being put into developing less energy-intensive desalination methods.

The salt that is removed from the seawater, and other chemicals which are utilised by the plant, are typically returned to the ocean (with about half of the volume of extracted water) in the form of a salty brine.⁵⁸ The brine water is heavier than seawater and therefore sinks to the seafloor, increasing salinity and reducing oxygen levels for bottom dwelling species. Chemicals that are used as anti-foulants and anti-scalants may also be discharged with the brine. Impacts are generally less in exposed high energy open-sea sites and greater in enclosed shallow sites⁵⁹ such as the Manukau Harbour. The location of any discharge is therefore something that would need to be carefully managed. In the future, it may be possible to reclaim the salt and heavy metals for other uses, avoiding the need to discharge them into the sea.

The intake and discharge pipes themselves can also have environmental impacts, although these are generally localised. In particular, marine life can be trapped against the intake screen or sucked into the treatment plant when seawater is pumped onshore. Such impacts can be reduced by using a subsurface intake which extracts seawater from beneath the seafloor, although this can be a costly solution.

Although desalination plants have their environmental challenges, they may be one way to provide vital public services in a manner that reduces reliance on ecologically stressed catchments and secure supplies that do not involve extensive damming of natural waterways. Some suggest that relying on the ocean as a potable water source might increase public awareness of the importance of a healthy marine environment more generally. If demand side incentives (eg more efficient use) cannot be relied on to maintain adequate supplies, any concerns with the technology will need to be considered against the alternatives. One is the reuse of wastewater—which is sometimes referred to as “purified-recycled water”. Highly treated wastewater has been deployed overseas (eg in Australia and Singapore) for various purposes. The process involves the use of ozone gas or ultraviolet light to remove viruses and bacteria and filtering the water through microscopic membranes to remove solids and trace contaminants. Although reuse means freshwater is being used more efficiently, the contaminants that are removed from it before reuse create a more concentrated stream of waste for disposal. There are also consumer perceptions to consider. Many people do not like the idea of drinking recycled wastewater, no matter how clean it is.⁶⁰ Against this option, it may be that desalination becomes a viable option as costs reduce, freshwater from waterways becomes more constrained and effective ways are found to address environmental impacts.

People use the sea in a variety of ways. It is of enormous value in supporting commercial activities like fishing, aquaculture, tourism, shipping and mining. Other uses of the marine area may also be deployed or expanded in the future, such as different forms of marine farming, energy generation, carbon sequestration and potable water supply (desalination).

Human use of the oceans is not just commercial. For one, it is extensively used for public purposes – as a receiving environment for stormwater/ floodwater and treated wastewater, as a space for defence and security facilities and operations, and as a blue highway for public transport. But as a society we also value it for many reasons that cannot always be subject to a dollar figure, including recreational, cultural and spiritual reasons. There are often no clear distinctions between those categories.

Māori were the first to use and protect the waters of Aotearoa New Zealand, and developed sophisticated fishing methods and knowledge about the timing of harvest.⁶¹ Some seabirds have been targeted for customary harvest as well as fish. Cultural practices with respect to the oceans have been long in development and remain at the heart of coastal Māori communities, so:⁶²

it is critical to manage these resources [mahinga kai] to allow people to continue gathering kai (food) in the way the ancestors did, and [it is] about mana and manaakitanga – the ability to welcome and host visitors by providing bountiful produce, as a demonstration of hospitality and respect.

In particular, there is no clear distinction in te ao Māori between cultural, commercial and recreational use of the oceans, or indeed between use and protection.⁶³ Kaitiakitanga encompasses all those things. Furthermore, whakapapa connections are not based on sharply delineated boundaries, and there can be difficulties when multiple mana whenua with overlapping rohe moana (district, region or area)⁶⁴ are pigeon-holed into Western processes that focus on particular resources or exclusive spaces. For example, there are around 26 different iwi and hapū groups with interests in the Hauraki Gulf, which is managed under multiple pieces of legislation. The process for delineating areas

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Umupuia, Hauraki Gulf

of customary marine title has also proved challenging where there are overlapping claims.⁶⁵

The marine environment is valued by all New Zealanders in a variety of ways; people enjoy swimming at the beach, sailing and water sports. Many people own or use boats. Estuaries and bays are highly valued for leisure and recreation, including in urban areas and holiday hotspots around the country. We like seeing the sea teeming with marine life; one study from 2008 recorded a staggering 375,000 annual visits to the Cape Rodney-Okakari Point marine reserve near Leigh.⁶⁶

Recreational fishing, in particular, is a core part of New Zealand culture. In a country with a population of around 5 million, in 2017-2018, people undertook around two million fishing trips a year and took home seven million fish and almost four million shellfish.⁶⁷ It is thought that around 600,000 people go fishing each year (whether from boats, surfcasting or diving), with a significant portion of the recreational catch being snapper and kahawai.⁶⁸ Most recreational fishing occurs along the north-eastern coast of the North Island, including in places close to large population centres like the Hauraki Gulf.⁶⁹ Many New Zealanders – including Māori – rely on recreational fishing as a source of food, it is not just a pastime.⁷⁰ However, the primary motivation of most recreational fishers is just that – sport and recreation.⁷¹

The marine environment is valued by Māori and non-Māori New Zealanders in ways that are not just commercial. It is a food basket, a playground and a place to find spiritual renewal.

Humans do not necessarily need to “use” the sea directly in order to value it. Many New Zealanders rely on the sea indirectly – for example, those who buy items that need to be shipped from overseas, the millions who enjoy eating kai moana, or those operating land-based service businesses in communities that are reliant on marine activities like fishing and tourism. Such people have a direct stake in our oceans, as much as those who are out on the waves.

Others may also appreciate the sea from a distance. A sea view, for example, can dramatically increase the value of a property. Such viewshafts can be an important consideration when it comes to urban planning and tourism ventures, even if people are not actually on the water. And even when the sea is out of sight it is not out of mind. To many people it has existence value. We like that we are never far from the

coast – its health is important to us. It informs our identity as a nation. The flipside of that is that it can distress us if the sea becomes degraded or inaccessible, even if we do not experience it on a day-to-day basis or harm cannot readily be seen.

The sea creates many benefits for people even if they do not use it directly, including amenity value and the flow of economic benefits from marine resource use. It also has considerable existence value.

More tangibly, the sea is a constant source of what some describe as “environmental services”. We often take these environmental services for granted because we do not see them, but we would be much worse off without them. For example, the sea provides local temperature regulation. It gives us a buffer against global warming by absorbing a significant portion of both heat and carbon dioxide from the atmosphere.⁷² Studies have also suggested that seabed sediments are the largest sink of carbon in the world.⁷³ Marine habitats (eg mangrove forests) trap sediment from land, and habitats also form a key part of the nutrient cycle. Marine filter feeders provide valuable filtration services to “clean” the water coming off the land.⁷⁴ Some habitats, such as mangrove and kelp forests, provide natural buffers against storm surges and erosion. They sequester carbon and produce oxygen. The sea not only provides kai moana directly in the form of fish; it also provides the habitats that shelter and support fish in their growth and development.

These ecosystem services, just like any other services we obtain, are all ways in which we *use* the marine environment even if they are not uses that involve us *intervening* in it. As such, by protecting oceans *from* environmental impacts that threaten ecosystem services, we are also “using” the marine environment in a very tangible sense, even if this is seldom given a dollar value. It is particularly important to note that we are “using” the marine environment’s assimilative and diluting capacity when we discharge or dump contaminants like sediment, wastewater and litter in it.

The sea and marine life within it provide extensive ecosystem services, including temperature regulation, water purification, carbon sequestration, oxygen production, food, flood management and nutrient cycling.

2.5 Conflict in the marine environment

It is not surprising that human uses of the marine environment can, and increasingly are, coming into conflict. That is particularly the case where activities need to be spatially defined, or where there is a shared resource being used in different ways. People may wish to use the same “resource” as others (eg fish), or they may use them in ways that have impacts on each other (eg excluding fishers from protected marine space).

Commercial activities can conflict with each other, but another notable tension has been between commercial, recreational and customary uses. That is particularly prominent in the case of wild fish harvest activities, where all three types of users have a stake in a shared resource but use it for quite different reasons. But tension is not limited to fishing; for example, conflicts have arisen between seabed mining operators and surfers concerned about mining impacts on wave action and surf breaks.⁷⁵ There are concerns about access to the coastal marine area for recreation, spiritual connection and enjoyment when spatially fixed activities like inshore aquaculture are authorised. One can foresee many more such conflicts as human uses diversify and technology develops. Often these conflicts can be worded in terms of protecting the environment from

particular kinds of use but underlying them are very human concerns about who gets to enjoy what from the ocean.

There may also be potential tensions between Māori as commercial operators under a Western capitalist system and more traditional Māori as kaitiaki of their rohe moana.⁷⁶ This is arguably a product of the system heading in two directions – first, the “Westernisation” of the management framework (eg through the creation of quota property rights and their use as currency to settle te Tiriti grievances) and the more recent and ongoing moves to accommodate more traditional Māori values and power-sharing through co-governance arrangements. This dynamic is still playing out. For example, commercial fishing plays an important role in the wellbeing and development of many iwi and hapū, but some Māori communities want to see constraints put on commercial operators (including those using settlement quota) to help preserve their local marine areas and fish stocks.

A spotlight on tensions in the moana

Conflicts and tension are not only between Māori and other interests but also between competing interests within Māori society. For example, Annette Sykes has been critical of the proliferation of a “Māori elite” model where a select few people monopolise the seats at the table with the Crown and control assets and resources.⁷⁷ This can clash with a desire for ground-up or local forms of control or management. Another tension that can occur is between the interests of iwi who hold commercial fishing quota and customary fishers or kaitiaki who wish to use or protect the resource for different purposes; some have argued that fishing under the quota management system (QMS) has depleted abundance and biodiversity, making it more difficult to acquire kai moana.⁷⁸ For better or for worse, the fisheries settlement has entrenched iwi within the QMS system of private property rights.⁷⁹ Tensions are also arising as the courts determine which group has mana whenua status or claims to tikanga-based rights (eg for customary marine title) over marine space, although there has been recognition that such rights can overlap and exist in harmony rather than be exclusive or in competition.⁸⁰

It is possible to see the tension between human activities on the one hand, and environmental protection on the other, as another manifestation of conflicting “uses”. For example, some may wish to “use” the marine environment as a protected area, while others may wish to use it for

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Oyster farm, Mahurangi harbour

fishing. But this undersells the importance of environmental protection, in assuming that it is just another interest group to be accommodated. The reality is that ecosystem health is vital for all humanity, not just those who have taken it upon themselves to advocate for its protection. Those are quite different things.

Conflicts can arise in the marine environment where different activities seek to use the same space or have impacts on each other. However, it would be wrong to frame environmental protection as just one more “use” to be accommodated or negotiated.

As outlined above, the oceans have clear value for humans. But do people need to value the sea for it to *have* value? That is a deeper normative question (see Chapter 7), and depends on one’s worldview. But it would be remiss of us to leave the impression that the oceans are there only for people. While we are unique in many ways, human beings are ultimately one species alongside thousands of others that belong in the oceans more than we do. To many, therefore, the oceans and the life they support have intrinsic value in the same kind of way that a human life does. This notion of intrinsic value is also akin to te ao Māori, where te moana is linked to people through whakapapa and whanaungatanga (relationship, kinship, sense of family connection).

Te moana has intrinsic value, not just instrumental value. For Māori, its value lies in whakapapa and whanaungatanga – it is not just a resource to be used, but rather an ancestor to be treasured and looked after.

2.6 Problems in the marine environment

The government’s environmental reporting makes for grim reading. It focuses on what we might call “biophysical” problems: traditional “environmental” issues concerning pollution, ecosystem health and the depletion of resources. However, it is important to note that there is no sharp distinction between the biophysical, the metaphysical, and the social, including (but not exclusively) in te ao Māori. What impacts on the health and mauri of te moana also impacts us in many ways.

The problems identified in this section can all be described as problematic *outcomes* – ones that can be observed in a physical sense “on the ground”

so to speak. There are also quite different problems with the marine management system *itself* (how it is structured and how it operates); these are looked at in Chapter 3.

Marine ecosystems, and the life within them, are being negatively impacted in many ways. While it is hard to generalise with respect to the thousands of ecosystems and habitats across the country, the available data for specific marine species is concerning at a national level. Some improvements have been observed in recent years (whether because of actual improvement or better data),⁸¹ but the overall trend is one of decline.

First and foremost, biodiversity is in crisis.⁸² Aotearoa New Zealand is a global hotspot for marine biodiversity⁸³ – of the 12,820 marine species that have been scientifically assessed, over half occur nowhere else in the world, though this figure is likely to be much higher given that many species are yet unknown or unassessed. Direct human-induced pressures causing biodiversity decline include exploitation, pollution, invasive species, changes in land and sea use, and climate change.

A recent national conservation status assessment found that 90 percent of seabirds and 80 percent of shorebirds are at risk of, or threatened with, extinction.⁸⁴ Based on a 2019 assessment, 10 out of 45 assessed species of marine mammals are in the same category, with 30 assessed species of marine mammals classified as data deficient.⁸⁵ Orca, Bryde’s whale, Hector’s dolphin and the southern elephant seal are in particularly dire straits. The threatened and nationally vulnerable Hector’s dolphin is estimated to have a population of only 15,000.⁸⁶ But the Māui dolphin (a subspecies of the Hector’s) is on the brink of extinction. It is estimated that this iconic animal has only 57 adults left⁸⁷ making it one of the most threatened dolphins in the world.

These trends have not occurred by chance, or through natural change. We, as human beings, are responsible. Marine mammals, for example, are directly impacted by a wide range of human activities, many of which are still occurring and even increasing. Although hunting of marine mammals is currently prohibited,⁸⁸ other activities indirect mortality in the course of other activities remains lawful, such as when they are legally kill animals directly, such as when they are inadvertently caught in fishing nets or are struck by ships. While the trend may be one of slow improvement, the entanglement of one Māui and 29 Hector’s dolphins in the decade ending 2018 is still cause for concern, given the small populations that remain and slow reproduction rates.⁸⁹ Could one reasonably expect that figure to be zero?⁹⁰

Bryde's whales living in the Hauraki Gulf have been subject to frequent ship strike in the past, and although this has reduced in recent years due to voluntary speed reductions,⁹¹ increased numbers of ships⁹² are linked to an increased likelihood of collision with marine mammals.⁹³ Shipping and other underwater activities, like mining and seismic surveying, can also interfere with marine mammals and other marine life through noise pollution,⁹⁴ and people's activities on beaches (such as driving and dog-walking) can threaten animals like shorebirds, especially during vulnerable life stages such as breeding and nesting.⁹⁵

We can also look with concern at figures for seabird bycatch – the non-target populations that arguably suffer most from commercial fishing activity. Although numbers are reducing, it is estimated that over 4000 birds were caught accidentally in fishing gear during the 2016/2017 fishing year. They include iconic, long-lived and vulnerable species like albatross (which have very low reproductive rates).⁹⁶ These figures are concerning enough, but it is hard to establish the real number due to low observer coverage on commercial fishing boats and a lack of meaningful surveillance of recreational fishers. Fishing can also imperil the food sources of some seabirds through fishing down “bait” fish.⁹⁷

Fishers take a range of other species that are not targeted, including non-QMS fish species and invertebrates such as deepwater corals. In 2017, 65,000 tonnes of marine life were estimated to have been caught as bycatch in offshore fisheries, of which 24 per cent (16,000 tonnes) was legally discarded.⁹⁸ Some fisheries have much greater quantities of bycatch than others. It is very significant in the scampi trawl fishery where the non-QMS bycatch can be double the weight of targeted species harvested.⁹⁹

Marine mammals and other animals can be impacted indirectly by human activities as well. The toxoplasma gondii parasite has been identified as a potentially serious threat to (particularly) female Māui and Hector's dolphins, and was identified in nine dolphins that died between 2007 and 2018.¹⁰⁰ The parasite travels from cat faeces into the dolphin's habitat through run-off, rainwater and wastewater and infects the dolphins when they consume contaminated water or fish.

Modern society's rapacious, industrial-scale fetish for creating disposable plastic products has also led to significant amounts of it being consumed by marine animals. Once ingested, plastics can block the digestive systems of birds, induce reproductive problems, cause poisoning, and block the uptake of nutrients.¹⁰¹ It is disturbing that something as innocent-sounding as a child's lollipop stick can end up as one of the main marine plastic pollutants noted in national level environmental reporting.¹⁰²

Of course, iconic and threatened/protected species like whales and dolphins do not exist in isolation. They are part of wider marine ecosystems, and rely on healthy habitats comprised of less well-known species and physical features. Unfortunately, these too are suffering, which not only puts further pressure on threatened species, but also flirts with the possibility of wider ecosystem decline and collapse. The extinction of a specific species is morally abhorrent, but the collapse of entire marine ecosystems is a disaster – not only for people with a strong environmental conscience but also for society as a whole.

The Ministry for the Environment has recorded that many biogenic habitats are under threat or degrading, including seagrass meadows, kelp forests, bryozoan thickets, corals, shellfish beds and tubeworm mounds.¹⁰³ The Parliamentary Commissioner for the Environment has recently investigated our estuaries, finding that many are in a poor state across a variety of habitat types.¹⁰⁴ Deep-sea habitats also face threats, but the proximity to land and heavy use by people means nearshore marine environments face the greatest pressures and rates of change. For example, environmental reporting has pointed out that green-lipped mussels in Ōhiwa Harbour declined in number from 100 million to half a million in just one decade. They were also:¹⁰⁵

once a dominant habitat growing on soft sediments in areas like the Firth of Thames, Hauraki Gulf, and the Kaipara Harbour. By the end of the 1970s, they were considered mostly ecologically extinct from soft sediment environments.

Our oceans are facing a number of problems. In particular, a number of species are declining, threatened or endangered, and ecosystems are experiencing significant stress. We are faced with the prospect of species extinctions and exceeding ecological tipping points, with flow on effects for human health and wellbeing.

While ecosystem degradation is not uniform across the country, enough is known to conclude that it is a reality. What is sometimes harder to establish is what exactly is causing it, and in what measure, in a particular location. Pressures on ecosystems come from all directions; there is no simple chain of causation. Yet wherever we look, there are human-induced pressures such as fishing, invasive species, plastics, sediment, nutrients, chemical pollution, tourism, coastal development and climate change. These pressures compound each other in the marine space.

Fishing

Although Aotearoa New Zealand is often held up as a poster child for sustainable fisheries management, it is not abundantly clear that we deserve that distinction, and in any case that status is relative to other countries, some of which are struggling even more. In some areas we are doing well. Reporting points out that 83 percent of *routinely assessed* fish stocks were, in 2020, considered within safe limits.¹⁰⁶ However, the remaining 17 percent were considered overfished, and nine stocks were considered to be collapsed.¹⁰⁷ Periodic collapse and rebuild of even this number of stocks – which are part of wider ecosystems – is not ideal. For example, East Coast tarakihi fish stocks were thought to be less than 16 percent of their virgin biomass in a 2019 stock assessment.¹⁰⁸ Although these stocks are now (hopefully) entering a period of rebuild, the fact that stocks were allowed to get so low, more than 30 years after their introduction into the QMS, is indicative of a system that is far from perfect.

It is an even more dire situation with some shellfish stocks. The biomass of historically important scallop beds in the Hauraki Gulf and Spirits Bay are now only five and two percent respectively of what they were when the last surveys were undertaken a decade or more ago, and most of the area has now been closed to fishing.¹⁰⁹ Moreover – and most significantly – many stocks are *not* routinely assessed, which is a hugely important caveat.¹¹⁰ And for those that are, there are questions about the adequacy and currency of the information on which assessments are based.¹¹¹ The ecological reality for many stocks remains murky.

Local depletion also matters. Stocks are assessed across very large areas (quota management areas (QMAs)), and while they may be deemed healthy overall, that may not be the case in particular places. Fish populations are part of local ecosystems and provide food and other values for local communities. In other words, it matters *where* fish are present within the areas in which stocks are managed. Communities and tangata whenua have responded to such concerns by calling for rāhui and closures to rebuild local kai moana resources.¹¹²

The point in time that fishing occurs in relation to the life cycles of fish is also important, as is the size and age of the fish that are caught. As the Ministry for the Environment has pointed out:¹¹³

Fishing changes the population structure of a species as well as reducing the overall number of fish. Fishing changes behaviour, leads to different size or sex ratios, and can affect population genetics. Population changes can have cascading effects through the food

web by affecting the dynamics of predation, food availability, and competition for food and habitat.

Cascading impacts of harvesting can be seen where the depletion of adult snapper and crayfish (which eat kina) results in high kina numbers, which in turn feed on kelp. In some cases, this can cause kelp to disappear altogether, tipping an ecosystem into a completely different state (described as a “kina barren”). This affects not just the kelp, but the myriad of other marine species that rely on the kelp as both habitat and a food source.¹¹⁴ The phenomenon has been notable in the Hauraki Gulf and Bay of Islands. Luckily we know that, in some cases, the system can slowly recover once fishing pressure is removed.¹¹⁵

How we harvest fish also matters. Commercial fishing methods like bottom trawling and dredging not only remove vast quantities of marine life, they can also damage the underlying biogenic habitat (eg living reef structures), change the non-living structure of the seabed, and resuspend sediment that then blocks light and smothers remaining benthic organisms. Recovery from such activities, if it happens at all, takes long periods of time; in the inner Hauraki Gulf, mussel beds have still not re-established despite dredging having not occurred for half a century.¹¹⁶ Although there has been a small downward trend over the past decade or so, large tracts of the country’s marine area are still subject to trawling. The area affected each year comprises some 90,000 square kilometres (equivalent to around 80 percent of the entire North Island). New areas are still being trawled meaning that the cumulative trawl footprint is ever increasing.¹¹⁷



Kina barren, Bay of Islands

Raewyn Peart

A spotlight on seafloor-damaging fishing methods

Some fishing gear and practices can cause physical damage to the marine environment, as a side effect of harvesting fish and shellfish. This is particularly the case when heavy fishing gear is dragged across the seafloor. Three types of fishing methods undertaken in Aotearoa New Zealand involve this practice: bottom trawling, Danish seining and dredging.

Bottom trawling is the most commonly used fishing method in the country, being used to catch just under half the country's total commercial harvest.¹¹⁸ It typically involves dragging a large net across the seafloor with associated equipment. This includes two large heavy doors which keep the mouth of the net open, bobbin chains which weigh down the leading edge of the net and roll along the seabed, and ropes and wires that connect the gear together and to the fishing vessel. Because the catch efficiency of trawling relies on close and persistent contact with the seabed, bottom trawling can have substantial adverse impacts on the seabed and sensitive habitats.¹¹⁹

Danish seining involves dragging weighted lines across the seabed in a circular formation to herd fish into the net. Because the gear dragged across the seabed is lighter than that used for bottom trawling, and the area impacted is smaller, the physical damage is less. However, it can still be significant.

Dredging involves dragging a steel box or bag across the seafloor to harvest shellfish living within the sediment. The area impacted by dredging is much smaller than with trawling, but the intensity of disturbance in affected areas is greater.¹²⁰ It does not just impact biogenic habitat – it can alter the geological structure of the seabed too.

The physical disturbance caused by dragging gear over the seabed results in a range of ecological impacts. In the first instance, organisms that grow above the seabed such as mussels, corals, sponges and bryozoans are physically damaged. This is significant because the three-dimensional benthic habitat these species create provides settlement sites for larvae, nursery areas for juvenile fish, and habitat and food sources for a wide range of organisms. Bottom contact fishing methods also suspend sediment in the water column, reducing water clarity, smothering

organisms when the sediment settles and, as heavier particles fall to the seafloor faster than fine particles, potentially changing the seafloor surface to a fine mud which (in coastal areas) is more susceptible to suspension through wave action.

Overall, benthic communities decrease in density and diversity as a result of such disturbance.¹²¹ In some cases, bottom contact methods can push the seabed over a “tipping point”, where it changes to a different (and less diverse) stable state and will not naturally recover. That includes where there has been physical alteration of the seabed itself, making recolonisation and recovery by benthic species much harder. Local examples of where dredging has helped “tip” a marine system into a different state include the Hauraki Gulf (with mussel dredging),¹²² Tasman and Golden Bays (with scallop dredging)¹²³ and the loss of epifaunal reefs in Fouveaux Strait (with oyster dredging).¹²⁴

Fishing activities are placing stress on marine ecosystems in a variety of ways, including through removal of fish from ecosystems and damage to habitats.



Scallop dredge, Whitianga harbour

Raewyn Peart

Invasive species

Other types of sea-based activities are also impacting on marine ecosystems. Invasive non-indigenous species – of which there are now upward of 200 in Aotearoa New Zealand – can predate on, compete with or crowd out indigenous species, fundamentally changing the nature of habitats and the species they support. They can also have significant impacts on human activities like aquaculture and fishing. Most commonly, and despite best biosecurity efforts, these unwanted visitors arrive on the hulls of ships or in ballast water. As shipping increases, so does the risk of both incursion and subsequent spread around our shores.¹²⁵ Once established, it has proved extremely difficult to eradicate some invasive species. A brown seaweed called *Undaria pinnatifida* has proved particularly problematic, as has the sea squirt *Puyra doppelgangera*.¹²⁶

Invasive species pose threats both to marine ecosystems and human activities in the marine environment.

Plastics

As well as being a problem from the viewpoint of particularly valued species like birds and marine mammals, and from an aesthetic point of view (they comprise the majority of litter around our coasts),¹²⁷ plastics from both marine and land-based activities are impacting on broader ecosystem health. They include micro-plastics, which are now well entrenched in the marine food chain, having been found in finfish and shellfish.¹²⁸ We do not fully understand the long-term impacts that the presence and ingestion of microplastics will have on ecosystems or human health – after all we are at the very top of the marine food web, the point where plastics have accumulated most. But some studies have made concerning findings; for example, microplastics have been linked to poor nutritional outcomes and mortality in plankton¹²⁹ and altered reproductive behaviour in shellfish like oysters.¹³⁰

In other words, microplastics (and even smaller “nano-plastics”) put additional pressure on parts of the marine food chain that are vital to healthy and stable ecosystems. Plastics are not just a problem when we discard them (eg as disposable packaging or single use nappies). They are more pervasive than that, which suggests human society may need to rethink its entire relationship with plastic products rather than just treat them as a waste stream that can be managed. For example, a significant source of plastic pollution comes from tiny fibres of what has become ubiquitous polyester and other synthetic clothing, which enters the sea

from people’s washing machines via wastewater flows. One study has suggested that up to 700,000 fibres could be released from an average six kilogram wash load of acrylic fabric.¹³¹ Some of these microplastics are small enough to bypass processing and enter directly into the marine environment.¹³² However, even particles that are removed during wastewater processing may make their way to the marine environment eventually via the disposal or use of wastewater sludge.¹³³

Plastics pose a threat to marine life in a number of ways. Perhaps of greatest concern are the risks posed by microplastics, which are now pervasive in the marine environment and the ultimate effects of which are not well understood.

Sediment

Some land-based inputs are not in themselves problematic, but it is their *extent* or *rate* that is having an adverse effect. For example, the land has always produced sediment, with significant volumes entering the coastal environment even from land that has full indigenous cover, particularly during and after storm events. Some marine environments were, prior to human settlement, already defined by a soft sedimentary seabed, and these support healthy ecosystems; erosion is a natural process. Catchments and estuaries also have highly variable natural sediment accumulation rates across the country.¹³⁴

However, the past one hundred and fifty years has seen an explosion in the volume and rate of sediment entering the marine environment. It has been noted that “New Zealand has one of the highest rates of sediment runoff in the world; equivalent to around 35 million truckloads of sediment entering the sea annually.”¹³⁵ It has accumulated, especially in some estuaries, at an unprecedented rate. Environmental reporting has noted, for example, that:¹³⁶

Inter-tidal sedimentation rates have generally increased and become highly variable since European settlement. In estuaries and harbours across the Waikato region, historical sediment accumulation rates were less than 0.5 millimetres per year. After European settlement, rates became unstable, reaching almost 200 times historical rates.

This is far from an isolated example.¹³⁷ We cannot place the blame at a door of a single sector or industry; contributions come from many sources, including agriculture, horticulture, commercial forestry and urban

development. Earthworks, often involving mass clearance of vegetation and deep disturbance of the soil, have had an impact disproportionate to the land area they occupy.¹³⁸

Irrespective of where it comes from, large amounts of deposited sediment can smother, stress and kill benthic life. Suspended sediment (contained in the water column) can impact the amount of light reaching photosynthetic species on the seabed such as seaweed,¹³⁹ and impact fish spawning and survival.¹⁴⁰ Food webs are affected by all of this, in negative and often unpredictable ways. And (as described later), especially in urban and intensive farming environments, sediment can also bring with it an increasing, and increasingly varied, confection of chemical contamination flowing from land and waterways.¹⁴¹ The seas get two types of pollutants for the price of one.

Planting trees has undoubtedly helped to stabilise soil that would otherwise have flowed to the sea as a result of land clearance and land use change, but the purpose of the tree matters – if every 30 years we allow extensive clear felling within a catchment (of plantation trees planted all at the same time), then such planting will have been of questionable value from a marine perspective (large bursts of sediment can be more harmful

than slow release). And while local conditions matter for how much sediment ends up where (including how energetic the coastal environment is; the slope of land; the intensity and frequency of precipitation in catchments; mitigation measures; the nature of the soil; and the presence of sediment-blocking habitats like mangroves), it is evident that in many places these factors are either not well understood or are not being tailored to the needs of our marine ecosystems. Marine life can only take so much stress.¹⁴²

What resilience remains in marine ecosystems will, in many places, be further strained by a changing climate. In some places this may see heavier and more frequent rainfall (and therefore more sediment running off the land from a variety of activities) coupled with other climate-related stresses on ecosystems (see further below).

Sedimentation is a significant problem, especially in estuaries. A lot of sediment comes from land, including through urban development, forestry operations and agricultural activities.



Sediment-laden seawater at the Tukituki River mouth

Nutrients

Nutrients are a vital part of marine productivity. Indeed, the presence of nutrients explains why some areas have an abundance of marine life. However, the concentrations of nutrients entering the sea from agriculture, horticulture and (in places) aquaculture, have in some locations gone beyond what marine ecosystems have become accustomed to or are able to cope with while maintaining a stable or productive state. Intensification of agriculture over the last decade or two has exacerbated impacts.

While environmental reporting indicates that total phosphorus levels have decreased in two thirds of monitored coastal and estuarine sites,¹⁴³ and over four fifths of sites have low concentrations of chlorophyll-a (meaning that “the effects of pollution are considered to be minimal”),¹⁴⁴ many monitored sites have negative trends for total nitrogen (35 percent of sites), ammoniacal nitrogen (41 percent), and dissolved oxygen (40 percent).¹⁴⁵ This is of concern, as seawater is generally “nitrogen-limited” when it comes to producing marine life so it can be affected by increased nitrogen inputs.

Substantial and largely unchecked expansion and intensification of urban and agricultural activity, particularly in some catchments that are intensely used, has caused eutrophication – excessive nitrogen enrichment – in a number of estuaries. This can cause algal blooms, reducing oxygen levels which can kill fish, and throw food webs out of balance. Blue-green algae (cyanobacteria) can be toxic to marine life and people.¹⁴⁶

Catchments are a substantial source of nutrients entering the marine environment. While nutrients from land are an important part of the natural cycle, an excessive amount entering some places is having adverse impacts, including through the eutrophication of estuarine environments and embayments.

Chemicals and pathogens

Many other kinds of contamination come from catchments. While some overall measures of chemical pollution at a national level show signs of improvement,¹⁴⁷ it is difficult to generalise, not least because we understand so little about the long-term impacts of some pollutants. Pharmaceutical and cleaning products, antibiotics, hormones and so forth – the list is growing as we continue to concoct new chemicals – are entering our seas with potential impacts on ecosystem and human health. For example, studies have linked medicinal waste to problems with feeding, immune response and habitat attachment in shellfish.¹⁴⁸

We understand a lot more about some chemicals and pathogens due to their immediate and obvious impacts on human health. Faecal matter and the presence of disease-causing pathogens is routinely monitored and leads to action like the closure of beaches.¹⁴⁹ Wastewater overflows – where raw sewage discharges into the oceans – can occur within high density urban areas and near recreational beaches and kai moana beds.¹⁵⁰ While pastoral catchments record *E.coli* levels over a dozen times higher than indigenous forested areas, that figure is much greater in urban waterways. Much of this ultimately reaches the sea.



Wastewater pollution in Akaroa harbour

Raewyn Peart

Aside from the occasional overflow event, people may be used to thinking of wastewater as being “dirty” but acceptable (as long as it is treated). New Zealanders are also arguably conditioned to think of stormwater as just a natural and therefore “clean” phenomenon (we are simply channelling it away from where it could cause damage). Neither of those things are true from the perspective of the marine area.

Stormwater refers to the runoff of rainfall from impermeable surfaces such as roofs, driveways, footpaths and roads. In urban areas, runoff usually enters an urban stormwater network which typically pipes the water into the sea. It is not “treated”, it is simply diverted. And yet, especially in urban areas (where hard infrastructure like gutters and drains collect and channel it), its main effect is to gather significant contamination together from many surfaces into one channel, and release it into a single spot in a river, onto a beach or into a marine area. This is not just flood control; it is also waste disposal.

Stormwater can be contaminated by multiple sources including construction sites, motor vehicles, domestic properties, domestic animals and spills (see the spotlight below). Most would not give this a second thought as they hose down their chemically cleaned car, or watch as rainwater washes heavy metals off their roof.



Stormwater outlet, Titahi Bay, Wellington

A spotlight on stormwater

Stormwater carries with it a wide range of contaminants including “litter, sediment, nutrients, metals, fuels, oils, polycyclic aromatic hydrocarbons (PAHs), legacy pesticides (such as DDT, lindane, dieldrin and chlordane), legacy synthetic compounds (such as PCBs), newer emerging organic contaminants (EOCs) including pharmaceuticals and pesticides, and pathogens”.¹⁵¹ It is a cocktail of contaminants.

Stormwater usually carries sediment, which can be at high levels when earthworks are being undertaken within a catchment. The sediment, in turn, carries with it other toxicants which are bound into the soil particles. Stormwater is also often contaminated with raw sewage, which can occur when wastewater systems overflow, or when accidental or illegal connections are made between wastewater and stormwater pipes. This can result in the presence of pathogens and elevated levels of nitrogen and phosphorus. Domestic and feral animals can also introduce pathogens into the system such as toxoplasma from cats which is affecting the Māui dolphin (see earlier). In addition, litter is often carried down stormwater pipes and can result in plastic fragments entering the marine environment.¹⁵²

Roads are a major source of stormwater contaminants due to residues left by vehicles and road building materials themselves. Vehicles deposit particles of copper from brake linings and zinc from tyres on roads. Engines and exhaust systems also deposit particles of oil, grease and fuel. Exhaust gases, tyre wear, oil leaks and the wear of tar binders and asphalt on roads all contribute to the accumulation of PAHs. Historically there have been elevated levels of lead in stormwater due to its use as an antiknock additive in petrol, although this use has been banned since 1996.¹⁵³

Heavy metals also come from a wide range of other sources due to their extensive usage in the fabrication of building materials. As well as being deployed as a decorative building material, copper is used for pipes and wires, in the manufacture of metal alloys and metal plating, and as a wood preservative and fungicide. Zinc is used widely in galvanised iron roofs and as an alloy.¹⁵⁴ In Figure 2.2 below, we outline some of the contaminants that are often found in stormwater and their impacts on marine environments.

Type	Example	Effect
Litter	Plastic bags, containers, nurdles	Mortality to marine life, transport of other chemicals and organisms, visual
Sediment	Total suspended solids	Visual (fish), mortality to marine life (burial), reduction in photosynthesis and primary production
Nutrients	Nitrogen, phosphorus, nitrate	Algal blooms, oxygen depletion, toxic effect
Microbial contaminants	Pathogenic bacteria, viruses	Risk to human health when drinking freshwater, bathing and eating shellfish
Biodegradable organic materials	Oxygen depletion in rivers, lakes and coastal environments, grease	Fish death, odours
Trace organic materials	Fuels and oils, PCBs, polycyclic aromatic hydrocarbons (PAHs), solvents, detergents, other emerging organic contaminants (EOCs)	Toxic effect, aesthetics, bio-accumulation in the food chain
Metals	Mercury, lead, cadmium, chromium, copper, nickel	Toxic effect, bioaccumulation

Figure 2.2: Typical constituents present in stormwater discharges with associated effects¹⁵⁵

Wastewater and stormwater can contain a cocktail of chemicals and pathogens that pose threats to marine life and ecosystems. People do not fully understand the potential impacts that some contaminants and novel chemicals could have.

Tourism

Although it can be hard to define exactly what a “tourist” activity is, international and domestic visitors to an area can have significant impacts. Tourism contributes to cumulative environmental pressures above and beyond a single activity’s impact. That can be the case even if tourist facilities like resorts are themselves “green” – the reality is that a larger number of people in a place will cause impacts, especially in sensitive environments.

Infrastructure issues can contribute to environmental degradation. If there is not infrastructure to support tourists’ needs, this can lead to issues with wastewater and litter. Conversely, the existence of infrastructure itself can be damaging through the physical alteration of the coast. Increasing infrastructure to meet demand can in turn result in increased numbers of tourists, drawn by the presence of amenities. This can lead to a phenomenon called “recreational succession”, where as an environment becomes more degraded by tourists, people who value environmental quality move on to other locations, and the tourist population skews towards those with lower environmental standards.¹⁵⁶

Direct interaction of tourists with the marine life and habitats can also be damaging. Marine species can be affected by ship strike, recreational fishing, and noise pollution. Habitats can be directly damaged by people walking on the intertidal zone or dropping anchors onto the seabed. Urban lights can confuse hatchlings and increase mortality rates. Vehicles and pedestrians on the beach can disturb and trample nests.

Contribution to climate change is another concern for marine tourism particularly due to the impact of greenhouse gases on our oceans (see below). For most international tourists, Aotearoa New Zealand is a long-haul destination.¹⁵⁷ A tourism strategy that encourages greater visitor numbers will increase flight emissions and consequent global warming. These costs can be hidden as emissions from international flights and are not reflected in the country’s carbon budget.¹⁵⁸

Coastal development

Some of Aotearoa New Zealand’s most degraded marine environments are estuarine, particularly those with large catchments flowing into them.¹⁵⁹ That reflects the significant impact that land-based activities are having upstream. Population increases and economic growth have seen development explode along great swathes of our coastlines. This urban and infrastructure development produces significant amounts of sediment, much of which ends up in the sea. Contaminated runoff from

coastal roads and industrial sites also enters the marine environment. We are even starting to see coastal landfills being eroded to the point that they disgorge their contents directly onto beaches or into the sea. We cannot even begin to know what is in there – there are often no records at all.¹⁶⁰

Development and reclamations remove an entire physical part of the marine environment in some places. They also often push human activities and hard structures like concrete walls, wharves and port infrastructure right up to (and into) the sea, leaving little or no inter-tidal habitat for shorebirds and other creatures to live and breed.¹⁶¹ Even where some space is left, such as by setting back residential development, the risk is that rising sea levels¹⁶² and the temptation to provide hard defences like seawalls and groynes to protect private property will mean that this ecologically valuable “middle ground” will disappear over time. Conversations about managed retreat versus protection on land are,

therefore, as important for the health of the marine environment as they are to the safety of people and property.

Coastal development poses risks to the marine environment, including through the clearance of land and generation of sediment, the consequent runoff of chemicals from dense human settlement, and the physical removal or change of coastal habitat.

Climate change

While rising sea levels are one important manifestation of climate change, they may not be the most concerning from a marine perspective. There is an important physical and chemical relationship between sea and atmosphere which means that, as the planet warms, our oceans are warming too. The oceans absorb a significant amount of global heat.¹⁶³ The Ministry for the

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Exposed landfill, Kaiaua

Environment reports that over the last 40 or so years, the average increase in temperature of our seawater was 0.2 degrees per decade. That is significant and the speed of increase is likely to accelerate in the future.

With a warming ocean comes a number of potential impacts on marine life and ecosystems. Some are direct – for example, some species' growth and reproductive cycles may be impacted,¹⁶⁴ including key species like plankton that form the basis of the food chain.¹⁶⁵ Mobile species like finfish may move elsewhere (an important thing to remember for the agility of fisheries management and its implications for equity in fishing rights). Other species better adapted to increased temperatures may take their place, with uncertain consequences for habitats and food webs. Species not endowed with the ability to move, may simply disappear. Impacts may not be universally negative – for example, it is predicted that an increase in phytoplankton around the Chatham Rise may stimulate fisheries.¹⁶⁶

Yet a warmer environment may also mean some species, will no longer have the resilience to continue the fight against other stressors such as pollution, invasive species or severe storm events. Scientists have reported that in some places bull kelp, decimated by an unusually long period of warm waters during 2017-18, failed to recover in the face of competition from the invasive seaweed *Undaria*.¹⁶⁷ That could become a concerning pattern; new species have been observed making forays into inshore waters as it warms.¹⁶⁸

Climate change does not manifest uniformly across time or space. In some places,¹⁶⁹ an increasing risk of devastating marine heatwaves year on year, or an increasing frequency of severe storms, raises the prospect of ecosystems tipping over ecological thresholds abruptly rather than adapting or shifting (or declining) slowly. Some have pointed to patchy or depleted shellfish beds being more vulnerable to the physical disturbance of storms and extreme wave events – put simply, fewer and more spaced out individuals lack safety in numbers.¹⁷⁰ A lack of biological diversity in a community generally erodes resilience.¹⁷¹

Climate change does not just manifest in a warming ocean. Our oceans are also growing more acidic, a chemical reaction that occurs as the water column absorbs ever increasing amounts of atmospheric carbon dioxide.¹⁷² (Incidentally, nutrient runoff from land can also contribute to ocean acidity.)¹⁷³ Alarming, it is thought that the oceans have already absorbed up to 50 percent of global emissions since the industrial revolution.¹⁷⁴ As a result, they have already become 30 percent more acidic,¹⁷⁵ with Aotearoa New Zealand measurements showing a 7.1 percent increase in the past 20 years.¹⁷⁶ That process is continuing; oceans are

estimated to be absorbing over a quarter of the carbon dioxide in the atmosphere,¹⁷⁷ and it is predicted that pH levels around Aotearoa New Zealand may decrease by as much as 0.4.¹⁷⁸

Acidification of the oceans adds a whole additional layer of impacts on marine ecosystems. In particular:¹⁷⁹

Acidification of seawater causes major problems for marine calcifying organisms, both large (eg calcareous macroscopic seaweeds, shellfish) and small (eg individual coral polyps, microscopic phytoplankton) because the lower pH both impairs the ability to build a shell, and dissolves existing calcareous shell. Scientists have established that ocean acidification will affect all primary producers – from microscopic phytoplankton to giant kelp forests, as well as higher trophic levels, including coral reefs, shellfish and fish.

One study found that the shell of the tiny sea butterfly – said to be the “ocean’s canary in the coal mine”¹⁸⁰ – dissolved in just 45 days when placed in sea water with pH levels anticipated for the year 2100.¹⁸¹ In short, acidification may have potentially enormous impacts on the primary productivity¹⁸² of the marine environment. *Our Marine Environment* has pointed out that edible and farmed shellfish like oysters, pāua and mussels, which are valuable elements of ecosystems and have significant commercial value, are particularly vulnerable.

In short, climate change has impacts on multiple related fronts; a warming and acidifying ocean may directly reduce underlying ecosystem resilience (because species struggle to survive in those conditions), but climate change can also exacerbate the events that can end up sending those already weaker ecosystems over the edge (eg increasing frequency of storms and wave energy, the movement of invasive species, and increasing large scale sediment runoff events from land).

It is not the role of this report to go into detail about where anthropogenic greenhouse gas emissions are *coming from*. Suffice it to say, most are from land, rather than marine activities. Indeed, most are from other countries.¹⁸³ Notwithstanding this, there are opportunities and imperatives to reduce the discharge of greenhouse gases at sea. The examples that may come to mind most readily are emissions from shipping and fishing vessels, which rely on fossil fuels. But research is now highlighting the impact that bottom trawling can have on greenhouse gas emissions. One recent international study, published in *Nature*, found that bottom trawling produces as much carbon dioxide globally as the entire aviation industry, through releasing it from the seabed into the water column.¹⁸⁴

The marine environment has the capacity to sequester significant amounts of carbon if we look after it – it has been suggested that it may even be higher than our terrestrial forests.¹⁸⁵ Yet warming of the ocean will reduce its capacity to absorb atmospheric carbon, and there is the risk that habitat change due to climate and other stressors may result in the sequestration capacity of marine plant life and other photosynthetic organisms being reduced or lost. In other words, we might lose a significant carbon sink without realising it, contributing to a vicious feedback loop of warming that impacts the oceans even more. Restoring the biodiversity of marine environments has the potential to mitigate climate change; degrading it has the potential not just to forego that opportunity but also to significantly add to emissions by removing carbon sinks.

Climate change poses potentially the greatest threat to our marine environment, through warming seas and acidification. Its impacts are also highly unpredictable. Our use of the marine environment has the potential to reduce or increase the emission of greenhouse gases and the ocean's sequestration potential.

Cumulative impacts

Human pressures on the marine environment interact in unpredictable ways. Stressors from land, sea and climate interact with natural processes and features in particular places over time to produce highly complex

and uncertain outcomes. It is these cumulative impacts that many have pointed out as the real crux of the problem we face.¹⁸⁶

Uncertainty about cumulative effects comes from several angles – the inherent uncertainty about the future, a lack of complete information on important indicators, uncertainty as to how people will behave, and uncertainty as to how localised combinations of stressors and features will play out. But uncertainty is not just about difficulties in predicting the future – it is also about a lack of information about the impacts that have *already* been felt. And in some cases authorities have been forced to admit we know less than we thought we did previously – for example, some taxa have been moved to a status of “data deficient” where they were previously classified.¹⁸⁷ Causal links are often uncertain, too. Although authorities know that many habitats are declining and that species are threatened, and it can be said with confidence that many human activities are contributing to this, it is much more tricky to identify precisely what activities are to blame and to what extent.

The overall effect of cumulative effects is that ecosystems become less diverse, they may shift suddenly to less productive states, they may no longer support species that people value, and they will often fail to provide ecosystem services. The case of the Firth of Thames is a cautionary tale about the cumulative effects that multiple pressures can have on the marine environment, and how impacts are often one step (or more) ahead of our understanding.



Mussel farm, Coromandel

A spotlight on cumulative effects in the Firth of Thames

By all accounts, the Firth of Thames was once a thriving, highly productive and biodiverse marine area. When early spar ships and traders visited the area in the late 1700s they reported immense kahikatea forests and a Waihou River that “abounds in salmon, flounders, breams, soles, and many other fish; also great quantities of crabs, clams, etc”.¹⁸⁸ The extensive catchment draining into the Firth contained the country’s largest natural wetland system which had developed over thousands of years.¹⁸⁹ Sadly, subsequent impacts of forest clearance, wetland drainage, fishing and other human activities have significantly degraded the Firth’s marine environment, and these impacts have been cumulative over time and across different stressors.

From the late 1880s up until the 1920s the landscape of the Hauraki Plains was fundamentally transformed with the felling of trees, drainage of the wetland and conversion of much of the land to dairy farms. This meant that the natural drainage services provided by the wetland, which had filtered sediment and nutrients out of the runoff before it entered the Firth, were critically compromised. The steep land on the Coromandel Peninsula was also logged and mined, further increasing sediment flows.¹⁹⁰ Later flood control works, which straightened the Waihou River and reduced flooding, further served to channel contaminants from the land directly into the sea with little filtering buffer.¹⁹¹

Extensive hard rock mining around Thames and elsewhere on the Coromandel Peninsula took place between the 1890s and 1950s. During this period, substantial quantities of mine waste were dumped directly into the Firth of Thames and mine tailings were also discharged into the Waihou and its tributary rivers which drain into the Firth. These mine tailings typically contained lead, zinc, copper and arsenic. Recent investigations have identified significantly enhanced levels of lead and zinc in sediments across a widespread area of the southern Firth of Thames. Although the levels measured are not high enough to currently pose a threat to marine life, there could be threats to the ecosystem if they were resuspended in the water column by such activities as dredging or bottom trawling.¹⁹²

Much of the seafloor of the Firth of Thames used to be encrusted with thick green-lipped mussel beds. These are thought to be one of the most valuable habitats in the Hauraki Gulf due to the critical role they play within the local ecosystem. “They provide attachment surfaces for algae and immobile invertebrates, refuge for small mobile vertebrates, foraging areas for adult fish and probably act as important habitats for juvenile fish”.¹⁹³ They also filter large quantities of seawater, with the Firth of Thames mussel beds estimated to have potentially filtered the entire water volume of the Firth in less than a day.¹⁹⁴ During the early 1900s, a commercial mussel industry developed. Initially the mussels were hand picked off the rocks, but then steel dredgers were used to harvest them from the seabed. By the early 1960s the stocks had collapsed, the mussel reefs were gone and the seafloor reduced to soft mud. Despite the cessation of fishing, the mussel beds have never recovered, indicative of a marine ecosystem that has gone over an ecological tipping point.¹⁹⁵

These mussel reefs have, in part, been replaced by extensive mussel farms in the Firth of Thames, where mussels are grown on ropes suspended in the water column. However, locating mussels within the water column, and regularly harvesting them, does not provide the same ecological services as permanent natural beds located on the seafloor. Monitoring has indicated that the farms have had only minor impacts on the Firth’s marine ecosystems, with a small reduction in phytoplankton levels, small impact on water temperature and no impact on the clarity of the water (which suggests the mussels are not improving water clarity through their filtering activity, possibly due to the breakdown of the pseudofaeces which binds the filtered sediment once released from the organism).¹⁹⁶

At the same time as the mussel reefs were being exploited, accelerated sediment flows from the Hauraki Plains were also impacting the seabed of the southern Firth of Thames. The area had historically featured gently sloping muddy-sand flats that were largely free of mangroves. But around the mid-1940s, there was a marked shift in the composition of the sediment which changed from sand to mud. This was followed by a rapid expansion of mangroves which were colonising the changed environment. The mangroves are efficient land builders and have since expanded around a kilometre into the Firth. Mangroves

play an important role in the marine ecosystem, but in the Firth of Thames they have replaced “ecologically diverse and productive open intertidal sand and mud” and have reduced the area suitable for roosting shorebirds.¹⁹⁷ The excess flows of sedimentation are likely to be exacerbated by climate change with potential increases in the frequency and intensity of storms.¹⁹⁸

Further seabed impacts to the outer Firth of Thames were caused by bottom trawling undertaken by commercial fishers. For example, in the first half of the 20th century, fishermen reported an extensive bed of horse mussels on a deep water shelf extending from the north of Coromandel Harbour up to Port Jackson and across to Waiheke Island. Longliners fishing above the beds reported large catches, indicating the productivity of the beds. During the 1950s, many commercial fishers in the Hauraki Gulf converted to bottom trawling and “with the trawler boards and sweep wires, they knocked the top off every horse mussel and in about five years they killed the lot”.¹⁹⁹ Horse mussel beds are now known to play a very important role within the marine ecosystem, supporting higher diversity and total abundances of marine life and providing hard strata enabling other reef-forming species such as bryozoans and sponge gardens to establish.²⁰⁰ Harvesting will also have impacted the size and makeup of fish populations in the Firth of Thames, although the extent of such impacts has not been quantified for the area.

The conversion of a wetland ecosystem to intensive dairying not only led to elevated sediment run-off but also increased nutrient loads into the Firth of Thames. “The once clear waters of the Thames, based on historical accounts from European settlement, are now considered a degraded nutrient-enriched environment”.²⁰¹ This has increased plankton production, and therefore the ecological productivity of the water. However, there are warning signs that the enrichment may have gone too far. Water monitoring in the outer Firth of Thames has indicated seasonal oxygen depletion in the bottom waters. Typically levels of oxygen were found to be reducing to around 70 percent saturation (with well-aerated water being 100 percent). In one case the low levels remained for several weeks and on another occasion a drop to 40 percent was measured.²⁰² Such oxygen drops within seawater can have adverse effects on sedentary marine life. Excess nutrients have also made the seawater more acidic.

Despite all these pressures, which cumulatively have meant that the Firth of Thames is only a shadow of its former self in terms of diversity, productivity and resilience, the marine area is still ecologically significant in the context of the broader Hauraki Gulf. For example, recent research has identified the Firth as being an important snapper nursery area.²⁰³ It continues to support important snapper and flounder fisheries and a large aquaculture industry. It also hosts a large population of migratory shorebirds. Efforts are now underway to restore the mussel beds in the Hauraki Gulf, but these seem unlikely to benefit the Firth itself until the existing stressors, such as sediment flows, are reduced.

The marine environment faces many pressures such as land-based pollution, fishing impacts on habitats and climate change. Often a range of stressors are concentrated in the same place, and act cumulatively and unpredictably. Cascading impacts can send an ecosystem over a tipping point into a completely different state.



Raewyn Peart

Dairying on drained wetlands in the Hauraki Plains

2.7 Consequential social, cultural and economic problems

Biophysical problems can also be seen as social, cultural and economic problems. They are of concern not just because we need to retain ecosystems intact and save species from extinction, but also because we want to protect a sustainable marine economy, cultural practice and integrity, and social and recreational opportunities that flow *from* them. For example:

- The pollution and invasive species that affect ecosystems (and which are exacerbated by a changing climate) can equally threaten the viability of activities like aquaculture and wild fish harvest. That is an economic problem as well as an “environmental” one, because the resource becomes less productive or unfit for consumption or use. Cumulative impacts on fish reproduction and mortality can be put in the same boat, as can the risks that increased storm frequency and marine heatwaves pose to all sorts of marine activities and infrastructure.²⁰⁴
- The contaminants that flow into an estuary, or out of an outfall, may not push a habitat beyond an ecological tipping point, but may make people sick if they eat shellfish or swim. This is a public health problem as well as an “environmental” one.
- The disappearance of an inter tidal habitat is ecologically destructive, but it also excludes people from accessing te moana. That is a social problem as well as an “environmental” one.
- The localised depletion of some stocks – eg tarakihi and scallops – may be as much a problem for community access to a resource as it is for ecological health.

All of the above – indeed, all the problems outlined in this chapter – impact on Māori as mana moana, as kaitiaki, and as the human link in the integrated cosmogony of te ao Māori. In other words, biophysical problems are inextricably bound up with cultural belief and cultural practice, and cannot be pigeon-holed as “just” a series of environmental issues. That, if nothing else, automatically makes the health of te moana a pressing te Tiriti issue and of broader importance than just environmental protection. For example, “the decline in fisheries is impacting the passing on of stories and knowledge that was part of the communal experience of collecting, preparing and eating local foods. ... Younger generations now have less familiarity with the foods that are part of tribal tradition.”²⁰⁵

Threats to marine life and ecosystems have significant flow-on effects for people who rely on ecosystem services and who have spiritual and intangible relationships with te moana.



Raewyn Peart

Recreational fishing, Coromandel

2.8 Non-biophysical issues in our marine environment

A quite different set of issues, which is not concerned with the marine environment or ecosystem health per se, plays out *in* the marine environment. They can be described as “social” and “economic” issues, for want of a better term. In other words, such issues would still exist even if biophysical outcomes were optimal. It is often less clear whether, or the extent to which, these are problems because there is often a fierce contest of values (about whether or not something *should* change). It is also worth noting that in te ao Māori it is not always easy to distinguish between “biophysical”, “social” and “economic” issues. This is because the health of the moana is about a whakapapa relationship, just as concepts like fairness and justice are about relationships between people.

For one, there are valid questions about whether we, as a nation, are making the most out of our seas. Should we be seeking to extract more *value*, not just more *stuff*, from them? Is this something that people, and the management system, should be concerned with? It is an increasingly pertinent question as the marine environment becomes more congested, and as technology allows us to use it for different and potentially mutually exclusive things (eg offshore aquaculture, energy generation, deep-sea mining, nutraceuticals).

Another social issue is whether the value generated by our use of the marine environment is distributed fairly. This can manifest in various ways. For example, is it fair that most coastal occupation rights are determined on a first in, first served basis? And that private interests can use marine resources – which for the most part belong to no one²⁰⁶ – with no obligation to return part of their value through resource rentals or the like?²⁰⁷ Is it fair that customary title and rights are taking so long to resolve? And that there is a lack clarity over how access to shared stocks will be allocated between customary, recreational and commercial fishers? Despite full and final settlement of commercial fishing and aquaculture rights for Māori, te Tiriti issues continue on this front (eg how the settlement relating to aquaculture space will be implemented at a regional level;²⁰⁸ how the ongoing Treaty partnership between iwi and the Crown will be given effect to in fisheries management;²⁰⁹ whether tools such as taiāpure adequately enable rangatiratanga).²¹⁰

One of the more controversial issues has been the privatisation of fishing rights through the QMS (for a description of the QMS, see Appendix 1).²¹¹ Some have pointed to issues around the concentration of wealth, suggesting that quota are now inequitably distributed and excessively aggregated to large corporate entities. This potentially has implications



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Harvest of snapper, Hauraki Gulf

for how efficiently the market works in terms of access to the fishery and share of the value chain.²¹² As economist Dr Tim Denne has pointed out:²¹³

Value will flow to the owner of the scarcest resource, which for most fisheries is quota shares ... Where there is over-capacity in vessels and marine space is not limiting, value will always flow to the owners of quota. The value will be set by the discounted expected future willingness to pay for ACE [annual catch entitlement] and where there is no scarcity of vessels, this is expected to reflect the difference between the expected revenue from fish sales and the short or medium run marginal costs of fishing (labour, fuel and equipment replacement). Value will only sit with vessel owners when they become the scarce resource.

Another commentator puts it in less economically focused terms:²¹⁴

Quota landlords now hold all the power, the middlemen clip the ticket, and the fishermen doing the hard yards only earn a fraction of what the fish is worth.

Those doing the fishing usually do not hold quota,²¹⁵ and because of it are generally worse off. So too are local communities traditionally reliant on fishing and the income it brings, who have lost their commercial fishing fleets. Recent Cabinet papers have signalled concerns about whether those actually harvesting the fish are receiving a fair return for their efforts and an intention to tackle such issues in some shape or form.²¹⁶

A number of social and economic problems can be seen playing out in our oceans. There are questions as to whether we are making the best use of our commons, and whether the value generated through using it is being fairly distributed.

2.9 Concluding comments

In this chapter we have looked at Aotearoa New Zealand's marine environment, and the problems it is facing. It is not surprising that our marine ecosystems are under increasing stress. If anything, the most remarkable thing is that many have remained as resilient as they have. But there is no guarantee they will remain so in the future.

It is difficult to say which problems are more important and urgent than others, as most are interconnected. However, a decade ago, a group of scientists had a go at ranking threats to Aotearoa New Zealand's marine habitats. Across all habitats, after two key climate-induced changes (ocean acidification and seawater warming), bottom trawling was ranked as the greatest threat, followed by sediment, further climate change impacts (changes in currents and increased storminess), dredging harvest methods and the dumping of dredge spoils. Invasive species were also highly ranked.²¹⁷ Although not definitive (the assessment was largely based on expert knowledge rather than scientific data), it does serve to provide some indication of where an oceans management system may need to focus its efforts.

For some things, damage may have been instigated a long time ago and legacy effects are inevitable in the future. For example, marine habitats are slow to recover from trawling impacts and it takes time to replace

vegetation cover in a catchment. Elements of climate change now seem inevitable. We will need to play a long game in resolving such things, and measure our success accordingly. But this also reminds us that what we allow to happen now will have impacts that reverberate down the years. A decision to clear-fell forests (or plant in the expectation of doing so), or to overstock dairying land, or to allow novel chemicals and plastics into a waste stream from which they may not be filtered out, will be experienced by future generations. What we do now matters.

Leaving the status quo will inevitably result in worse environmental outcomes. Although there is a historical legacy of damage that we need to deal with, much damage has also occurred under the current management system and will continue to do so unless we change it. It is to our existing system that we turn in the following chapter, to look at what is going wrong with it.



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New Zealand sea lion, Otago Peninsula

Endnotes

- 1 Simon Winchester *Land: How the Hunger for Ownership Shaped the Modern World* (William Collins, 2021) at 7.
- 2 For example, if a problem is “too few marine reserves”, then objectives include “more marine reserves”.
- 3 Approximately 65 percent of New Zealanders live within five kilometres of the sea: Organisation for Economic Co-operation and Development (OECD) *Responding to Rising Seas: OECD Country Approaches to Tackling Coastal Risks* (OECD Publishing, Paris, 6 March 2019) as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
- 4 From a baseline defined in international law (under the United Nations Convention on the Law of the Sea), which has complexities around how to draw lines across harbours, embayments, estuaries and so forth. On international law, see Chapter 3.
- 5 This includes its underlying continental shelf/seabed. Under international law, the EEZ and its underlying continental shelf are defined in separate but overlapping ways. There is a further overlay of a “contiguous zone” between 12 and 24 nautical miles, which provides Aotearoa New Zealand with enhanced jurisdiction over particular matters like customs and sanitation.
- 6 Commission on the Limits of the Continental Shelf (CLCS) *Recommendation of the Commission on the Limits of the Continental Shelf (CLCS) in regard to the Submission Made by New Zealand 19 April 2006* (United Nations, 22 August 2008). See also United Nations Convention on the Law of the Sea 1833 UNTS 397 (opened for signature 10 December 1982, entered into force 16 November 1994), art 76(8).
- 7 Although it is important to note that jurisdiction over the extended continental shelf does not give corresponding jurisdiction over the water column above it, which remains the “high seas” under international law.
- 8 Te Ahukaramū Charles Royal “Kaitiakitanga – guardianship and conservation - Understanding kaitiakitanga” (24 September 2007) Te Ara - the Encyclopedia of New Zealand <www.TeAra.govt.nz/en/artwork/11542/gods-of-the-natural-world>.
- 9 Te Ahukaramū Charles Royal “Tangaroa – the sea - The importance of the sea” (12 June 2006) Te Ara - the Encyclopedia of New Zealand <<http://www.TeAra.govt.nz/en/tangaroa-the-sea/page-11>>.
- 10 *Mana whenua* means customary authority exercised by an iwi or hapū in an identified area: see Resource Management Act 1991, s 2.
- 11 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 2.
- 12 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 3.
- 13 Much of this section draws from EDS's previous publication: Lucy Brake and Raewyn Peart *Sustainable Seas: Managing the marine environment* (Environmental Defence Society, Auckland, 2015).
- 14 Greg Severinsen, Raewyn Peart and Bella Rollinson *The Breaking Wave: A conversation about reforming the oceans management system in Aotearoa New Zealand* (Environmental Defence Society, August 2021), available from <<https://eds.org.nz/wp-content/uploads/2021/11/Oceans-Reform-Paper.pdf>>
- 15 Cyanobacteria are important too – these are photosynthetic bacteria (they make their own food) but are not part of the plant kingdom.
- 16 Lucy Brake and Raewyn Peart *Sustainable Seas: Managing the marine environment* (Environmental Defence Society, Auckland, 2015).
- 17 Project AWARE Foundation *Aware: Our World, Our Water* (eBook ed, Kindle, 2009) at 2-10.
- 18 Lucy Brake and Raewyn Peart *Sustainable Seas: Managing the marine environment* (Environmental Defence Society, Auckland, 2015).
- 19 Lucy Brake and Raewyn Peart *Sustainable Seas: Managing the marine environment* (Environmental Defence Society, Auckland, 2015) at 14.
- 20 Paddy Ryan “Eels – Life cycle and breeding grounds” (24 September 2007) Te Ara – the Encyclopedia of New Zealand <<http://www.TeAra.govt.nz/en/eels/page-3>>
- 21 Dennis Gordon and others “Marine biodiversity of Aotearoa New Zealand” (2010) 5(8) PLoS One e10905.
- 22 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 10.
- 23 Ministry for the Environment *Environment New Zealand 2007* (ME 847, December 2007) at 316.
- 24 Lucy Brake and Raewyn Peart *Sustainable Seas: Managing the marine environment* (Environmental Defence Society, Auckland, 2015) at 10.
- 25 All animals are members of the Kingdom Animalia, also called Metazoa.
- 26 Any organism whose cells contain a nucleus and other structures enclosed within membranes.
- 27 Single-celled eukaryotes (organisms whose cells have nuclei) that commonly show characteristics usually associated with animals.
- 28 Dennis Gordon (ed) *New Zealand Inventory of Biodiversity Volume Three: Kingdoms Bacteria, Protozoa, Chromista, Plantae, Fungi* (Canterbury University Press, Christchurch, 2012)
- 29 Rāwiri Taonui, “Muriwhenua tribes” (8 February 2005) Te Ara - the Encyclopedia of New Zealand <<http://www.TeAra.govt.nz/en/muriwhenua-tribes>>. See also Department of Conservation “Cape Reinga/Te Rerenga Wairua heritage” <www.doc.govt.nz/parks-and-recreation/places-to-go/northland/places/te-paki-recreation-reserve/cape-reinga-te-rerenga-wairua/heritage/>.
- 30 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 10.
- 31 See Hauraki Gulf Forum *State of our Gulf 2020* (State of the Environment Report 2020, February 2020) at 6.
- 32 Ministry for the Environment *Environment Aotearoa 2019* (ME 1416, April 2019) at 88.
- 33 Rodney Yeoman, Doug Fairgray and Brian Lin *Measuring New Zealand's Blue Economy* (report prepared by Market Economics Ltd for Sustainable Seas and University of Auckland, published online, September 2019) at 5. The statistics presented in this report were prepared using data that were collected prior to any Covid-19 impacts.
- 34 Minister for Oceans and Fisheries and Minister for Conservation “Revitalising the Hauraki Gulf – Government Sea Change Strategy” (2 July 2021) at [13].
- 35 Statistics New Zealand *Environmental-economic accounts: Sources and methods* (Wellington, June 2019).
- 36 Statistics New Zealand “Marine economy: 2007-17 (updated 27 June 2019)” at Tables 1-2, on Statistics New Zealand “Environmental-economic accounts: 2019 – tables” (14 February 2019) <www.stats.govt.nz/information-releases/environmental-economic-accounts-2019-tables>.
- 37 Department of Conservation and Fisheries New Zealand *National Plan of Action – Seabirds 2020* (Fisheries New Zealand, November 2019) at 13. Credit: Tim Denne.
- 38 In the 2019-2020 fishing year. See Minister for Oceans and Fisheries “Fisheries system reform agenda” (2 July 2021) at [19].
- 39 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 3.
- 40 See Ministry for Primary Industries *Aquaculture Strategy* (2019); Ministry for Primary Industries *Aquaculture Strategy: 2021 Implementation Plan* (2019); and further Ministry for Primary Industries <www.mpi.govt.nz>.
- 41 Environmental Protection Authority *Decision on Marine Consent Application: Chatham Rock Phosphate Limited: to mine phosphorite nodules on the Chatham Rise* (February 2015).
- 42 New Zealand Steel “The History of Ironsand” <www.nzsteel.co.nz/new-zealand-steel/the-story-of-steel/the-history-of-ironsand/>.
- 43 See Taranaki Regional Council “Oil & gas – production” <www.trc.govt.nz/council/plans-and-reports/monitoring-reports/consent-compliance-monitoring-reports/oil-and-gas-production/>.
- 44 Statistics New Zealand “Marine economy: 2007-17 (updated 27 June 2019)” at Tables 1-2, on Statistics New Zealand “Environmental-economic accounts: 2019 – tables” (14 February 2019) <www.stats.govt.nz/information-releases/environmental-economic-accounts-2019-tables>.
- 45 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 39. Shipping accounted for 37 percent of the marine economy.
- 46 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 7.
- 47 Rodney Yeoman, Doug Fairgray and Brian Lin *Measuring New Zealand's Blue Economy* (report prepared by Market Economics Ltd for Sustainable Seas and University of Auckland, published online, September 2019) at 5. The statistics presented in this report were prepared using data that were collected prior to any Covid-19 impacts.
- 48 Tracey Neal “Whale Watch Kaikōura expands with unique new boat following post-quake growth” *RNZ* (online ed, 13 June 2019).
- 49 At 6.
- 50 For example, a recent Cabinet paper suggested that “there are also small scale individual fishers who are looking beyond traditional fishing approaches and using more selective harvesting practices. They are creating new markets that allow them to create far greater value while leaving more fish in the water.” Minister for Ocean and Fisheries “Fisheries Amendment

- Bill: Strengthening fishing rules and policies: landings and discards" (2 July 2021) at [22].
- 51 See Venture Taranaki and Elemental Group *Offshore Wind: A New Energy Opportunity For Taranaki Discussion Paper* (2020); Ian Mason and Giacomo Caleffi "Developing Offshore Wind in New Zealand: Technical, socio-economic and environmental issues in relation to a post-pandemic future" (presented to 14th OERC Symposium, University of Otago, November 2020); Sustainable Seas National Science Challenge "Energy from tidal currents – Kick-starting a new marine industry with huge potential" <www.sustainableseaschallenge.co.nz>.
- 52 Such as difficulties in construction and repair in a challenging marine environment, costs of connection to electricity grids, and the need to compete with alternative generating activities on land.
- 53 Jamie Gray "NZ Super to explore large-scale offshore wind energy generation off South Taranaki coast" NZ Herald (date unknown) <www.nzherald.co.nz/business/nz-super-to-explore-large-scale-offshore-wind-energy-generation-off-south-taranaki-coast/UUWMHW5G5P3SHLSBFZYSX7XLRM/>.
- 54 *Trans-Tasman Resources Ltd v Taranaki-Whanganui Conservation Board* [2020] NZSC 67 granted leave to appeal to the Supreme Court.
- 55 Barry Barton, Kimberley Jane Jordan and Greg Severinsen "Carbon capture and storage: Designing the legal and regulatory framework for New Zealand" (Centre for Environmental, Energy and Resources Law, University of Waikato, 2013) and EnviroStrat Ltd *Sustainable Seas, National Science Challenge – Transitioning to a Blue Economy: Scoping and Horizon Scanning* (December 2019) at 24. See further Sustainable Seas National Science Challenge "Building a Seaweed Sector" <www.sustainableseaschallenge.co.nz>.
- 56 Barry Barton, Kimberley Jane Jordan and Greg Severinsen "Carbon capture and storage: Designing the legal and regulatory framework for New Zealand" (Centre for Environmental, Energy and Resources Law, University of Waikato, 2013).
- 57 Jim Robbins "As water scarcity increases, desalination plants are on the rise" *Yale Environment* 360 (11 June 2019).
- 58 Water Corporaon "Desalination" <https://www.watercorporation.com.au/Our-water/Desalination>.
- 59 Mohamed A Dawoud and Mohamed M Al Mulla "Environmental impacts of seawater desalination: Arabian Gulf case study" (2012) 1(3) Int J Environ Sustain 22 at 27.
- 60 See Cecilia Tortajada and Pierre van Rensburg "Drink more recycled water" (2020) 577 Nature 26-28, which recounts residents around the world successfully opposing wastewater recycling schemes due to safety concerns.
- 61 Sustainable Seas National Science Challenge "Ngā Tohu o te Ao: Maramataka and marine management" <www.sustainableseaschallenge.co.nz/our-research/nga-tohu-o-te-ao-maramataka-and-marine-management/>.
- 62 Environment Canterbury "Mahinga Kai" <www.ecan.govt.nz/your-region/farmers-hub/fep/mahinga-kai/>.
- 63 This can cause clashes with respect to, for example, marine protected areas where some forms of use are forbidden or restricted.
- 64 See Te Aka Māori Dictionary "rohe" <www.maoridictionary.co.nz>
- 65 See *Re Edwards (Te Whakatohea (No.2))* [2021] NZHC 1025.
- 66 Louise Hunt *Economic Impact Analysis of the Cape Rodney Okakari Point (Leigh) Marine Reserve on the Rodney District* (Department of Conservation, Investigation number 4052, June 2008) at 9.
- 67 J Wynne-Jones and others *National Panel Survey of Marine Recreational Fishers 2017-2018* (Fisheries New Zealand, New Zealand Fisheries Assessment Report 2019/24, July 2019).
- 68 John Holdsworth and others *Recreational Fishing in New Zealand: A Billion Dollar Industry* (New Zealand Marine Research Foundation, March 2016) at 3; and compare J Wynne-Jones and others *National Panel Survey of Marine Recreational Fishers 2017-2018* (Fisheries New Zealand, New Zealand Fisheries Assessment Report 2019/24, July 2019).
- 69 J Wynne-Jones and others *National Panel Survey of Marine Recreational Fishers 2017-2018* (Fisheries New Zealand, New Zealand Fisheries Assessment Report 2019/24, July 2019).
- 70 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 3.
- 71 Kim Walshe and Jo Akroyd *Motivations and Perceptions of Seawater Recreational Fishers In New Zealand* (Ministry of Fisheries, REC9802, 2000) at 1, as cited in Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018).
- 72 Whether these things are a "service" or just another form of pollution depends on its extent and one's perspective.
- 73 Enric Sala and others "Protecting the global ocean for biodiversity, food and climate" (2021) 592 Nature 397.
- 74 See for example, Revive our Gulf "Restoring Mussel Reefs off the Hauraki Gulf" at <www.reviveourgulf.org.nz>.
- 75 Todd Prodanovich "The Battleground Below" *Surfer* (online ed, 16 July 2018).
- 76 See for example, Fiona McCormack "Fish is My Daily Bread: Owning and Transacting in Māori Fisheries" (2010) 20(1) Anthropol Forum at 19-39. See further, Valmaine Toki "Adopting a Maori Property Rights Approach to Fisheries" (2010) 14 NZJEL 197.
- 77 Annette Sykes "The Politics of the Brown Table" (The Political Studies Department and the Bruce Jesson Foundation Lecture 2010, 27 October 2010) at 2.
- 78 Jonathan Dick and others "Listening to the Kaitiaki: Consequences of the loss of abundance and biodiversity of coastal ecosystems in Aotearoa New Zealand" (2012) 1(2) MAI Journal 117.
- 79 Michael De Alessi "The Political Economy of Fishing Rights and Claims: The Māori Experience in New Zealand" (2012) 12(2) Journal of Agrarian Change 390 at 409.
- 80 See *Ngāti Maru Trust v Ngāti Whātua Ōrākei Whai Māia Ltd* [2020] NZHC 2768; and *Re Edwards (Te Whakatohea) (No 2)* [2021] NZHC 1025.
- 81 For example, the New Zealand sealion went from nationally critical to nationally vulnerable between 2013 and 2019.
- 82 Department of Conservation *Te Mana o Te Taiao – Aotearoa New Zealand Biodiversity Strategy 2020* (Department of Conservation, Wellington, August 2020) at 12.
- 83 Introductory commentary from Department of Conservation *Biodiversity in Aotearoa: an overview of state, trends and pressures* (August 2020), Marine domain – Te Whaitua Moana, at 96 – 117.]
- 84 Statistics New Zealand "Conservation status of indigenous marine species - published October 2019" (17 October 2019) at www.stats.govt.nz/indicators/conservation-status-of-indigenous-marine-species> as reported in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 16.
- 85 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 16.
- 86 See Department of Conservation, "Hector's Dolphin" <www.doc.govt.nz>.
- 87 Elisabeth Slooten and Stephen M Dawson "Delays in Protecting a Small Endangered Cetacean: Lessons Learned for Science and Management" (2021) 8 Front Mar Sci 1 at 2.
- 88 Marine Mammals Protection Act 1978, s 4(1)(b); see also interpretation, s 2(a), 'take'. Although, technically, a permit can be issued to hunt marine mammals none are currently extant.
- 89 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 37.
- 90 Minister for Fisheries "Hectors and Māui Dolphin Threat Management Plan Review – Fisheries Measures" (23 June 2020) at [7].
- 91 Raewyn Peart *Bryde's whale voluntary protocol case study* (unpublished report for the Sustainable Seas National Science Challenge, Environmental Defence Society, 2017).
- 92 Including cruise ships, although Covid-19 has drastically reduced their number and cast doubt over the future of the cruise ship industry.
- 93 S Behrens and R Constantine *Large Whale and Vessel Collisions in Northern New Zealand* (International Whaling Commission Scientific Committee, SC/60/BC/9, 2008), as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
- 94 Walker and others "Environmental Effects of Marine Transportation in "World Seas: An Environmental Evaluation" (2018), as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
- 95 Department of Conservation "Shorebirds start nesting in Eastern Bay of Plenty" (media release 6 October 2020) <www.doc.govt.nz>
- 96 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 37; see also Y Richard, E R Abraham and K Berkenbusch *Assessment of the risk of commercial fisheries to New Zealand seabirds, 2006-07 to 2016-17* (Fisheries New Zealand, New Zealand Aquatic Environment and Biodiversity Report 237, January 2020). A recent estimation which calculates seabird captures from trawl and longline fisheries found that over 3000 birds were captured in the 2017-18 year: E R Abraham and Y Richard *Estimated capture of seabirds in New Zealand trawl and longline fisheries, to 2017-18* (Fisheries New Zealand, New Zealand Biodiversity Report 249, October 2020).
- 97 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 4.
- 98 Fisheries New Zealand *Aquatic Environment and Biodiversity Annual Review 2019-20: A summary of environmental interactions between the seafood sector and the aquatic environment* (Ministry for Primary Industries, Wellington, June 2020) at 323.
- 99 Lynne Zeitlin Hale and Jeremy Rude (eds) *Learning from New Zealand's 30 Years of Experience Managing Fisheries Under a Quota Management System* (The Nature Conservancy, Arlington, 2017) at 106.

- 100 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 38.
- 101 On the impacts of plastics on marine life, see Murray Gregory “Environmental implications of plastic debris in marine settings—entanglement, ingestion, smothering, hangers-on, hitch-hiking and alien invasions” (2009) 364(1526) *Philosophical Transactions of the Royal Society B: Biological Sciences*; Chris Wilcox, Erik Van Sebille and Britta Denise Hardesty “Threat of plastic pollution to seabirds is global, pervasive, and increasing” (2015) 112(38) *PNAS*, both as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
- 102 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
- 103 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 5.
- 104 Parliamentary Commissioner for the Environment *Managing our estuaries* (August 2020).
- 105 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 22.
- 106 Fisheries New Zealand *The Status of New Zealand's Fisheries 2020* (February 2021) at 1.
- 107 Fisheries New Zealand *The Status of New Zealand's Fisheries 2020* (February 2021) at 8.
- 108 *Royal Forest and Bird Protection Society v Minister of Fisheries* [2021] NZHC 1427.
- 109 See submission prepared by the Environmental Defence Society entitled “Review of Sustainability Measures for Selected Fish Stocks - April 2022 Round” (8 February 2022) available from www.eds.org.nz; and Fisheries New Zealand *Review of Sustainability Measures for New Zealand Scallops (SCA 1 & SCA CS) for 2022/23* (Discussion Paper 2021/30, December 2021).
- 110 There are many reasons why a stock may not be routinely assessed including a lack of scientific data to inform an assessment, limited capacity within the fisheries management system to undertake assessments (requiring prioritisation), and the presence of some purely “administrative” stocks which are not fished and therefore are not assessed or managed.
- 111 Office of the Prime Minister's Chief Science Advisor *The future of commercial fishing in Aotearoa New Zealand* (February 2021) at page 137 and following.
- 112 See for example, the decision by Ngati Paoa to place a rāhui over shallow coastal waters surrounding Waiheke Island (within one nautical mile from the shore), Hauraki Gulf, Te Aorewa Rolleston “Two-year rāhui for Waiheke Island waters to protect kaimoana” RNZ (online ed, 30 January 2021).
- 113 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 40.
- 114 Nick Shears and Russell Babcock “Marine reserves demonstrate top-down control of community structure on temperate reefs” (2002) 132 *Oecologia* 131 as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 22.
- 115 Recovery of kelp has occurred on the rocky reef systems within the marine reserves at Leigh and Hahei. We only know that these systems can recover because of such reserves, which serves to demonstrate their importance in understanding the impacts of fishing activity on the broader marine environment.
- 116 Malcom Clark and others “Little Evidence of Benthic Community Resilience to Bottom Trawling on Seamounts After 15 Years” (2019) 6 *Front Mar Sci*, as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 40; LJ Paul *A history of the Firth of Thames dredge fishery for mussels: Use and abuse of a coastal resource* (Ministry for Agriculture and Fisheries, New Zealand Aquatic Environment and Biodiversity Report 94, 11 April 2012); Sean Handley and others *A 1,000 year history of seabed change in Pelorus Sound/Te Hoiere, Marlborough* (NIWA, April 2017); both as cited in Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 4.
- 117 Office of the Prime Minister's Chief Science Advisor “The future of commercial fishing in Aotearoa New Zealand” (February 2021)
- 118 At 89.
- 119 Stephen Eayrs, Tony Craig and Katherine Short *Mitigation Techniques to Reduce Benthic Impacts of Trawling: MIT2019-02 A review for the Department of Conservation by Terra Moana Limited* (Terra Moana, April 2020) at 25 and following.
- 120 Fisheries New Zealand *Aquatic Environment and Biodiversity Annual Review 2019-20: A summary of environmental interactions between the seafood sector and the aquatic environment* (Ministry for Primary Industries, Wellington, June 2020) at 397-398.
- 121 At 411; Lynne Zeitlin Hale and Jeremy Rude (eds) *Learning from New Zealand's 30 Years of Experience Managing Fisheries Under a Quota Management System* (The Nature Conservancy, Arlington, 2017) at 106.
- 122 LJ Paul *A history of the Firth of Thames dredge fishery for mussels: Use and abuse of a coastal resource* (Ministry for Agriculture and Fisheries, New Zealand Aquatic Environment and Biodiversity Report 94, 11 April 2012).
- 123 Keith Michael and others *A summary of information and expert opinion to help rebuild shellfish fisheries in Golden and Tasman Bays* (NIWA, November 2015).
- 124 H John Cranfield, Keith Michael and Ian Doonan “Changes in the distribution of epifaunal reefs and oysters during 130 years of dredging for oysters in Foveaux Strait, southern New Zealand” (1999) 9(5) *Aquat Conserv* 461.
- 125 John Darling, Leif-Matthias Herborg and Ian Davidson “Intracoastal shipping drives patterns of regional population expansion by an invasive marine invertebrate” (2012) 10(2) *Ecology and Evolution* 2557; and Hanno Seebens and others “Predicting the spread of marine species introduced by global shipping” (2016) 113(20) *PNAS* 5646; both as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 40.
- 126 See for example, Department of Conservation, pest seaweed *Undaria pinnatifida* at <www.doc.govt.nz/nature/pests-and-threats>.
- 127 “Citizen science data collected at 44 sites showed more than 60 percent of beach litter was plastic.” Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 6.
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- 129 Zara Botterel and others “Bioavailability and effects of microplastics on marine zooplankton: a review” (2019) 245 *Environmental Pollution* 98; Matthew Cole and others “Microplastic Ingestion by Zooplankton” (2013) 47(12) *Environ Sci Technol* 6646; both as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
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- 132 Royal Society Te Apārangi *Plastics in the Environment: Te Ao Hurihuri -The Changing World* (2019) at 27.
- 133 At 27.
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- 135 Minister for Oceans and Fisheries “Fisheries system reform agenda” (2 July 2021) at [34].
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- 138 For example, “earthworks accounted for 24 percent of all sediment from the Te Awarua-o-Porirua Harbour catchment, despite accounting for just 1 percent of the land area.” Greater Wellington Regional Council and others *Te Awarua-o-Porirua Harbour and Catchment Sediment Reduction Plan* (2015) as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 26.
- 139 See Ministry for the Environment and Statistics New Zealand “Our marine environment 2019” (ME 1468, October 2019) at 51.
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- 141 Barry Robertson and Leigh Stevens *Porirua Estuary: Fine Scale Monitoring 2014/15* (Greater Wellington Regional Council, 2015), as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
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- 147 “Coastal water quality is variable but generally improving nationally...” Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 6.
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- 150 See for example, Jamie Morton “Sewage overflows surged by 379% last year – Water New Zealand report” *New Zealand Herald* (online ed, 12 April 2018).
- 151 Mike Stewart and others *Literature review of the risks and adverse effects from discharges of stormwater, wastewater, industrial and trade waste, and other hazardous substances in Otago* (report prepared by Streamlined Environmental Ltd for Otago Regional Council, Report ORC1601-FINAL-v2, 10 February 2017) at 31.
- 152 Mike Stewart and others *Literature review of the risks and adverse effects from discharges of stormwater, wastewater, industrial and trade waste, and other hazardous substances in Otago* (report prepared by Streamlined Environmental Ltd for Otago Regional Council, Report ORC1601-FINAL-v2, 10 February 2017) at 31.
- 153 Earl Shaver and Alastair Suren *Assessing Impacts of State Highway Stormwater Runoff on Stream Invertebrate Communities* (New Zealand Transport Agency, October 2011) at 5; Mike Stewart and others *Literature review of the risks and adverse effects from discharges of stormwater, wastewater, industrial and trade waste, and other hazardous substances in Otago* (report prepared by Streamlined Environmental Ltd for Otago Regional Council, Report ORC1601-FINAL-v2, 10 February 2017) at 31.
- 154 Mike Stewart and others *Literature review of the risks and adverse effects from discharges of stormwater, wastewater, industrial and trade waste, and other hazardous substances in Otago* (report prepared by Streamlined Environmental Ltd for Otago Regional Council, Report ORC1601-FINAL-v2, 10 February 2017) at 31.
- 155 Figure adapted from Mike Stewart and others *Literature review of the risks and adverse effects from discharges of stormwater, wastewater, industrial and trade waste, and other hazardous substances in Otago* (report prepared by Streamlined Environmental Ltd for Otago Regional Council, Report ORC1601-FINAL-v2, 10 February 2017) at 31.
- 156 At 59.
- 157 Raewyn Peart and Cordelia Woodhouse *Te Manahuna-Mackenzie Basin and Landscape Protection* (Environmental Defence Society, Auckland, 2021) at 18.
- 158 See United Nations Framework Convention on Climate Change (UNFCCC) “Emissions from fuels used for international aviation and maritime transport” <<https://unfccc.int/topics/mitigation/workstreams/emissions-from-international-transport-bunker-fuels>>.
- 159 Bruce Dudley, John Zeldis and Olivia Burge *New Zealand Coastal Water Quality Assessment* (report prepared by NIWA for the Ministry for the Environment, February 2017). Such sites have disproportionately high concentrations of nitrogen and faecal bacteria.
- 160 See Nikki Macdonald “More than 300 old dumps at risk of coastal erosion and flooding” *Stuff Media* (online ed, 29 March 2021).
- 161 See Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 6 and 36.
- 162 Both from melting polar ice and from thermal expansion.
- 163 See Environment Protection Agency (United States) “Climate Change Indicators: Ocean Heat” <www.epa.gov>.
- 164 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 51-53.
- 165 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 6.
- 166 Matt Pinkerton, Philip Sutton and Simon Wood *Satellite indicators of phytoplankton and ocean surface temperature for New Zealand* (Ministry for the Environment, CR 387, June 2019), as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 51.
- 167 Mads Thomsen and others “Local Extinction of Bull Kelp (*Durvillaea* spp.) Due to a Marine Heatwave” (2019) 6 (March) *Front Mar Sci* 1, as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
- 168 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 51.
- 169 See for example, Terry P Hughes and others “Global warming and recurrent mass bleaching of corals” (2017) 543 *Nature* 373.
- 170 Heather Hunt and Robert Scheibling “Predicting wave dislodgment of mussels: variation in attachment strength with body size, habitat, and season” (2001) 213 *Mar Ecol Prog Ser* 157.
- 171 Tom Oliver and others “Biodiversity and Resilience of Ecosystem Functions” (2015) 30(11) *Trends in Ecology and Evolution* 673; Simon Thrush and others *A strategy to assess trends in the ecological integrity of New Zealand's marine ecosystems* report prepared by NIWA for the Department of Conservation, NIWA Client Report HAM2011-140, 2011); both as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 22.
- 172 It is, however, interesting to note that our oceans are not themselves acidic in the scientific sense of the word. Their average pH sits firmly towards the alkaline end of the scale at around 8.1.
- 173 University of Otago “What is ocean acidification?” <www.otago.ac.nz/oceanacidification/whatisoa/index.html>
- 174 University of Otago “What is ocean acidification?” <www.otago.ac.nz/oceanacidification/whatisoa/index.html>
- 175 University of Otago “What is ocean acidification?” <www.otago.ac.nz/oceanacidification/whatisoa/index.html>
- 176 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
- 177 Nicolas Gruber and others “The oceanic sink for anthropogenic CO₂ from 1994 to 2007” (2019) 363 *Science* 1193.
- 178 This is more significant than it sounds, since pH is measured on a logarithmic scale. Each unit on the pH scale represents a tenfold change in acidity. Thus “a decrease of 0.1 pH units represents a 26% increase in the relative acidity of ocean water”: Woods Hole Oceanographic Institution “The pH Scale” <www.whoi.edu>.
- 179 University of Otago “What is ocean acidification?” <www.otago.ac.nz/oceanacidification/whatisoa/index.html>
- 180 Hannah Waters “Amazing Sea Butterflies Are the Ocean's Canary in the Coal Mine” *Smithsonian Magazine* (online ed, 14 May 2013).
- 181 Pacific Marine Environmental Laboratory (PMEL) Carbon Dioxide Program at National Oceanic and Atmospheric Administration (NOAA) “What is Ocean Acidification?” Center for Environmental Visualisation <www.pmel.noaa.gov/co2/story/What+is+Ocean+Acidification%3F>
- 182 A measure of productivity in terms of energy, not in the narrower commercial sense we often deploy for “primary production” on land.
- 183 See Environmental Protection Agency (United States) “Global Greenhouse Gas Emissions Data” <www.epa.gov>.
- 184 Karen McVeigh “Bottom trawling releases as much carbon as air travel, landmark study finds” *The Guardian* (online ed, 17 March 2021).
- 185 A MacDiarmid and others “New Zealand Marine Ecosystem Services” in JR Dymond (ed) *Ecosystem services in New Zealand – conditions and trends* (Manaaki Whenua press, Lincoln, 2013), as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
- 186 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 54 and following.
- 187 At 17.
- 188 Robert McNab (ed) “Extracts from the journal of Lieut Governor King of Norfolk Island 1791-96” in *Historical Records of New Zealand: Vol II* (John Mackay, Wellington, 1914).

- 189 Hauraki Gulf Forum *State of our Gulf* (State of the Environment Report, August 2011) at 35.
 190 At 35.
 191 Raewyn Peart *The Story of the Hauraki Gulf* (David Bateman, Auckland, 2016) at 116.
 192 Sandy Boehnert and others “Historic development of heavy metal contamination into the Firth of Thames, New Zealand” (2020) 40 Geo-Marine Letters 149.
 193 Hauraki Gulf Forum *State of our Gulf* (State of the Environment Report, August 2011) at 36.
 194 MA Morrison and others *Linking marine fisheries species to biogenic habitats in New Zealand: a review and synthesis of knowledge New Zealand* (Ministry for Primary Industries, New Zealand Aquatic Environment and Biodiversity Report 130, 2014).
 195 Raewyn Peart *The Story of the Hauraki Gulf* (David Bateman, Auckland, 2016) at 244.
 196 Raewyn Peart *Farming the Sea* (Environmental Defence Society, Auckland, 2019) at 87.
 197 Hauraki Gulf Forum *State of our Gulf* (State of the Environment Report, August 2011) at 35-36.
 198 Seachange Stakeholder Working Group *Sea Change Tai Timu Tai Pari Hauraki Gulf Marine Spatial Plan* (Hauraki Gulf Forum in partnership with others, April 2017) at 261.
 199 Raewyn Peart *The Story of the Hauraki Gulf* (David Bateman, Auckland, 2016) at 233.
 200 Tara Anderson and others *Review of New Zealand's Key Biogenic Habitats* (report prepared by NIWA for the Ministry for the Environment, Client Report 2018139WN, January 2019).
 201 MA Morrison and others *A review of land-based effects on coastal fisheries and supporting biodiversity in New Zealand* (Ministry of Fisheries, New Zealand Aquatic Environment and Biodiversity Report 37, 2009).
 202 Malcom Green and John Zeldis *Firth of Thames water quality and ecosystem health: A synthesis* (NIWA, TR 2015/23, April 2015).
 203 MA Morrison, J McKenzie and R Bian *Pre-recruit (0+) snapper (Chrysophrys auratus) beam trawl and beach seine surveys of East Northland and the Hauraki Gulf (SNA 1)* (Fisheries New Zealand, New Zealand Fisheries Assessment Report 2019/72, November 2019).
 204 New Zealand King Salmon *NZK Annual Report 2018* (3 October 2018); Sanford Limited *2018 Annual Report* (15 November 2018); both as cited in Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019).
 205 Johnathan Dick and others “Listening to the kaitiaki: consequences of the loss of abundance and biodiversity of coastal ecosystems in Aotearoa New Zealand” (2012) 1(2) MAI Journal 117 at 120-123.
 206 With the exceptions of Crown owned minerals and privately held parts of the seabed.
 207 Office of the Prime Minister's Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021) at 3 and 194; nor, for example, is recreational fishing licenced.
 208 See Ministry for Primary Industries *Delivering on the Māori Commercial Aquaculture Settlement* (Guidance Note 6 of 6, October 2012); Ministry for Primary Industries “Māori commercial aquaculture claims settlement” (30 September 2021) <www.mpi.govt.nz/fishing-aquaculture/fishing-aquaculture-funding-support/maori-commercial-aquaculture-claims-settlement/>; and Te Puni Kōkiri *Te Whakataunga Ahumoana: The Aquaculture Settlement* (Information Sheet 5 of 6, May 2007) at 1.
 209 Te Ohu Kaimoana *Māori Fisheries Strategy* (February 2017) at 52.
 210 See Anne-Marie Jackson “Erosion of Māori Fishing Rights in Customary Fisheries Management” 2013 21 Waikato LR 59.
 211 The way the market operates has given rise to many issues – for example, in our publication *Voices from the Sea* we discussed the inequities and te Tiriti implications of what are known as section 28N rights under the Fisheries Act (see Peart, above n 11, at 12).
 212 Minister for Oceans and Fisheries “Fisheries system reform agenda” (2 July 2021) at 3.
 213 T Denne *Review and analysis of the QMS* (unpublished paper commissioned by Environmental Defence Society, 2021).
 214 Legasea “LegaSea is not anti-commercial fishing” (9 December 2019) <www.legasea.co.nz/2019/12/09/legasea-is-not-anti-commercial-fishing/>.
 215 Those who rely on purchasing annual catch entitlement (ACE) on a year-by-year basis form around 80 percent of the inshore fleet: Minister for Oceans and Fisheries “Fisheries system reform agenda” (2 July 2021).
 216 Minister for Oceans and Fisheries “Fisheries system reform agenda” (2 July 2021) at 5 and 6.
 217 A MacDiarmid and others *Assessment of anthropogenic threats to New Zealand marine habitats* (Ministry of Agriculture and Forestry, New Zealand Aquatic Environment and Biodiversity Report 93, 2012) at 40.

3 The current oceans management system



Bean Rock lighthouse

3.1 Introduction

The existing oceans management system should not be the *conceptual* starting point for change. Some reform options may involve upending assumptions underpinning the current system. The scale of problems we face means it is legitimate to at least contemplate revolution, not evolution. However, describing the current system is still important for a number of reasons.

1. It gives a practical sense of what the boundaries of the oceans management system are, making the definition in Chapter 1 more tangible.
2. It sets out the practical starting point from which any further reforms – whether smaller scale or transformational – would occur, and gives a sense of how much change would be involved in getting from here to there.
3. It enables the diagnosis of problems with the system, which will need to be addressed in the future.
4. It can highlight opportunities for small-scale or targeted reforms (eg where there is already a legal framing for tools that have not been used to their full potential).

We provide a more detailed summary of the existing system in Appendix 1. The system is complex and comprises much more than just our statute book. It also includes thousands of tools – from national direction and regional coastal plans, to catch limits and the QMS, to product stewardship schemes and waste levies, to mining permits and resource consents. An important element of the system is the existing te Tiriti settlements for fisheries and aquaculture.

When we speak of an “oceans management system”, however, it is important to remember there is actually more than one system in operation. The ways in which we manage our oceans are not limited to legislation, institutions and formal tools like regulations; that system is largely Western in its origins, construction and operation. Before it existed, the system in Aotearoa New Zealand was that of tangata whenua, based on tikanga.

Tikanga is more than just a te ao Māori worldview (see Chapter 7) or set of metaphysical values. It is also a practical system of resource management. The point is that tikanga should not be treated as just another “principle”



Pou, Moturua Island

to be given recognition alongside things like precaution, inter-generational equity, ecosystem-based management and so forth. It is an entire system in its own right, and comes with its own normative principles (eg kaitiakitanga, mauri, mana, utu and so forth).

A spotlight on tikanga

Centuries before Europeans reached the waters of the Pacific Ocean and the shores of Aotearoa New Zealand, Pacific peoples had developed rich understandings of, and practices within, the marine environment. Marine spatial management has historic roots in policies, processes and practices that nations have been applying to marine spaces for many centuries, including Pacific nations' use of space and time to protect and regulate marine resources.¹

There are many common threads that run through Pacific understandings and practices connected to the marine environment; they have been described as “dialects of each other”.² The relationship to Tangaroa, an ocean deity/ancestor, can be found across the region.³ Rāhui has also been used by many different cultures.⁴ The practice of rāhui is often underpinned by the concept of tapu, a prohibition that can be applied to places that must be left alone.⁵

Māori tikanga and kawa developed from the tradition of Pacific ancestors who settled on groups of small islands integrally connected to the sea.⁶ The wider cultural practices, rituals and world view of Māori were intimately entwined with the oceans because of these ancestral connections. But they were reinforced by the realities of life in Aotearoa New Zealand where the sea provided significant food sources and major transport routes.

As a Western system of marine management dominated Aotearoa New Zealand after European settlement, tikanga was eroded. Yet, although it is often constrained by the extent to which Māori have control over resources and their management, tikanga is still very much alive and operates on a day-to-day basis in Māori communities. Indigenous scholarship has also made strides in recent years to recover or better recognise traditional methods of managing the marine environment as well as traditional knowledge or wisdom (mātauranga).⁷

A spotlight on the Marine and Coastal Area (Takutai Moana) Act

A recent decision of the High Court in *Re Edwards*⁸ made some poignant comments on the intersection between the dual systems of oceans management in Aotearoa New Zealand. This was in the context of claims to the foreshore and seabed. The importance of the decision is, for present purposes, not its specific ruling but its reminder to be aware of the difference between tikanga itself and the hybrid constructs that seek to formalise it in a Western system.

The Marine and Coastal Area (Takutai Moana) Act 2011 (MACA Act) provides an avenue for tangata whenua to establish customary marine title over areas of the foreshore and seabed. In ruling on various legal matters, the Court followed the approach of the Court of Appeal in the earlier *Ngati Apa* decision whereby “the existence and extent” of customary rights was not to be gauged by “applying common law concepts but [rather] from applying tikanga.”⁹ This builds on previous decisions which recognised the importance of tikanga as *part* of the common law of Aotearoa New Zealand.

In short, the Court found that the connections between Māori and te moana are strong, and they are not presumed to have been eroded by Western interventions like raupatu (confiscation without right), occupation, or the exercise of resource consents. It found that “in terms of tikanga, the confiscation of lands and destruction of property would not have severed the connection with the takutai moana.”¹⁰ With respect to existing resource consents, it was held that “[n]othing in the RMA shows an intention to extinguish Māori customary rights.”¹¹ While activities relating to infrastructure “may well amount to substantial interruption”, whether this has occurred is to be determined by an examination of the facts in each case “not by applying a presumption”.¹²

When addressing an opening submission of counsel that “there was very little that was ‘customary’ left in the concept of customary marine title” in the legislation, the Court noted that the restoration of customary rights under the MACA Act is “given legal expression in accordance with the Act”.¹³ The upshot is that, although the MACA Act refers to reinstating pre-existing customary entitlements and translating “inherited” rights into “legal” rights and interests, the “specific rights actually conferred

by the Act are much narrower and more limited than the customary title and rights that Māori would have enjoyed and exercised in the foreshore and seabed as at 1840”.¹⁴

This means that, although the *existence* of new legal rights under the MACA Act is to be determined by (at least in part) looking at tikanga, the extent of such rights falls short of actually enabling tikanga to be exercised. It could therefore be argued that the legislation continues to *constrain* the ability to do so. However, the decision does make clear that the MACA Act does not *create* or *replace* tikanga, which continues to exist alongside legislative constructs – as it has always done.¹⁵

As mentioned in Chapter 1, we are not attempting to *reform* tikanga. We are not the ones to undertake such an exercise. In any case, tikanga evolves through practice over time, and is not amenable to sharp or formal “reforms” such as legislative amendment or case law. That said, while our work is concerned with the “formal” system of legislation, institutions and so forth, it is also about the extent to which this accommodates te ao Māori through: (1) te Tiriti o Waitangi obligations; and (2) considering where the system can best *intersect* with a parallel system based on tikanga.

Tikanga Māori is a system of oceans management in its own right, and it intersects with the system of legislation, institutions and formal tools. While it is the latter that we are seeking to reform (and not tikanga itself), part of that involves thinking about where the systems intersect and how the formal system can provide for and be informed by tikanga.

3.2 Describing the current system

We refer readers to Appendix 1 for a more extensive summary of the current system, but some high-level observations can usefully be made here. Most obviously, there is a great deal of complexity. This is reflected in the multiplicity of statutes that exist within the oceans management system, and which span it and other systems. There is not a single place like an “Oceans Act” where we can go for all things marine. We have separate frameworks for resource management, conservation, fisheries, transport, climate change, biosecurity, mining, and many other things. These are matched by an equally diverse range of institutions that administer and operate them and hundreds, if not thousands, of tools that operate under them.¹⁶

In Figure 3.1, we outline the key statutes that form the core of the current oceans management system, and show their spatial application.

Raewyn Peart



Cockle harvesting, Kawakawa Bay

Family of statutes	Statute	Spatial application		
		Land/freshwater	Territorial sea	EEZ and extended continental shelf
Resource management	Resource Management Act 1991			
	Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012			
Fisheries	Fisheries Act 1996			
	Treaty of Waitangi (Fisheries Claims) Settlement Act 1992			
	Māori Commercial Aquaculture Claims Settlement Act 2004			
	Māori Fisheries Act 2004			
	Fisheries (Quota Operations Validation) Act 1997			
Shipping	Maritime Transport Act 1994			
Biosecurity	Biosecurity Act 1993			
Conservation	Conservation Act 1987			
	Marine Reserves Act 1971			
	Wildlife Act 1953			
	Marine Mammals Protection Act 1978			
	Hauraki Gulf Marine Park Act 2000	(islands and catchments)	(specific area)	
	Fiordland (Te Moana o Atawhenua) Marine Management Act 2005		(specific area)	
	Sugar Loaf Islands Marine Protected Area Act 1991		(specific area)	
	Kaikōura (Te Tai o Marokura) Marine Management Act 2014		(specific area)	
Climate change	Climate Change Response Act 2002			
Mining	Crown Minerals Act 1991			
	Continental Shelf Act 1964			
Other	Heritage New Zealand Pouhere Taonga Act 2014			
	Marine and Coastal Area (Takutai Moana) Act 2011			
	Submarine Cables and Pipelines Protection Act 1996			

Figure 3.1: Key statutes that form the core of the current oceans management system, and their spatial application

There are also statutes that do not address marine management specifically (in the sense of activities occurring in the oceans), but which regulate or guide human activities that can have consequential impacts on it. Examples include the Land Transport Management Act 2003, Urban Development Act 2020, Building Act 2004, Litter Act 1979 and Waste Minimisation Act 2008. Similarly, the RMA deals with pollution from catchments, coastal development and noise (all of which impact on the marine space), and various conservation laws protect and manage species that travel between the sea and land/freshwater.

Finally, there is a variety of statutes that establish various aspects of the system’s architecture, but do not directly restrict or influence human activities. Some of these are marine focused, but others are not (where, for example, they create institutions or processes that span land and sea). They include the Territorial Sea, Contiguous Zone, and Exclusive Economic Zone Act 1977 (not to be confused with the EEZ Act), Environmental Reporting Act, Local Government Act 2002 (and related local government legislation), Environmental Protection Authority Act 2011 and Environment Act 1986 (see Figure 3.2).

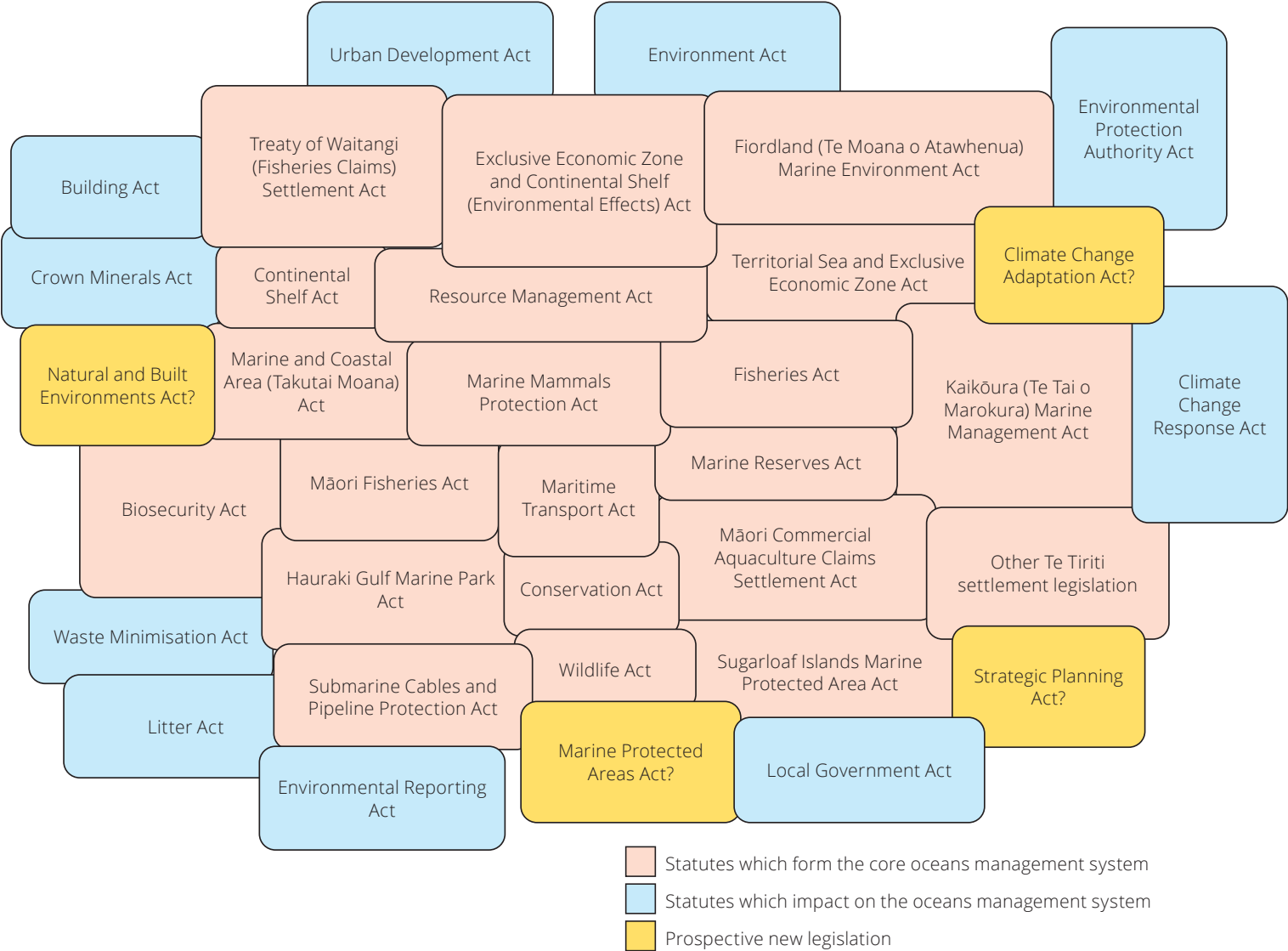


Figure 3.2: Statutes which form the core of, or impact on, the oceans management system

The array of statutes in the current system means that they interact with each other in complex and sometimes unclear ways. Such interactions are too numerous to list here, and are explored through a more critical lens when we look at legislative design options in Chapter 11. However, a few are worth noting.

- There is an overlapping and ultimately unclear relationship between the RMA (which controls activities having impacts on the marine environment) and the Fisheries Act (which manages fish stocks and the impacts of fishing on the marine environment). The Court of Appeal has recently made a significant ruling on that relationship in the *Motiti* case (see Chapter 11).¹⁷
- There is a sharp spatial distinction between the jurisdiction of the RMA (out to 12 nautical miles) and the EEZ Act (beyond 12 nautical miles). Cross-boundary activities are framed under the latter (and there are processes to coordinate these across the statutes), but that relationship can still be unclear where effects, rather than activities, span the artificial jurisdictional line. A similar spatial distinction is the basis for the relationship between the Crown Minerals Act 1991 and the Continental Shelf Act 1964.¹⁸
- Marine biosecurity functions are discharged under both the RMA and Biosecurity Act 1993.
- The MACA Act intersects with processes under the RMA and conservation legislation, including through special input into planning processes and a requirement for additional permission rights to be obtained.
- Generally, conservation laws form an additional layer on top of more general frameworks like the RMA and EEZ Act (ie restrictions under both will apply in a particular space). However, some interface in more complex ways (eg where provisions under one, like the Hauraki Gulf Marine Park Act 2000, are deemed to be a New Zealand Coastal Policy Statement (NZCPS) under the RMA).
- Marine-focused te Tiriti settlement legislation stands alone, but also intersects with broader legislation like the Fisheries Act and the RMA¹⁹ (often through the inclusion of specific amendments within the latter).
- The rate of depletion of minerals (including oil and gas), and decisions about its allocation, are carved out from the RMA and EEZ Act (and placed in the Crown Minerals Act and Continental Shelf Act) but these

Acts retain jurisdiction over the effects of mining activities on the marine environment beyond the resource itself.

- Environmental jurisdiction (especially relating to the discharge of pollutants) is split between the Maritime Transport Act 1994 and the EEZ Act, despite a substantial shift of functions from the former to the latter a few years ago. In essence, the Maritime Transport Act governs discharges from most ships as well as oil spill preparedness and response, while the EEZ Act covers all other discharges and dumping in the EEZ.
- The RMA provides for councils to monitor and report on the state of their marine environment, but other legislation does not have such provisions (eg the EEZ Act and Fisheries Act), and at a national level more integrated reporting is done under the auspices of the Environmental Reporting Act.
- The Climate Change Response Act 2002 requires the creation of emissions reduction plans, but tools under other legislation like the RMA and EEZ Act (and potentially the Fisheries Act) could be used to achieve them. It is still unclear how this relationship will play out in practice.

Of particular importance is te Tiriti settlement legislation relating to fisheries and aquaculture (see spotlights below).



Salmon farm, Marlborough Sounds

Raewyn Peart

A spotlight on the Māori fisheries te Tiriti settlement

When the QMS was introduced in 1986, the creation of private property rights in fisheries was challenged by tribes of the Far North Muriwhenua, who took a claim to the Waitangi Tribunal.²⁰ The basis of the a claim was that the QMS failed to recognise the prior protected rights of Māori to fisheries under te Tiriti.²¹ In 1988, the Waitangi Tribunal released its report, concluding that the QMS fundamentally conflicted with the principles and terms of te Tiriti.²² The Tribunal recommended that Muriwhenua Māori be given quota in exchange for giving up the fishery rights that were protected under te Tiriti.

An interim settlement recognising Māori fishing rights was implemented through the Māori Fisheries Act 1989. This provided for the establishment of the Māori Fisheries Commission and the transfer of \$10 million and 10 percent of existing quota from the Crown to the Commission.²³ A final settlement was implemented

by the Treaty of Waitangi (Fisheries Claims) Settlement Act 1992.²⁴ As part of this settlement, the Crown provided \$150 million to Māori to purchase a half share in the fishing company Sealord Products Limited²⁵ and promised that 20 percent of all new quota species brought into the QMS would be given to Māori. The Māori Fisheries Commission became the Treaty of Waitangi Fisheries Commission.²⁶ The Commission was responsible for managing fisheries assets on behalf of Māori and facilitating their allocation and transfer to iwi.

The allocation process did not commence until the enactment of the Māori Fisheries Act 2004, which set out a formula for classifying and allocating new quota based on determinations of iwi population and coastal entitlements. Te Ohu Kai Moana Trust was established to replace and fulfil the role of the Treaty of Waitangi Fisheries Commission.²⁷ Most quota has now been allocated to iwi.



Longlining, Hauraki Gulf

A spotlight on the Māori aquaculture te Tiriti settlement

The Māori Commercial Aquaculture Claims Settlement Act 2004 provides for the full and final settlement of Māori claims to commercial aquaculture arising after 21 September 1992.²⁸ Under that Act, iwi are to receive a representative 20 percent of the value of all aquaculture space that has been consented since 21 September 1992. The settlement recognises rights in aquaculture under te Tiriti, while customary rights are addressed under separate legislation.²⁹

The Settlement Act established the Māori Commercial Aquaculture Settlement Trust to receive regional aquaculture settlement assets from the Crown.³⁰ Te Ohu Kaimoana is responsible for the allocation and transfer of settlement assets to iwi.³¹ Entitlements to aquaculture assets are determined and allocated on a regional basis.³² Iwi are required to register as an “iwi aquaculture organisation” in order to receive assets from Te Ohu Kaimoana.³³

The form of settlement obligations has varied over time in response to legislative changes to aquaculture management itself. To date, there have been three distinct phases in the settlement process.³⁴

1. Space that was consented for aquaculture between 21 September 1992 and 31 December 2004 was treated as “pre-commencement space”. During this period, settlement assets could take the form of cash, authorisations to develop space, or the transfer of existing marine farms. In practice, iwi received cash to meet these settlement obligations.
2. Space that was consented for aquaculture between 1 January 2005 and 30 September 2011 was treated as “interim aquaculture management area space”. During this period, settlement obligations were to be fulfilled in the form of authorisations to develop 20 percent of designated aquaculture management areas (AMAs).
3. For space that has been consented after 1 October 2011 (“new space phase”), regional agreements between iwi and the Crown have been used to reach settlement. Under the “new space” mechanism, prospective aquaculture

development can be forecast and provided for in settlement obligations. Consequently, iwi have the option of receiving monetary settlement in lieu of (uncertain) future space. Under regional agreements, a mixture of assets can be delivered to iwi including cash, authorisations to develop space, or a combination of both.

Due to the constantly changing context around aquaculture, the settlement process is still ongoing.

We have many statutes in the current oceans management system. This system has not come into being all at the same time, and has emerged and changed over decades in an organic and piecemeal fashion. The statutory frameworks in the current system, and the tools and processes under them, interact with each other in complex ways.

A description of the system’s legislative frameworks (see Appendix 1), and the connections between them, does not present the whole picture or complexity of the system. An institutional lens is important too, because various institutions have roles under multiple statutes or have mandates that go beyond legislation. These can have quite different characteristics – central, regional or local (eg the Environmental Protection Authority (EPA), regional councils, iwi/hapū); independent or accountable (the Parliamentary Commissioner for the Environment, various Ministers of the Crown); or focused only on the sea or on subjects that span the land-sea divide (Maritime New Zealand, the Environment Court). They can be designed to interact with each other in many different ways (co-operatively, in tension, or hierarchically).³⁵ We explore various maritime institutions further in Chapter 12 when we consider future institutional design options.

Some of the diversity of institutions in the current system, and where they fit in terms of three of their key characteristics (centralisation, independence³⁶ and degree of focus on marine issues), is shown in Figure 3.3. Of particular note is the recent establishment of a ministerial portfolio for oceans and fisheries, and associated Oceans Secretariat, signalling a stronger leadership role within the political sphere of central government for taking action on oceans in a holistic manner.

	Centralised		Devolved
Independent	Higher Courts	Iwi	Hapū/whanau
	Environment Court		
	Waitangi Tribunal		
	Parliamentary Commissioner for the Environment		
	Climate Change Commission		
	Te Ohu Kai Moana		
	Environmental Protection Authority	Hauraki Gulf Forum	Fiordland Marine Guardians
	Conservation Authority		Kaikōura Guardians
	Maritime New Zealand	Harbour masters	
	Statistics New Zealand	Conservation boards	
	Heritage New Zealand Pouhere Taonga		
	Government departments (including Department of Conservation, Ministry for the Environment, Ministry for Primary Industries/Fisheries New Zealand, Ministry of Transport, Ministry of Business, Innovation and Employment, Te Puni Kokiri, Treasury) and working partnerships (eg Oceans Secretariat)		
	Ministers of the Crown (including Oceans and Fisheries, Conservation, Environment, Energy)	Regional councils	Territorial authorities
Accountable			

Figure 3.3: Institutions within the current marine management system, and where they fit in terms of centralisation, independence and degree of focus on marine issues

Figure 3.3 by no means captures all the institutions in the oceans management system. Dozens if not hundreds of others exist. However, there are several interesting features of the existing system when looked at through an institutional lens. For example, there are relatively few entities focused exclusively on the marine area (although divisions within some institutions may have this focus). More common are those whose concerns span land and sea. There is also a much wider range of institutions at a central than local level, reflecting that central bodies have a wide spatial coverage (the whole country) but are more noticeably divided across subject areas (eg climate change mitigation, conservation) or tasks (eg policy creation vs enforcement). At the more devolved level, institutions (eg councils) tend to have a broader range of functions integrated into a single entity, but their spatial coverage is narrower.

It is also worth keeping in mind the non-statutory features of the current system. These arise partly because institutions typically have broader mandates and powers of general competence beyond the specific roles and duties placed on them by statutes like the RMA and Fisheries Act. For example, Ministers, councils and government departments are particularly adept at creating strategies, policies and programmes, and supporting

initiatives through funding and investment. It is not always easy to say that all these things are part of the “system” – for example, some are more comfortably regarded as political manifestos for *changing* the system,³⁷ while others are short-term grants of money – but some are significant as parts of the system itself. For example, *Te mana o te taiao – the Aotearoa New Zealand biodiversity strategy 2020* is a non-statutory document, but one required of the government to discharge the country’s obligations under international law.³⁸ Various funds are also significant, such as the *Jobs for Nature* investment programme. The *Marine protected areas policy and implementation plan* (2005) and *Marine protected areas: classification, protection standard and implementation* (2008), despite their age, remain the latest policy developed specifically for MPAs in Aotearoa New Zealand.³⁹ These are also non-statutory.

A feature of the current oceans management system is its considerable complexity. We have many separate statutes and institutions. While there may be scope for rationalisation and simplification, the system will, to some extent, always be complex, because it manages complex issues.

3.3 International law

Aotearoa New Zealand's domestic oceans management system operates within a broad framework of international law, which is primarily treaty-based.⁴⁰ We provide a more extensive account of international legal obligations in Appendix 2.

Some international treaties are multilateral (to which many countries have signed up), while others are regional (eg regional fisheries agreements).⁴¹ Some are bilateral, such as treaties establishing borders between EEZs. It is not always clear whether treaties are about "resource management", as some (such as the International Convention for the Prevention of Pollution from Ships) or "MARPOL" can cover a wide variety of matters that overlap (eg health and safety requirements for ships, navigation).

Perhaps most importantly, the United Nations Convention on the Law of the Sea (UNCLOS) lays out basic jurisdictional matters by creating a series of zones (see Figure 3.4). Basically, the further from shore one gets, the weaker the jurisdiction of the coastal state becomes and the greater the freedoms enjoyed by other states or ships.⁴²

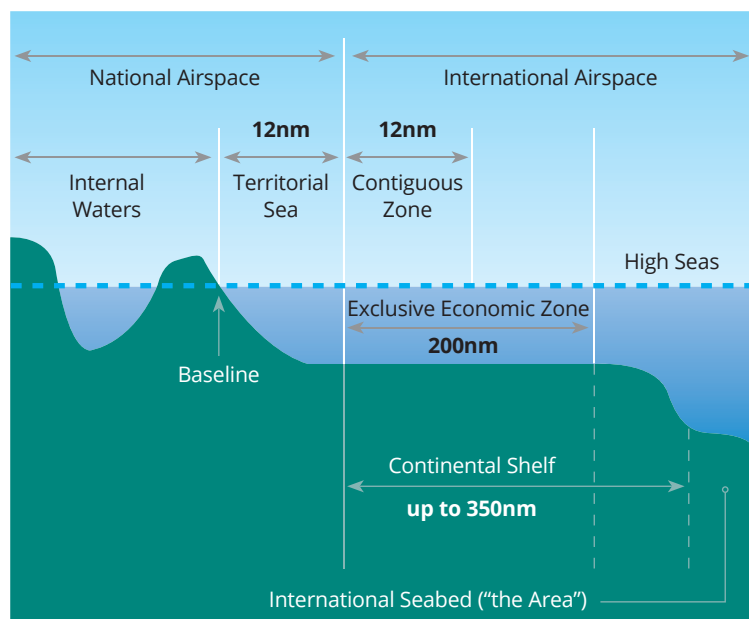


Figure 3.4: Maritime zones under UNCLOS
(The breadth of each zone is shown in nautical miles (nm)).

Aotearoa New Zealand's sovereignty only extends out to the edge of the territorial sea. Within the EEZ the country has more limited sovereign rights and the Area is not part of its territory. These rights include the

ability to explore, exploit, conserve and manage natural resources (including marine life, oil, gas and minerals) as well as the ability to economically exploit the zone for activities such as energy production. Other countries retain the freedom of navigation and overflight within the EEZ and can lay submarine pipes and cables within it. Beyond the EEZ, sovereign rights over the continental shelf are even further restricted to the exploitation of minerals, gas and other non-living resources within the seabed and subsoil and sedentary species on the seabed (excluding the harvest of mobile fish).

Alongside its jurisdictional matters, UNCLOS contains substantive environmental obligations, although most are high level. It requires the adoption of laws and regulations to "prevent, reduce and control pollution" arising from a range of activities including land-based sources, seabed activities, structures, vessels, dumping and the atmosphere.⁴³

A particularly interesting consequence of UNCLOS *conferring* rather than just *recognising* coastal state jurisdiction in the EEZ is that this jurisdiction is said to be *contingent upon* meeting the treaty's substantive environmental obligations. Although there is no meaningful enforcement mechanism, and it would be hard to establish non-compliance, it remains that a failure to protect the marine environment could, in theory, result in the forfeiture of sovereign rights over large parts of the ocean, not just being in breach of obligations.

For fisheries there is an expectation under UNCLOS that states will pursue "optimum utilisation" in the EEZ.⁴⁴ This requires the government to grant surplus catch to other states if national fishers cannot harvest the total allowable catch (TAC) set in any given year.⁴⁵ This is designed to achieve greater efficiency and equity between states in terms of access to marine resources, but it has potential environmental ramifications. What would the consequences be if, for example, Aotearoa New Zealand were to transition from fishing to aquaculture? The extent of this normative obligation is unclear as is its enforceability.

While UNCLOS provides a broad framing, other international conventions provide more specificity in some areas of marine management. Many of these are designed to manage inter-state relations (eg ships coming from elsewhere) and issues related to "common" areas like the high seas. Although they tend to be less detailed, others are intended to bind states in terms of their approach to more substantive issues within their own marine territory. We provide more detail of these in Appendix 2, and a summary of key agreements is provided in Figure 3.5. Of particular interest are strategic aspects like the Aichi Biodiversity Targets.

Convention	Obligations
United Nations Convention on the Law of the Sea 1982 (1996)*	• Exert sovereignty over the territorial sea, EEZ and extended continental shelf. • Protect and preserve the marine environment. • Protect and preserve rare or fragile ecosystems as well as the habitat of depleted, threatened or endangered species. • Determine the TAC and, taking into account the best scientific evidence, ensure that stocks are not endangered by overexploitation. • Maintain and restore populations of harvested species at levels which can produce the maximum sustainable yield (MSY). • Promote optimum utilisation of living resources in the EEZ by determining the country's capacity to harvest the living resources of the EEZ. • Prevent and control marine pollution.
United Nations Convention on Biological Diversity 1992 (2005)	• Establish a system of protected areas and areas where special measures are needed to conserve biological diversity. • Develop guidelines to select, establish and manage these areas. • Regulate and manage biological resources to ensure conservation and sustainable use. • Promote the protection of ecosystems, natural habitats and the maintenance of viable populations of species in natural surroundings. • Promote environmentally sound and sustainable development in areas adjacent to protected areas. • Rehabilitate and restore degraded ecosystems and promote the recovery of threatened species. • Regulate, manage and control the risks associated with the use and release of living modified organisms resulting from biotechnology. • Prevent the introduction of, control or eradicate those alien species which threaten ecosystems, habitats or species. • Endeavour to provide the conditions needed for compatibility between present uses and the conservation of biological diversity and the sustainable use of its components. • Respect indigenous and local community knowledge. • Protect threatened species and populations. • Where a significant adverse effect on biological diversity has been determined, regulate or manage the relevant processes and categories of activities. • Integrate consideration of conservation and sustainable use of biological resources into national decision-making. • Avoid or minimise adverse impacts on biological diversity. • Protect and encourage customary use of biological resources in accordance with traditional cultural practices to the extent they are compatible with conservation and sustainable requirements.

Aichi Biodiversity Targets 2010 (2010)	• Effectively integrate biodiversity values in development, planning processes, national accounting and reporting systems. • Eliminate incentives and subsidies that harm biodiversity and develop and apply those that incentivise conservation and sustainable use. • Governments, business and stakeholders to have plans to achieve sustainable production and consumption and keep impacts of the use of natural resources within safe ecological limits. • Halve the rate of loss of habitats or bring it to zero and reduce degradation and fragmentation. • Manage and harvest all fish and invertebrate stocks sustainably. • Manage agriculture, aquaculture and forestry sustainably, ensuring conservation of biodiversity. • Bring pollution to levels that are not detrimental to ecosystem functioning. • Identify invasive alien species and pathways and control and eradicate priority species. • Minimise the anthropogenic pressures on vulnerable ecosystems impacted by climate change and ocean acidification. • Conserve and manage 17 percent of terrestrial and inland water and 10 percent of coastal and marine areas. • Restore and safeguard ecosystems that provide essential services, including services related to water and those that contribute to health, livelihood and wellbeing. • Restore 15 percent of degraded ecosystems. • Adopt a National Biodiversity Strategy and Action Plan. • Respect and integrate traditional knowledge and customary use. • Fully integrated community engagement at all levels.
Cancun Declaration on Mainstreaming the Conservation and Sustainable Use of Biodiversity for Well-being 2016 (2016)	• Integrate policies, plans and programmes and legal and administrative measures and budgets for the conservation, sustainable use, management and restoration of biological diversity and ecosystems. • Incorporate biodiversity values in national accounting and reporting systems. • Strengthen institutional support and capacities for biodiversity mainstreaming. • Promote conservation, sustainable use, management and restoration of biodiversity as a basis for achieving resilient, sustainable and inclusive cities and human settlements, and climate change adaption and mitigation. • Promote sustainable growth as reducing the ecological footprint, combating land degradation and desertification, and addressing social inequality. • Increase and strengthen ecologically representative and well-connected systems of protected areas and other effective area-based conservation measures. • Facilitate the active and effective involvement of all relevant actors and stakeholders. • Strengthen indigenous peoples and local communities' capacities to implement the Convention on Biological Diversity by respecting their rights and customary, sustainable use of biodiversity and fair and equitable sharing of benefits arising from their traditional knowledge and practices. • Improve the regulatory framework for private sector activities, enhance incentives, and promote tools for the conservation and sustainable use of biodiversity.

Cancun Declaration on Mainstreaming the Conservation and Sustainable Use of Biodiversity for Well-being 2016 (2016) (continued)	• Promote sustainable agriculture. • Adopt a holistic integrated view and assessment of ecosystems and the interlinkages between agriculture and biodiversity. • Use integrated and cross-sectoral planning processes to reduce inefficiencies and increase productivity whilst avoiding negative impacts on ecosystems and associated biodiversity. • Conserve and cultivate native varieties. • Prevent agricultural pollution. • Control pests and diseases. • Promote sustainable consumption and production patterns. • Integrate the ecosystem approach into fisheries policies, programmes and plans. • Establish actions for the conservation and sustainable use of fishery resources to ensure the long-term viability of the fishing sector. • Conserve marine, coastal and inland water ecosystems, recognising their role as carbon sinks. • Enhance actions to reduce pollution, including noise and plastic materials. • Promote and encourage aquaculture that uses native species. • Prevent, control and eradicate invasive alien species. • Develop strategies to reduce unregulated and unreported fishing and illegal trade. • Strengthen the implementation of the <i>FAO Code of Conduct for Responsible Fisheries</i>. • Promote sustainable forest management as a dynamic and evolving concept for all types of forest. • Emphasise the relevance of forests as carbon sinks and their critical role for developing strategies for climate change adaption and mitigation. • Design and promote incentive packages for restoration, conservation and sustainable use. • Promote participation of the private sector in the development of production chains to reduce deforestation and degradation. • Promote the <i>International Agreement on Forests</i>. • Adopt practices for sustainable blue and green infrastructure.
Convention on International Trade in Endangered Species of Wild Fauna and Flora 1973 (1989)	• Protect approximately 5,800 species of animals and 30,000 species of plants from trade through a system of permits and certificates.

United Nations Agreement for the Implementation of the provisions of UNCLOS relating to the Conservation and Management of Straddling Fish Stocks and Highly Migratory Fish Stocks 1995 (2001)	• Adopt measures to ensure the long-term sustainability of straddling fish stocks and highly migratory fish stocks and promote the objective of their optimum utilisation. • Ensure that such measures are based on the best scientific evidence available and are designed to maintain or restore stocks at levels capable of producing MSY. • Assess the impacts of fishing, other human activities and environmental factors on target stocks. • Adopt conservation and management measures for species belonging to the same ecosystem with a view to maintaining or restoring populations of such species above levels at which their reproduction may become seriously threatened. • Minimise pollution, waste, discards, catch by lost or abandoned gear, catch of non-target species (both fish and non-fish species), and impacts on associated or dependent species. • Protect biodiversity in the marine environment. • Take measures to prevent or eliminate overfishing and excess fishing capacity and to ensure that levels of fishing effort do not exceed those commensurate with the sustainable use of fishery resources. • Collect and share complete and accurate data concerning fishing activities. • Promote and conduct scientific research and develop appropriate technologies in support of fishery conservation and management. • Implement and enforce conservation and management measures through effective monitoring, control and surveillance.
FAO Code of Conduct for Responsible Fisheries 1995 (1995)	• Adopt clear and well-organised fishing policies that have been developed in cooperation with all the groups with an interest in fisheries. • Establish new regional fisheries organisations or strengthen existing organisations that aim to cover the cost of conservation, management and research activities for their members. • Minimise negative impacts on the environment of fishing and fishing processes in ways that reduce waste and preserve the quality of fish caught. • Ensure fishers keep records of their fishing operations. • Have enforceable laws with procedures for determining and punishing violators – punishment for violations could include fines or even the removal of fishing licences if violations are severe. • When developing fisheries policies consider the costs and benefits of fishing and the environmental and social impacts of fishing and use the best scientific information available whilst taking into account traditional fishing practices and knowledge. • When information is absent take the precautionary approach to setting fishing limits. • Encourage people and organisations to share their views on fishing issues, and particular attention should be given to the needs of local people. • Prohibit dynamiting, poisoning and other destructive fishing practices. • Avoid overfishing and ensure the size of the fishing fleet should not be too large for the natural supply of fish. • Understand the effects of fishing gear on the environment (impacts on coral reefs, for example) before using a new fishing gear.

FAO Code of Conduct for Responsible Fisheries 1995 (1995) (continued)	• Ensure fishing methods and gear are selective and designed to minimise waste and promote high survival rates for escaping fish. • Ensure gear minimises the catching of fish species that are not wanted (non-target or bycatch fish) or that are endangered. • Phase out fishing gear and fishing methods that are not selective or which cause high levels of waste. • Protect important fish habitats such as wetlands, mangroves, reefs and lagoons from destruction and pollution. • Where natural disasters harm fisheries resources take emergency conservation and management measures when necessary. • Conserve genetic diversity and minimise negative effects of farmed fish on wild fish populations while increasing supplies of fish for human consumption. • Avoid disputes and conflict between different users of resources. • Ensure that the livelihoods of local communities are not negatively affected by aquaculture developments. • Establish procedures for monitoring and assessing the environmental effects of aquaculture. • Monitor the types of feed and fertiliser used in farming fish. • Take into account local communities and their ways of living and opinions in the coastal planning process. • Carry out fisheries practices in a way that avoids conflict among fishers and other users. • Support fisheries research efforts, monitor the conditions of fish and their habitat and gather data on the effects of different types of fishing gear on target populations and the environment generally.
United Nations Framework Convention on Climate Change 1992 (1993)	• Adopt national policies to mitigate climate change through limiting anthropogenic (human-induced) emissions of greenhouse gases and protecting and enhancing greenhouse gas sinks and reservoirs. • Report detailed information on greenhouse gas inventories, national actions and projected human-induced greenhouse gas emissions and removal by sinks, according to timeframes set in the Convention. • Take into account climate change considerations in relevant social, economic and environmental policies and actions. • Promote, and cooperate in, relevant scientific and technological research and exchange information in such areas (including transferring technology to developing countries). • Provide additional financial resources to meet the agreed full costs incurred by developing countries in complying with their obligations under the Convention. • Promote public awareness of, and education about, climate change issues.
Kyoto Protocol to the United Nations Framework Convention on Climate Change 1997 (2005)	• Reduce greenhouse gas emissions to 1990 levels. • Submit an annual inventory of greenhouse gas emissions to the Convention. • Formulate, implement and publish regular updates to national and regional programmes that contain measures to mitigate climate change and facilitate adequate adaptation to climate change. • Cooperate internationally in relation to policies and measures (including scientific and technical research and development) and facilitating public awareness and access to information on climate change.

Paris Agreement to the United Nations Framework Convention on Climate Change 2015 (2016)	• Prepare, communicate and maintain successive nationally determined contributions and pursue domestic measures to achieve them. • Communicate nationally determined contributions every five years and ensure each represents a progression beyond the previous one. • Regularly report on emissions and how they are tracking to meet the target. • Engage in adaptation planning which involves submitting and periodically updating an adaptation communication of priorities, implementation and support needs, plans and actions. • Provide financial support to assist developing countries' mitigation and adaptation efforts.
International Convention for the Prevention of Pollution from Ships 1973 (1998)	• Prevent pollution of the marine environment from oil and oily matter, harmful substances carried in packaged form, sewage and garbage from ships. • Prevent air pollution from ships.
Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter 1972 (1975) (and subsequent Protocol)⁴⁶	• Prohibit the dumping of all wastes except: dredged material; sewage sludge; fish waste or material resulting from industrial fish processing operations; vessels and platforms or other man-made structures at sea; inert, inorganic geological material; organic material of natural origin; bulky items primarily comprising iron, steel, concrete and similar unarmful materials; and carbon dioxide streams from carbon dioxide capture processes.

* Years in parentheses indicate when Aotearoa New Zealand became a party to the agreement

Figure 3.5: Summary of international legal obligations related to the oceans



Port Nelson

Raewyn Peart

Although it is not concerned specifically with the marine area or the environment, another important element of international law is the United Nations Declaration on the Rights of Indigenous People (UNDRIP). This stands alongside te Tiriti o Waitangi as a foundation for debates about the Crown's obligations to mana whenua in a future system, and has led to recent documents like *He Puapua* and *Matike Mai* (see Chapter 4) which go beyond the principles of te Tiriti into a broader exploration of constitutional issues and power sharing possibilities.

A spotlight on UNDRIP

UNDRIP is a comprehensive human rights statement on the rights of indigenous peoples. It was adopted on 13 September 2007 with 144 states voting in support, 11 abstentions and four opposed.⁴⁷ Aotearoa New Zealand was one of the states that initially opposed the Declaration and it was not formally endorsed by the government until 20 April 2010.⁴⁸

UNDRIP covers a broad range of individual and collective rights and freedoms. It expressly recognises that indigenous peoples have the right to be free from discrimination⁴⁹ and to self-determination.⁵⁰

The self-determination article is widely recognised as the most important right in the Declaration,⁵¹ not least because it encapsulates “the right to autonomy or self-government in matters relating to their internal and local affairs”.⁵² The meaning of self-determination and how the right can, or should, be realised, has been contentious.⁵³ During negotiations on the Declaration, states expressed concern at the potential for self-determination to provide for the veto of national legislation or even secession.⁵⁴ Commentators have suggested that the right to self-determination provides opportunities for pluralism and shared authority between Māori and the government (see Chapter 12).⁵⁵

The broader meaning of self-determination is informed by numerous articles in the Declaration. Article 19 requires a state to “consult and cooperate in good faith” with indigenous peoples and to obtain their consent before implementing legislation that may affect them. Article 26 provides for the rights of indigenous peoples to land and resources that were “traditionally owned, occupied or otherwise used or acquired”; and imposes a duty on states to give legal recognition and protection to such lands.

Article 28 expressly provides indigenous peoples with a right to redress, in the form of compensation, for land or resources that were confiscated, taken, occupied, used or damaged without their prior consent. The Declaration also recognises broader rights in respect of the conservation and protection of the environment,⁵⁶ and the importance of maintaining and strengthening spiritual connections to traditional lands and resources.⁵⁷

The original reluctance of the government to support UNDRIP stemmed from uncertainty around the implications of Māori rights to traditional lands and resources, associated rights of compensation, and the potential for Māori to veto legislative changes.⁵⁸ The government was concerned that the nature of the rights affirmed by the Declaration were incompatible with Aotearoa New Zealand's constitutional and legislative arrangements.⁵⁹ It was unclear how the existing rights of Māori under te Tiriti would influence implementation, and there were concerns that the nature of Māori rights in land and natural resources could impact private property interests at a national scale.⁶⁰

The government's eventual endorsement of UNDRIP in 2010 was therefore subject to express caveats.⁶¹ The government emphasised the important role of te Tiriti in protecting existing rights and providing for settlements; and the desire to retain existing institutions and processes that provided for Māori involvement. It still remains unclear how the Declaration is to be implemented by government.

Although this project is not exploring ways in which international law itself might be reformed (and we note the importance of current negotiations around marine protection on the high seas and an international plastics treaty),⁶² such law is important for two reasons: it forms a key part of the current system; and provides obligations which a reformed system will need to fulfil. One important objective of reform will be to ensure that Aotearoa New Zealand is complying with international law. Such obligations also suggest interesting options for domestic reform, including the formalisation of international targets in legislation or institutional mandates.

It is worth noting that international law is constantly evolving. Sometimes this is reflected in formal changes to existing treaties, or the creation of new ones. However, it can also take the form of “soft law”. A reformed system needs to be cognisant of commitments that the country has made

on the international stage through resolutions and declarations. Some of these have been noted above. Indeed, there are opportunities for the country to be a leader on the world stage in this area, given the size and importance of the ocean space we are responsible for.

Aotearoa New Zealand has a number of international obligations and expectations with respect to the marine environment. Although it has broad jurisdiction for “resource management” to the edges of the EEZ, important substantive obligations exist with respect to biodiversity, pollution, fishing and climate change, alongside more general obligations to Māori via the UNDRIP.

3.4 Problems with the current system

The focus of the remainder of this chapter is on describing issues with the current system. Because the system is so vast, we tackle these by themes – categories of systemic problems that arise across multiple frameworks – rather than identifying every single complaint about every statute or institution. This approach inevitably makes the problem definition incomplete (and more specific issues will no doubt need to be addressed through a reform process). We hope, however, that they can be fitted into the categories we provide.

Types of problem

The current system has many problems. Most obviously, if we accept that the outcomes described in Chapter 2 are problems (eg threatened species, habitat loss, social inequities),⁶³ then the existing system that presides over them must, itself, be problematic to a degree. After all, to prevent such problems is a core reason for having the system in the first place.⁶⁴ However, problems with the oceans management *system* are a subtly different thing to the biophysical and social outcomes described in Chapter 2 and are therefore worth looking at separately. This is for two reasons.

First, for some things, the system may be expected to make a *contribution* to solving the problem rather than entirely “fixing” it. In other words, the system can be said to be working “well” even if some degree of problem remains. For example, even the best oceans legislation will not be able to prevent warming and acidifying seas. This does not make climate change any less of a problem, but it does mean that the system has not necessarily failed or is in need of an overhaul if climate change occurs. Other measures, including planting trees on land,⁶⁵ private action, and mitigation measures in other countries, are required to contribute as well. Similarly,

for many problems (eg sedimentation from catchments), it might be years before action results in meaningful improvements. The natural world may recover slowly. A well-functioning system may be one in which *trends* are positive or improvements are *predicted*, not one in which everything is perfect straightaway.

Secondly, the system itself can create an entirely *new* range of problems. Even if (in a hypothetically perfect future) the negative outcomes described in Chapter 2 disappeared, the system could still be problematic because of how it operates (who is involved, how long it takes, how costly it is, the values that underpin it) can be as important as the measurable results it generates. That is particularly the case from a te Tiriti perspective. As has been seen in the case of the proposed Rangitāhua/Kermadec Islands Ocean Sanctuary (see spotlight in Chapter 7), process can be as important as outcome.

Problems with the existing system are distinct from the biophysical and social problems outlined in Chapter 2. This is because (1) the existence of a problematic outcome does not *always* mean the system behind it is broken, and (2) a poorly designed system can itself create a wider range of problems through *how* it operates, not just through the tangible outcomes it produces or does not produce.



Forestry harvesting above marine farms in Port Underwood

Raewyn Peart

There is, of course, substantial overlap between biophysical and system problems. Most obviously, the existing oceans management system is squarely responsible for most of the poor biophysical outcomes and trends discussed in Chapter 2. Their very existence indicates a serious systemic failure:⁶⁶

the most significant problem is that the system has failed to achieve what it was always clearly meant to: notably in its establishment and defence of many environmental bottom lines and associated limits on human activities.

The system has not prevented year on year increases in greenhouse gas emissions that are acidifying the seas. It has not prevented the continuation of land uses that have smothered coastal habitats and estuaries in sediment. And it has not prevented the often irretrievably harmful impacts on benthic environments caused by bottom contact fishing methods. The list goes on.

In particular, only patchy progress has been made on biodiversity protection in Aotearoa New Zealand. In 2019, in its sixth national report under the Convention on Biological Diversity, the Department of Conservation noted the country is progressing “at an insufficient rate”.⁶⁷ Work is required to improve national coordination, the integration of protected areas into broader ecosystem management, and information on the social, cultural, economic and ecological values associated with the marine environment.⁶⁸ Summarising deficiencies further, *Biodiversity in Aotearoa: an overview of state, trends and pressures 2020* identified the need to better understand links between biodiversity and ecosystem function in marine communities; there is no comprehensive picture of marine ecosystem integrity, and no national guidelines for consistent assessment of coastal waters.⁶⁹

The existing system has failed to achieve many of the outcomes that it was always intended, and expected, to achieve. This is its most significant problem.

Other problems, however, have arisen because the system was not really designed to do some things it arguably should (or at least should *now* be expected to) do. For example, it has not resolved allocative questions in a way that is most “fair”, provided meaningfully for targets for environmental enhancement, or driven the deployment of spatial protections in a coherent way.⁷⁰ Part of these deficiencies may be the

result of the market-led ethos of the era during which much of the system was put in place (including the RMA and Fisheries Act), where there was a drive for efficiency over equity and where the government was seen as a trusted manager rather than a change maker to be held accountable for continuing progress towards something *better*.⁷¹

The existing system was not really designed to achieve some outcomes that it may now, in a modern context, be expected to achieve.

Another set of issues with the system has arisen through the manner in which it has evolved over time. It has grown much larger, more complicated, and fragmented than it used to be. Layers of complexity have been added, things have been carved out, and amendments have transformed what were once reasonably coherent statutory schemes into chaotic behemoths that must interact with dozens of other statutes.

The way the system has evolved over time means it has gradually become less coherent and more complex.

Finally, a distinct set of issues reflect inherent tensions that may never be resolved or “fixed” to the satisfaction of all parties (eg the tension between participation, efficiency and timeliness of decision-making, or between the interests of commercial and recreational fishers in a shared resource).⁷² While the balance may need to be shifted in one way or another as times change, we need to be aware of the difference between true system failures and the system simply performing one of its key roles – resolving disputes.

The oceans management system will not always please everyone, but this does not necessarily mean it is broken or problematic at a systemic level. One of its key roles will always be to resolve disputes, and that will produce a perception of winners and losers.

Below, we consider some more specific characteristics of the existing system that can be regarded as problematic. These are cross-cutting characteristics, rather than a comprehensive list of things that might be wrong with particular statutory frameworks or individual decisions.

Weak environmental limits

Not all elements of the current system are fundamentally broken. Some have had notable successes. For example, we have seen the rebuild of a number of stocks under the QMS.⁷³ A significant leap forward was also made in the *King Salmon* litigation when the Supreme Court held that the RMA, through the NZCPS, can set environmental limits that cannot then be undermined in lower-level planning and consenting decisions.⁷⁴ The ban on products containing microbeads through the Waste Minimisation Act, and prohibitions on marine dumping established under RMA regulations, are also forms of a “limit”.

However, overall, the existing system can be said to lack strong environmental limits in the marine area. By this, we mean there are few lines in the sand beyond which impacts are unacceptable and which trigger strict prohibitions on activities or their effects, not just mitigation or trading off of values. As highlighted by the Supreme Court in the *King Salmon* case, “if there is no bottom line and development is possible in any coastal area no matter how outstanding, there is no certainty of outcome ...”.⁷⁵

The lack of such limits can be seen in a number of core frameworks:

- The Marine Reserves Act is powerless to impose area-based protections for reasons other than scientific research, and there is no compulsion to use its tools (ie limits do not exist until a reserve is established). Marine reserves are also unable to be created in the EEZ.
- The RMA lacks consistent national level regulatory limits for activities impacting the marine environment (eg a National Environmental Standard (NES) for wastewater discharges). In fact, national regulation goes so far as to require development that will inevitably impact on the marine area to be permitted (the National Policy Statement (NPS) on Urban Development) or prevents the establishment of controls to prevent impacts (eg the NES for Plantation Forestry and sedimentation).⁷⁶
- Limits under the RMA can exist in theory but not be realised in practice; for example, the NZCPS requires avoidance of adverse effects on areas set aside for protection of indigenous biological diversity under other legislation, such as marine reserves, but has proved insufficient to prevent the consenting of harmful activities higher up in catchments (which, for example, caused the degradation of the Long Bay-Okura Marine Reserve).⁷⁷ Te Whanganui-A-Hei (Cathedral Cove) Marine Reserve has also been impacted by sediment coming out of Whitianga Harbour.⁷⁸
- No firm policy-based limits exist in the EEZ (since policy statements have not been created). Regulations that have been made apply only to a relatively narrow range of activities and provide largely for consenting pathways by which discretion is exercised (eg for deep sea mining) rather than prohibitions on activities or impacts when a limit is reached.
- Although specific place-based restrictions and regulations abound under the Fisheries Act,⁷⁹ these cannot really be regarded as systemic limits, as core sustainability measures (eg protecting habitats of importance to fisheries or preventing destruction of benthic ecosystems through practices like dredging and bottom trawling) are discretionary and have been sparsely used.⁸⁰ Similarly, there are questions as to whether benthic protection areas in the EEZ can be regarded as a true environmental limit, given they are mainly located in areas where the activity they seek to manage (bottom trawling) cannot take place in any event due to physical limitations (ie the protected areas are too deep to trawl).⁸¹
- The TAC set for fish stocks under the Fisheries Act looks like a firm limit. However, there are hidden issues. There is, in practice, no overall limit for the recreational take component. Predicted catch is estimated and controlled through things like bag limits, but there is no line at which recreational fishing must stop. Bag limits do not keep the catch within a harvest cap if people fish more frequently or twice the number of people fish than expected. Provisions allowing for legal discarding in the commercial fishery, where harvested fish are not counted against Annual Catch Entitlement (ACE), arguably turn the TAC into a soft rather than a hard limit.⁸² Further, it is legal for a commercial fisher to harvest above his or her allocation through the payment of deemed values.⁸³
- The TAC is not always set at a level that achieves the outcome sought by a limit (ie the retention of an unfished biomass that leads to MSY),⁸⁴ whether because of a lack of information or a stock assessment being outdated.⁸⁵ Catch limits have not prevented the decline of some fish stocks in practice (eg East Coast tarakihi), because they have been set at the wrong level.⁸⁶ This is illustrated by the fact that, for some stocks, recorded catch is much *lower* than the allowable catch, a potential indication of stock depletion.⁸⁷ And catch limits for single stocks do not necessarily reflect an ecosystem-based limit (ie to support dependent species or avoid trophic cascades), or a limit to protect highly valued species.⁸⁸

- There is a lack of comprehensive limits to protect threatened or valued species. For example, the Wildlife Act does not protect marine life unless it is specifically listed in a schedule to the Act, and very few marine species are listed.⁸⁹ Although there are firm prohibitions on hunting seabirds and other protected marine wildlife under the Act, a statutory loophole allows an uncapped number of them to be killed in fishing gear without recourse to prosecution.⁹⁰ Similar provisions apply to marine mammals under the Marine Mammals Protection Act.⁹¹ There is statutory provision for the development of population management plans for protected marine species, which are designed to impose limits on total mortality caused by fisheries bycatch, but these are not mandatory; are not triggered by the status of a species; and have proven difficult to progress (none have been finalised since the insertion of the provisions in 1996).⁹²

Overall, the system has many tools that *can* impose limits, but often these are not mandatory. Furthermore, limits set in one place can, in practice, be undermined by exceptions or allowances in other places. The process for setting limits can be highly politicised and, in the absence of strong legislative backing, liable to capture by vested interests in terms of the *level at which* they are set. While we have no shortage of strong sounding environmental principles (including in the Fisheries Act,⁹³ RMA⁹⁴ and EEZ Act⁹⁵), in practice they have not often translated into actual regulatory limits on human activity. The *King Salmon* decision highlights some of the inherent shortcomings of the RMA when it comes to the establishment of limits (see spotlight).

A spotlight on *King Salmon*: Limitations of environmental limits under the RMA

The *King Salmon* jurisprudence made a significant positive contribution to how environmental limits are recognised under the RMA, and its focus was the coastal environment. The legal position prior to the case was that decision-makers were to engage in reaching what was generally called an “overall broad judgment” when making decisions on plans and consents. That meant a decision-maker had recourse to Part 2 of the RMA in balancing the benefits and costs of a proposal, even if objectives and policies in lower planning instruments were much more specific, directive and protective. As the Randerson Panel said, this approach “allowed environmental limits in plans to be set aside on the basis of advancing [the] social, economic and cultural wellbeing” reflected in Part 2.⁹⁶

In *King Salmon*, the Supreme Court overturned that approach. It emphasised that, in certain circumstances, the RMA was about defending firm environmental bottom lines set under it, not weighing up many factors. Central to the decision was the fact that the NZCPS contained directive and firm provisions concerning the protection of the coastal environment. The Court made clear that national direction could impose firm, policy-based limits and that subsequent lower-level decisions would not be allowed to undermine them by referring back to the wider, more balanced set of considerations in Part 2 of the Act. That was a big step forward, because “the failure of the RMA to deal well with cumulative effects is ... partly rooted in the misinterpretation of its purpose statement”.⁹⁷

Since 2014, the Supreme Court’s message has been applied and refined through a number of other decisions. However, overall, this line of case law still falls short in a number of senses. It has since been made clear that a balancing approach (including in the consenting context) will often still need to happen (eg where there are no firm and directive policies in planning instruments to point to, or where there are multiple provisions that conflict). This is a real issue, because limits are scattered across national direction, regional plans, district plans, policy statements and consents. It is not always clear whether something is a true limit or not, or how flexible it is meant to be. Much depends on the (sometimes tortuous) unpicking of particular words in non-regulatory policies

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Common dolphins, Hauraki Gulf

(like “avoid”) and assessing how dozens of provisions in different documents interact. Even in a reasonably well developed tool like the NZCPS, many important policies are expressed in language much weaker than an obligation to “avoid” harm.⁹⁸

The Supreme Court also stressed that Part 2 is not an operative set of provisions; it sets an expectation that a cascade of subordinate instruments will impose strict protections.⁹⁹ But it does not itself demand that firm limits *are* generated through planning instruments; regional plans are not mandatory, rules are not required, and the government could make the NZCPS less protective if it wished. In the RMA, Parliament has above all created a *framework* for limit setting in the marine environment rather than directly creating limits. The absence of modern MPA legislation to fill some of that gap makes the system even shakier.¹⁰⁰

The important lesson from the Supreme Court is, essentially, that authorities can impose bottom lines if they consider Part 2 demands it (although there is no effective mechanism to ensure action is taken), and it is not permissible to undermine a higher-level authority’s (eg a Minister’s) decision to do so. However, even that may not be watertight. There are ongoing efforts to carve out exceptions to environmental limits for activities that are recognised as having benefits under the NZCPS, for example ports.¹⁰¹

There is an ongoing conversation as to whether the RMA’s proposed replacement – the Natural and Built Environments Act (NBA) – will take a more effective approach to setting environmental limits.¹⁰²

Parts of the existing system seem actively opposed to the setting of limits. For example, marine reserves can be regarded as a key tool in setting limits to prevent biodiversity decline. But under current law, if an objection is raised to a proposal for a new reserve, the Minister must uphold it if he or she is satisfied that the proposed marine reserve would:¹⁰³

- (a) interfere unduly with any estate or interest in land in or adjoining the proposed reserve:
- (b) interfere unduly with any existing right of navigation:
- (c) interfere unduly with commercial fishing:

- (d) interfere unduly with or adversely affect any existing usage of the area for recreational purposes:
- (e) otherwise be contrary to the public interest.

Not only are such criteria extremely broad and discretionary, they go against the imperative, under Aotearoa New Zealand’s international obligations, to increase the spatial protection of the marine environment. Instead, they are regarded as a “nice to have” as long as they do not disturb existing interests.

Finally, restrictions are often established too late in the policy cycle, and in too fragmented a fashion, to really be limits at all. For example, this can be seen in consent conditions on harvesting individual plantation forestry blocks rather than a strategic plan for staggering small coup harvesting and afforestation across a whole catchment; in single species stock assessments under the Fisheries Act which are blind to broader inter-species and habitat interactions; or in the long-standing assumption that regional rules have no place in protecting habitats like kelp forests from the trophic impacts of fishing activity (eg to prevent kina barrens).

While the current system is capable of imposing firm environmental limits, and does so in a number of cases, overall it can be said to lack a robust approach to environmental bottom lines. It is sometimes not clear whether a limit is non-negotiable or not, whether it must be set, or the reasons for which it is to be set in one place over another. This lack of clarity about where (and whether) limits are set can not only create issues for environmental wellbeing, but also uncertainty for businesses which can invest in consent applications only to have them turned down or subjected to unexpected conditions.

Outdated norms and values

There is a strong case that the norms that underpin our legislation do not reflect modern times and concerns. There is an inconsistent approach to te Tiriti across legislation, including the RMA’s requirement to take its principles into account¹⁰⁴ and the EEZ Act’s simple and presumptuous assertion that the provisions of the Act already embed the principles without needing further interpretation.¹⁰⁵ Older legislation is generally silent as to the impact of te Tiriti or its principles, although older marine conservation legislation is tied to the Treaty clause of the more modern

Conservation Act,¹⁰⁶ which has been given new life and potential by the Supreme Court in the *Ngai Tai* decision.¹⁰⁷ There are increasing calls to strengthen such requirements,¹⁰⁸ for example to “give effect” to the principles, or more directly to give effect to te Tiriti itself.¹⁰⁹ The formulation in the EEZ Act, where compliance with te Tiriti is essentially deemed to have occurred in the provisions of the Act rather than guiding subsequent decisions under it, has been subject to particular criticism.¹¹⁰

Many are of the view that a meaningful expression of such principles also needs to flow through to other design features (eg co-governance and the inclusion of mātauranga in decision-making), and not just be a general statement of principle.¹¹¹ Tikanga struggles to find recognition in legal processes focused on a narrow approach to fact finding and evidence,¹¹² although there have been recent efforts to incorporate it more into the mainstream of some statutory decision-making.¹¹³ Criticisms have also been made about processes not respecting te Tiriti, such as for the establishment of MPAs,¹¹⁴ and the length of time and inadequate resourcing for progressing claims for customary marine title under the MACA Act.¹¹⁵ Statutory definitions of wāhi tapu may not always reflect a Māori view of this concept, including the range of reasons that restrictions (eg rāhui) should be imposed in such areas.¹¹⁶

Aside from te Tiriti, the most egregious example of outdated norms might be the Marine Reserves Act, which proclaims that it is concerned only with scientific research.¹¹⁷ It reflects a time in which much deeper concerns like biodiversity protection and climate change were not high on the agenda. Additionally, it lacks statutory guidance for processes now central to environmental decision-making, such as how te Tiriti principles apply, and how public consultation should happen. But there are numerous other examples. The Wildlife Act does not make a clear distinction between the importance of protecting indigenous species and introduced ones, or between threatened and non-threatened ones.¹¹⁸ The Crown Minerals Act and Continental Shelf Act are concerned with maximising the economic benefits of mineral extraction, and do not temper that ambition with a recognition that keeping oil and gas beneath the seabed might be a better approach, or that alternative uses of subsurface space like carbon geo-sequestration might be considered.¹¹⁹ Even something as “modern” as the RMA is oriented towards passive management and mitigation of adverse effects rather than defending limits and striving for positive outcomes for the oceans. That no longer reflects what we are trying to achieve. Many statutes, including the Maritime Transport Act, do not have clear purposes at all. And on the institutional front, the EPA does not have a clear environmentally focused mandate, and it remains unclear what its position in the system should be.

While there are still debates to be had at the margins – for example, whether fish stocks should be managed under a principle of MSY, or what the purpose of the RMA’s replacement should be – our existing system overall reflects a patchwork of norms and objectives that do not sit well with each other or with society’s current values. This has, unfortunately, been the case for some time. The words of the Parliamentary Commissioner for the Environment over 20 years ago ring just as true today, in that:¹²⁰

values and fisheries management systems that permitted the desecration of Spirits Bay in Northland ... and ecosystems such as sea mounts, are simply not good enough to meet New Zealand’s needs in the first years of the 21st century.

The norms underpinning legislation in the current system, overall, are outdated and reflect the concerns of a time different to the present. That is particularly noticeable when it comes to te Tiriti o Waitangi, climate change, and the imperative to protect and restore the natural environment, not just mitigate impacts on it.

Fragmentation, gaps and overlaps

While the system has a number of large statutory schemes like the RMA and Fisheries Act, our package of legislation has overall developed in an ad hoc way. In many cases it provides bespoke workarounds to existing frameworks no longer fit for purpose. For example, special legislation for the Hauraki Gulf reflects not only the importance of this special place and the unique pressures it is under, but also the absence of a statutory regime under which multiple agencies and stakeholders can be compelled to work cooperatively together across silos. Similarly, place-based legislation establishing MPAs (eg in Fiordland, and forthcoming for the Hauraki Gulf to implement its non-statutory spatial plan) in part reflect shortcomings with more general frameworks like the Marine Reserves Act.¹²¹ In other words, there is a risk that existing fragmentation breeds even greater fragmentation.

Furthermore, while various options were on the table as to the ambition of new EEZ legislation, it ended up being a gap filling mechanism, to be squeezed into the existing landscape rather than remodelling it (eg through combining it with the RMA). And it is by no means clear that there is a sensible reason for the fragmentation of conservation legislation whereby marine mammals are protected under one framework and

other marine wildlife under another, or whether place-based protections under the much-maligned Marine Reserves Act should be separate to spatial measures to protect marine mammals under the Marine Mammals Protection Act. There is now talk of even more layers of place-based legislation, including for the creation of fit-for-purpose MPAs (eg in the Hauraki Gulf) and to connect the tools available under other frameworks.¹²² We have a system that is steadily accreting statutes, that is not entirely coherent, and where the new *modus operandi* appears to be to reach for the drafting pen when something needs to be done.

The extent to which this matters or not is an interesting question, and one that is explored in Chapter 11 when we consider legislative design. It is not necessarily the case that a single piece of marine legislation – one act to rule them all – would be desirable. What creates integration in one way might cause fragmentation in another. Fragmentation is also noticeable on the institutional front, which is explored in Chapter 12. This can be seen in gaps, overlaps and confusion about the respective roles of councils and Fisheries New Zealand, institutional boundaries between different

councils, and the ebbs and flows that have characterised the EPA's position in the system relative to councils, ministers and boards of inquiry.

Moreover, the solution is not necessarily legislative or institutional redesign; we could instead work on clarifying the relationship between the purposes (and therefore tools) of different statutes or between the mandates and duties of different institutions. Marine spatial planning may be one way forward (see Chapter 10). However, a more integrated legislative framing may enable more progress to be made and help ensure things do not fall between the cracks.

The current system is fragmented across legislation, institutions and tools. This can lead to gaps, overlaps, inefficiency and a lack of coordination of frameworks that need to work together to achieve positive outcomes.



Waikawa, Marlborough Sounds

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Complexity and inaccessibility

A fragmented system creates not just gaps, overlaps and uncertainty, but also causes complexity. Even those deeply involved in the system can find it complicated; it has become more and more inaccessible over time as more processes, carve outs and legislative layers have been added. For example, former Chief Justice Sian Elias put it well when she said that the RMA is “meant to engage communities, not alienate them” and bemoaned the “impenetrability” of the Act.¹²³ There has also been a proliferation of alternative planning and consenting processes. The RMA is twice as long as it used to be. Current decision-making processes for fisheries are slow, cumbersome and largely inaccessible to non-commercial fisheries stakeholders and the general public (with one estimate of eight years to change a TAC).¹²⁴

Other legislation is also complex. The Maritime Transport Act 1994 is a good example.¹²⁵ It has been criticised as a mess¹²⁶ with an incoherent structure overall.¹²⁷ The lack of coherence has been accentuated by the repeal of certain parts of the Act, as Maritime Rules¹²⁸ replaced primary legislative provisions as the means of regulating activity. The Act contains 17 separate interpretation provisions throughout its 29 Parts, and both general and specific provisions which are reinforced by voluminous regulations, Maritime Rules, and Maritime Protection Orders.¹²⁹



Port of Tauranga

The ways in which various pieces of legislation interact with each other is also complicated and often not apparent. It can be hard for one, not intimately acquainted with the law, to know what measures can be taken under the Marine Mammals Protection Act 1978, Fisheries Act 1996 and Wildlife Act 1953, and why the three are so different.

Complexity is not just limited to statutes. To understand how it works, one must be familiar with how markets operate (eg the QMS), the place of non-statutory strategies and policies, and an overwhelming array of regulations, plans, policy statements, existing use rights, orders, and so forth. There is also the multitude of Te Tiriti settlement legislation. The list grows every day. How all these elements of the system interact with each other is not always clear and often requires litigation to determine.¹³⁰ The system will always be complex, but it is apparent that it is much more complex than it needs to be.

The current system is extremely complex and confusing. Fundamental features, such as the interpretation of the purpose of the RMA and its relationship with the Fisheries Act 1996, frequently require resolution in the courts. While the system will always have a degree of complexity, it is by no means clear that the present high-level of complexity is necessary or desirable.

Unclear stewardship/leadership

Legislative and institutional fragmentation not only creates overlaps; it can also create a vacuum of leadership, where agencies struggle to work together or simply assume that responsibility lies elsewhere. A cornucopia of different tools that *can* be used does not automatically mean they *must* or *will* be used. The existing system is, generally, characterised by a strong trust that those responsible for marine outcomes will choose to use the available mechanisms effectively.

While there are positive signals – for example in the recent establishment of a new portfolio for fisheries *and* oceans as well as an Oceans Secretariat to support government marine initiatives (and facilitate sharing of resources), these are not legislated and could be easily undone. We have seen that happen in the abortive attempts to establish an oceans policy two decades ago. While marine issues are by no means simple, shaky leadership can be seen on many fronts. Central government has made little progress on establishing a coherent network of MPAs, or setting controls on damaging fishing methods. The mandatory NZCPS has existed

since the enactment of the RMA, but (in contrast to the enabling of productive activities like forestry) it has not led to national level regulations (an NES for the sea)¹³¹ or a meaningful policy framework for estuaries in the same way as (for example) urban development. This is not so much an issue with the RMA, but rather with the political will to use the RMA in the ways it was intended to be used.

Similarly, while regional councils have always had responsibilities under the RMA (and NZCPS) for habitat protection in the marine area, the extent to which that has manifested in practice has been low and varies around the country. And councils are only now, after the *Motiti* case, coming to terms with what the legislation has always envisaged – that they are expected to be active in controlling the impacts of fishing on the marine environment.¹³²

It is arguable that the current system lacks leadership and a framework of accountability for achieving outcomes.

A lack of strategy and agility

Generally speaking, the existing system can be said to lack both a future focus and agility to respond to environmental change in a timely way. The RMA, for example, talks about enhancement, but lacks a framework for setting targets and a mechanism for holding authorities to account for failing to meet them. National direction is, for the most part, optional and until recently (ie for freshwater) has not been concerned with establishing a pathway towards change. Even where such a pathway is created, amendments to regional coastal plans can take years to achieve and even longer to flow through to outcomes on the ground. Moreover, existing use rights on land (eg for sediment-inducing activities like agriculture, urban development and forestry) can be hard to change legally as well as politically, meaning the best strategies and policies in the world may struggle to be effective in practice. The lack of strategy is even more noticeable under the EEZ Act, where general statements of policy are not mandatory and have, as a result, not been created.¹³³ In short, the RMA and EEZ Act are about managing the status quo, not pushing towards something better.

Generally speaking, the same can be said about fisheries. Although there are notable examples of agility where risks are significant and clear (such as where closures were put in place following the Kaikoura earthquake in 2016), in other situations it is not so obvious. For example, there are ongoing questions about what tools like fisheries plans are actually intended to achieve (if anything) in a strategic sense, and if they are meant to be an industry-led self-management tool or an instrument for government and communities to steer commercial fishing in a different direction.¹³⁴ They are not mandatory and have a chequered history. Furthermore, the boundaries of QMAs are large and difficult to change,¹³⁵ despite the fact that localised depletion in parts of the areas is fairly common, and boundaries do not reflect the biological reality of stocks being managed.¹³⁶ As marine species venture into new areas as the effects of climate change are realised, the arbitrary nature of existing management approaches may be exacerbated. Stock assessments and catch limits rely on uncertain and constrained data and they do not keep up with the actual state of a fishery.¹³⁷ The biological and ecological state of fish stocks and marine ecosystems can change rapidly (especially in a changing climate), but this is not always reflected in the system that manages them.

A lack of strategy and agility is also observable in the marine conservation context. A change in the status of a protected marine species does not automatically trigger a regulatory or policy response (eg the creation of a population management plan or the provision of funding).¹³⁸ Conservation strategies and plans can routinely be out of date, and lack adequate weight when it comes to the consideration of

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Fishing trawler, Nelson

concessions.¹³⁹ Similarly, the deployment of MPAs has been left largely to political discretion¹⁴⁰ or tackled through bespoke legislation following collaborative processes or negotiation (eg in Fiordland and Kaikōura), rather than forming a core part of the system itself.¹⁴¹ *Te mana o te taiao – the Aotearoa New Zealand biodiversity strategy 2020* is a non-statutory document and its relationship with operational statutes is neither strong nor clear.¹⁴² All of this is in stark contrast to more modern legislation like the Climate Change Response Act, where there is a much stronger framework for establishing system-wide targets, stepping stones towards them, plans for driving that change, and an institutional framework to ensure transparency and accountability.

There are many other examples of ways in which the current oceans management system lacks strategic direction and the agility to respond to, and force, change. This is true not only where the imperative is environmental protection. For example, aquaculture proponents are struggling with fixed spatial consents that cannot move easily when conditions change (such as seawater warming in the Marlborough Sounds), and offshore aquaculture proponents are encountering a highly uncertain policy and regulatory environment. The lack of policy instruments and spatial planning (eg delineating what can happen where) under the EEZ Act has led to mining companies spending huge amounts seeking to obtain consent for activities in sensitive areas. Clearer policies about what is “off limits” (eg mining in benthic protection areas or parts of the Chatham Rise) might have saved all parties a great deal of time and effort.

There may be an even more fraught situation in the future if the country were to go down the path of deploying offshore wind at scale. There is no real marine policy framework (let alone a nationally consistent one) to support discretionary decisions for offshore wind, other than very general support in an NPS on renewable electricity generation (which largely parrots the provisions in the RMA itself). There is little guidance as to where facilities would be best located (or where they should be avoided), with the most suitable places in terms of wind energy potentially not matching other values and uses of the marine space (eg tourism, recreation, fishing and aquaculture). That uncertainty seems likely to exacerbate the already considerable difficulties that have beset wind farms onshore.

The current system lacks clear goals for what it is trying to achieve for the future and the agility to respond in a timely way to changes in environmental conditions.

Procedural fairness

How elements of the existing system operate is arguably unfair. This is not uniform across the system, and some complaints may be more about some people's or interests' dissatisfaction with where a balance has been struck. However, many complaints are understandable or at least deserve consideration. Some of these relate to the social issues described in Chapter 2, for which the system is directly responsible. For example, the Fisheries Act established the market for quota and separated quota ownership from the harvest of fish, which has resulted in the unequal distribution of value across the fishing sector (with a large share of the profit going to quota holders, who tend not to be those out at sea).¹⁴³ The RMA allows for coastal occupation rights to be allocated using structured tendering processes (to determine which use would be “best”), but more common is the default use of “first in first served” consenting.¹⁴⁴ Charges can be imposed on occupiers but tend not to be (and where they exist are designed inconsistently), while the harvesting of marine life under the Fisheries Act operates on a cost recovery rather than “royalties” basis. It is in the operation of the system – the ways in which its tools are used – that these issues arise. Other equity issues arising from processes under the existing system include the following.

- An ongoing uncertainty about when compensation should be forthcoming for an erosion of “rights” in the marine area (such as a diminution of value of quota due to spatial exclusions or restrictions on fishing methods). At present, such decisions are largely based on negotiation and political factors.
- The lack of strategy and agility within the system, outlined above, can have implications for equity as the environment changes. For example, the movement of fish between QMAs as the climate changes may cause fish stocks – and the value of quota – to decline in some places and increase in others (a windfall capital gain of sorts).
- There are relatively extensive rights for the public to participate in the development of instruments like national direction and regional plans under the RMA, but there are fewer formal opportunities to participate in the development of instruments, or decision-making, under the Fisheries Act (eg through hearings and merit appeals) although pre-engagement and public consultation does often occur as a matter of practice.
- The proactive involvement of iwi and hapū in marine management decisions and processes varies across the country and across

legislation, with much being discretionary (eg the transfer of powers under the RMA). In practice it can also vary due to differences in resourcing, with many Māori groups struggling to sustain involvement across multiple processes and lacking financial support.¹⁴⁵ Māori voices are saying they feel excluded or marginalised from decision-making processes and that the system does not reflect te Tiriti principles.¹⁴⁶

- The broader public can also struggle to participate in marine planning and consenting processes (eg under the RMA) due to a lack of resources or “burnout”. This can be perpetuated where matters need to be litigated in the formal and expensive setting of the courts. This is especially in frameworks like the Fisheries Act where there is recourse only to judicial review in the High Court, and no scope for appeals on the merits or substance of a decision in more accessible fora, such as the Environment Court.¹⁴⁷
- Some may be concerned with the “privatisation” of the marine area (eg for aquaculture, private ownership of coastal land) and its implications for public access.
- The operation of the system is also arguably inequitable from an eco-centric perspective. People have many rights, and these are often used to defend interests in legal and political processes. But despite the advocacy functions of institutions like the Department of Conservation, nature itself is seldom represented in a systemic way. Much is left to the efforts of private individuals and civil society, whose resources can be stretched and whose interests are not necessarily aligned with the “environment”.

In short, there is a degree of dissatisfaction with how the system operates. However, such things are not easy to resolve because there are frequently tensions between different interests, and the idea of “fairness” is often in the eye of the beholder. Perhaps most importantly, the system lacks a *clear* foundation when it comes to social equity and procedural justice, with complaints about equity continuing to arise and be addressed a case-by-case basis. This can be seen in the processes by which individual protected areas are being created, notably in the case of the proposed Rangitāhua/ Kermadec Ocean Sanctuary (see Chapters 9 and 10 and the spotlight in Chapter 7).¹⁴⁸

There are a number of ways in which the current system is arguably unfair, both in terms of substantive outcomes and the processes by which decisions are made. In particular, there is no overarching framework for how oceans management decision-making is to be conducted in a way that implements the principles of te Tiriti o Waitangi.

Information and funding

The oceans management system is information hungry, and the collection and use of information can be expensive. Some have pointed to problems with how information and funding operate in the current system. This is not unique to oceans. For example, the Parliamentary Commissioner for the Environment has pointed to the lack of a coherent research strategy for environmental issues more broadly (and the absence of forward planning for what our data needs might be)¹⁴⁹ and the existence of significant gaps.¹⁵⁰ Monitoring data and fundamental research is “cobbled” together in an opportunistic way from a range of sources to present in reporting, rather than being collected in a highly purposive and time-series fashion according to what is most useful to tackle pressing problems.¹⁵¹

Information is also not aggregated or stored in a way across institutions that is easily accessed, interrogated or used. Datasets do not speak well to each other,¹⁵² and research is frequently stored in an unstructured way, leading



Private property sign, Ponui Island

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to it being underutilised or unnecessarily replicated. Funding for research is not ring-fenced from other competing funding pressures,¹⁵³ and it can be intermittent and insecure.¹⁵⁴ Sometimes institutional knowledge disappears due to staff turnover. And almost across the board, the system does not allow for full input or consideration of mātauranga Māori.¹⁵⁵ Some have pointed to the lack of “data sovereignty” for Māori, and called for greater integration of mātauranga into broader research databases and datasets.¹⁵⁶

A lack of information is particularly important in the marine context, because we cannot easily see beneath the waves to observe what is going on first hand. We are heavily reliant on science (including indigenous knowledge). Environmental reporting is full of references to things we don't know.¹⁵⁷ This ranges from an incomplete understanding of the biology and life cycles of species and the condition of many fish stocks, to the extent of human activities in the marine environment and how these and other pressures impact on complex ecosystems (including cumulatively).¹⁵⁸ In particular, while topographical mapping exists, there is a paucity of data on the nature and extent of habitats in the marine environment.¹⁵⁹ Where data relates to specific industries, such as fishing, aquaculture or mining, there can be access issues due to commercial sensitivity.

Some progress has been made in deploying tools that make marine datasets speak better to each other, such as *SeaSketch* and the *National Aquatic Biodiversity Information System*.¹⁶⁰ Overall, however, it is still reasonable to conclude that:¹⁶¹

we simply do not know enough about the marine ecosystem, and our impacts on it, to manage it sustainably. It can be argued we will never know enough. That is not the central issue. We do know enough now to be aware that we should proceed with caution, like a ship sailing in poorly charted waters. Instead we are charging ahead as though the precision of our ecological knowledge matched the precision with which we can position a trawl net in the ocean or an oil well in the earth's mantle beneath our seas. Sustainable management of the marine environment will, in practical terms, be a process of continual research and improvement.

There are significant questions as to whether we are investing enough in research on the marine environment to understand it better, whether we are targeting the right things, and whether our institutional settings¹⁶² and funding tools¹⁶³ are adequate to improve our knowledge sufficiently. The Parliamentary Commissioner for the Environment has pointed out that is difficult to tell how much actually *is* being invested.¹⁶⁴ The Prime Minister's Chief Science Advisor has also recently pointed to a number

of issues about data and information in decision-making in the context of commercial fishing.¹⁶⁵ A heavy burden can be placed on applicants for consent in the marine space, given the lack of research and information. Combined with a precautionary principle (see Chapter 7), this has resulted in activities like mining being declined consent despite significant investment being made.

The way in which information is generated, funded, stored, shared and used in the current system has a number of issues.

Compliance with international law

While many international legal obligations are expressed in a relatively high-level way (as summarised earlier in Figure 3.5), it is not clear that Aotearoa New Zealand's current system is doing enough to comply with our international obligations (or at least expectations or commitments).

There are general obligations to protect the marine environment under UNCLOS (and it is hard to establish that we are breaching those in any specific sense), and there are more focused requirements and commitments under treaties like the Convention on Biological Diversity.¹⁶⁶ Some have questioned whether Aotearoa New Zealand is meeting those, particularly with respect to slow progress in creating MPAs and linking protective measures to the conservation status of marine species.¹⁶⁷

There are questions to be asked as to whether we are living up to international expectations with respect to oceans management, especially when it comes to marine biodiversity protection.

3.5 Concluding comments

Above, we have identified a number of problems and challenges with respect to the marine environment itself and the existing management system. A future system could reasonably be expected to “fix” these. However, the very concept of a “problem” is subjective. What is problematic for one person may not be for another. Even if there is broad consensus, the reasons behind why something is seen to be a problem (and therefore the legitimacy of different solutions to it) can differ. And solving one problem might create a completely different one, making a quick “fix” out of reach. These considerations are important when it comes to looking at objectives in Chapter 7.



Awaawaroa Bay, Bay of Islands

Endnotes

- 1 Charles N Ehler "Two Decades of progress in Marine Spatial Planning" Marine Policy 132 (2021) 104134 at [3].
- 2 At 38.
- 3 Anne-Marie Jackson, Ngahuia Mita and Hauiti Hakopa *Hui-te-ana-nui: Understanding kaitiakitanga in our marine environment* (report prepared for Sustainable Seas National Science Challenge, Te Koronga, University of Otago, July 2017) at 38.
- 4 See Tamatoa Bambridge (ed) *Rāhui: Legal pluralism in Polynesian traditional management of resources and territories* (ANU Press, Australia, 2016).
- 5 Charles N Ehler "Two Decades of progress in Marine Spatial Planning" Marine Policy 132 (2021) 104134 at [3].
- 6 See for example, Foss Leach *Fishing in pre-European New Zealand* (New Zealand Journal of Archaeology, Wellington, 2021) at 311.
- 7 See Benjamin Dimitrios Hanara and Anne-Marie Jackson *Tangaroa Ara Rau: Tangaroa the Atua of Human Movement* (Ngā Pae o te Māramatanga, ngā Ākonga Report 18INT01, 2019).
- 8 *Re Edwards (Te Whakatohea (No 2))* [2021] NZHC 1025.
- 9 *Re Edwards (Te Whakatohea (No2))* [2021] NZHC 1025 at [37]; *Attorney-General v Ngāti Apa* [2003] 3 NZLR 643 (CA). See further *Re Edwards* at [70], [71] and [120]. This approach also aligns with the MACA Act, s 99 which provides for the Court to refer to the Māori Appellate Court or pūkenga for opinion or advice on tikanga. Churchman J also referred to "Kupe's Law" and "Cook's Law" and noted that the Courts have started to engage in an analysis of the relationship between the "first and second laws of Aotearoa New Zealand and their impact on the current legal system" at [69].
- 10 *Re Edwards* at [206].
- 11 *Re Edwards* at [227].
- 12 *Re Edwards* at [230]. For the effect of reclamation on customary marine title and protected customary rights claims see [231]–[250]; and regarding third-party structures and use of third-party use and occupation, see [251]–[269].
- 13 *Re Edwards* at [32]. Churchman J here referred to the MACA Act, s 6(1) and Preamble (4).
- 14 Per Churchman J in *Re Edwards* at [33].
- 15 See generally Dame Anne Salmond "Iwi vs Kiwi [Series]" *Newsroom* (online ed, 13 July 2021); Dame Anne Salmond "He Puapua and a Forgotten Promise" *Newsroom* (online ed, 12 July 2021) both <www.newsroom.co.nz>.
- 16 Depending on how one defines a "tool".
- 17 *Attorney-General v The Trustees of the Motiti Rohe Moana Trust & ors* [2019] NZCA 532.
- 18 Although the latter is little more than a shell through which key parts of the former are deemed to apply beyond the coastal marine area.
- 19 See Treaty of Waitangi (Fisheries Claims) Settlement Act 1992; Maori Fisheries Act 2004; Maori Commercial Aquaculture Claims Settlement Act 2004; Fisheries Act 1996, s 5(b); and Resource Management Act, ss 165E and 165K.
- 20 Waitangi Tribunal *Report of the Waitangi Tribunal on the Muriwhenua Fishing Claim* (Wai 22, 1988) ("the Muriwhenua claim").
- 21 Waitangi Tribunal *Report of the Waitangi Tribunal on the Muriwhenua Fishing Claim* (Wai 22, 1988) at 1.
- 22 Waitangi Tribunal *Report of the Waitangi Tribunal on the Muriwhenua Fishing Claim* (Wai 22, 1988) at 239 to 240.
- 23 See Māori Fisheries Act 1989, sss 4 and 45.
- 24 Treaty of Waitangi (Fisheries Claims) Settlement Act 1992.
- 25 Treaty of Waitangi (Fisheries Claims) Settlement Act 1992, ss 5 and 7.
- 26 Treaty of Waitangi (Fisheries Claims) Settlement Act 1992, s 14.
- 27 Māori Fisheries Act 2004, ss 31–32.
- 28 Māori Commercial Aquaculture Claims Settlement Act 2004, s 6.
- 29 Associated customary rights are addressed under the MACA Act. See Māori Commercial Aquaculture Claims Settlement Act 2004, s 6A.
- 30 Māori Commercial Aquaculture Claims Settlement Act 2004, s 34.
- 31 Māori Commercial Aquaculture Claims Settlement Act 2004, s 38.
- 32 Māori Commercial Aquaculture Claims Settlement Act 2004, s 45.
- 33 Māori Commercial Aquaculture Claims Settlement Act 2004, s 31.
- 34 These phases are identified and described by Raewyn Peart *Farming the sea: Marine aquaculture within resource management system reform* (EDS, Auckland, 2019) at 10.
- 35 See generally Lucy Brake and Raewyn Peart *Sustainable Seas: Managing the marine environment* (Environmental Defence Society, Auckland, 2015).
- 36 A question is "independent from what?". We are using this term in the sense of independence from government. Accountability is also used in this sense – being accountable to a broad electorate. This is explored in Chapter 12 (institutional design).
- 37 Such as proposals for new legislation, an oceans vision and so forth.
- 38 See Department of Conservation *Te Mana o Te Taiao - Aotearoa New Zealand Biodiversity Strategy 2020* (August 2020).
- 39 The Department of Conservation website still lists "New Marine Protected Areas Act" from 2016, and identifies a range of benefits the proposed reforms would bring, see <www.doc.govt.nz/get-involved/have-your-say/all-consultations/2016/new-marine-protected-areas-act/>. We discuss marine protected areas in Chapter 9.
- 40 Customary international law is also relevant, but in the environmental space it tends to be relatively high-level compared to treaties.
- 41 On regional fisheries agreements, see Appendix 2.
- 42 The maritime boundaries of Aotearoa New Zealand are confirmed by the Territorial Sea, Contiguous Zone and Exclusive Economic Zone Act 1977 and the Continental Shelf Act 1964 in accordance with UNCLOS.
- 43 UNCLOS, arts 207, 208, 210, 211 and 212.
- 44 UNCLOS, art 62(1).
- 45 UNCLOS, art 62(2).
- 46 Which effectively replaces the parent Convention for states parties.
- 47 Claire Charters *The Rights of Indigenous Peoples Under International Law and Their Domestic Relevance in Aotearoa/New Zealand* (December 2016) paper accessed from SSRN <http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2887664>
- 48 Jacinta Ruru "Finding Support for a Changed Property Discourse for Aotearoa New Zealand in the United Nations Declaration on the Rights of Indigenous Peoples" (2011) 15:4 LCLR 951.
- 49 Declaration on the Rights of Indigenous Peoples GA Res 61/295, A/Res/61/295 (2007), art 2.
- 50 Declaration on the Rights of Indigenous Peoples GA Res 61/295, A/Res/61/295 (2007), art 3.
- 51 Claire Charters "Te Tiriti o Waitangi, the UN Declaration on the Rights of Indigenous Peoples and Constitutional Change: What is the Role of Judges and Adjudication?" (International Association of Women Judges Annual Hui, 7 May 2021) [text available from <www.SSRN.com>].
- 52 Declaration on the Rights of Indigenous Peoples GA Res 61/295, A/Res/61/295 (2007), art 4.
- 53 Claire Charters *The Rights of Indigenous Peoples Under International Law and Their Domestic Relevance in Aotearoa/New Zealand* (December 2016) paper accessed from SSRN <http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2887664>
- 54 Jacinta Ruru "Finding Support for a Changed Property Discourse for Aotearoa New Zealand in the United Nations Declaration on the Rights of Indigenous Peoples" (2011) 15:4 LCLR 951.
- 55 Claire Charters "Te Tiriti o Waitangi, the UN Declaration on the Rights of Indigenous Peoples and Constitutional Change: What is the Role of Judges and Adjudication?" (International Association of Women Judges Annual Hui, 7 May 2021) [text available from <www.SSRN.com>].
- 56 Declaration on the Rights of Indigenous Peoples GA Res 61/295, A/Res/61/295 (2007), art 29.
- 57 Declaration on the Rights of Indigenous Peoples GA Res 61/295, A/Res/61/295 (2007), art 25.
- 58 Jacinta Ruru "Finding Support for a Changed Property Discourse for Aotearoa New Zealand in the United Nations Declaration on the Rights of Indigenous Peoples" (2011) 15:4 LCLR 951 at 956–958.
- 59 Jacinta Ruru "Finding Support for a Changed Property Discourse for Aotearoa New Zealand in the United Nations Declaration on the Rights of Indigenous Peoples" (2011) 15:4 LCLR 951 at 958.
- 60 Jacinta Ruru "Finding Support for a Changed Property Discourse for Aotearoa New Zealand in the United Nations Declaration on the Rights of Indigenous Peoples" (2011) 15:4 LCLR 951 at 958.
- 61 Jacinta Ruru "Finding Support for a Changed Property Discourse for Aotearoa New Zealand in the United Nations Declaration on the Rights of Indigenous Peoples" (2011) 15:4 LCLR 951.
- 62 See United Nations "Intergovernmental conference on marine biodiversity of areas beyond national jurisdiction" <www.un.org/bbnj/>; and Helen Briggs "Plastic pollution: Green light for 'historic treaty'" *BBC* (online ed, 2 March 2022) <www.bbc.com/news/science-environment-60590515>
- 63 Whether they *are* problems, and the *extent* to which they are problematic, may depend on worldviews.

- 64 Again, with the caveat that not all will agree on what the rationale for the system is (see Chapter 6).
- 65 We might expect more afforestation on land than sequestration in the sea, meaning the role of the oceans management system is far from obvious.
- 66 Greg Severinsen *Reform of the Resource Management System: A model for the future Synthesis report* (Environmental Defence Society, Auckland, December 2019) at 70.
- 67 Department of Conservation *New Zealand's latest National Report under the Convention on Biological Diversity: Reporting period 2014-2018* (Department of Conservation, Wellington, 2019) at 79.
- 68 At 78.
- 69 Department of Conservation *Biodiversity in Aotearoa: an overview of state, trends and pressures 2020* (Department of Conservation, Wellington, August 2020) at 98-99.
- 70 See Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation Synthesis Report and Next Steps* (Environmental Defence Society, Auckland, February 2019).
- 71 See generally Jane Kelsey *The New Zealand Experiment: A World Model for Structural Adjustment?* (Bridget Williams Books, Wellington, 1997).
- 72 On such issues more generally, see Quentin R Grafton "Responding to the 'Wicked Problem' of Water Insecurity" (2017) 31(10) *Water Resour Manag* 3023.
- 73 See Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at xi and xii.
- 74 *Environmental Defence Society Inc v The New Zealand King Salmon Co Ltd* [2014] NZSC 38, [2014] 1 NZLR 593.
- 75 *Environmental Defence Society Inc v The New Zealand King Salmon Co Ltd* [2014] NZSC 38, [2014] 1 NZLR 593 at [137]
- 76 Although it allows "stringency" where more stringent controls can be imposed by councils to achieve the objectives of the NZCPS.
- 77 See Keep Okura Green Incorporated *Hauraki Gulf and the Long Bay Okura Marine Reserve. Case Study: Weiti Development* (July 2017). The Department of Conservation is currently undergoing research on the area: Department of Conservation "Long Bay-Okura Marine Reserve sentinel site research" <www.doc.govt.nz/nature/habitats/marine/type-1-marine-protected-areas-marine-reserves/marine-sentinel-site-programme/sentinel-site---long-bay-okura-marine-reserve/>.
- 78 See Suzannah Dodd and Kay Vopel *Proposal for monitoring sedimentation in the Te Whanganui-a-Hei Marine Reserve* (Auckland University of Technology, Auckland, July 2010) at 2 and A Schwarz, R Taylor, J Hewitt, N Phillips, J Shima, R Cole and R Budd *Impacts of terrestrial runoff on the biodiversity of rocky reefs* (Fisheries New Zealand, New Zealand Aquatic Environment and Biodiversity Report 7, 2006).
- 79 See V A Froude and R Smith *Area-based restrictions in the New Zealand marine environment* report prepared by Pacific Eco-Logic Ltd for the Department of Conservation, Wellington, 2004).
- 80 Some seamounts, for example, have been closed.
- 81 As much as 82 percent of the benthic protection areas is too deep to be trawled with current technology: see J Leathwick, K Julian and M Francis *Exploration of the use of reserve planning software to identify potential Marine Protected Areas in New Zealand's Exclusive Economic Zone* (NIWA, June 2006) at 29 and A Reiser, L Watling and J Guinotte "Trawl fisheries, catch shares and the protection of benthic marine ecosystems: Has ownership generated incentives for seafloor stewardship?" (2013) 40 *Marine Policy* 75. Seamounts have been identified by the United Nations as specific habitats that need protection; however, only half of all seamounts are protected: Ministry for Primary Industries "Benthic protection areas" (16 November 2020) <www.mpi.govt.nz>.
- 82 But see proposed measures to change settings: Minister for Ocean and Fisheries "Fisheries Amendment Bill: Strengthening fishing rules and policies: landings and discards" (2 July 2021).
- 83 Anecdotal evidence suggests that more people are engaging in recreational fishing since Covid-19. There is an argument that recreational fishing is essentially self-moderating, as people do less of it if it becomes harder to catch fish (ie when stocks decline), but that is by no means clear.
- 84 Fisheries Act 1996, s 13(2).
- 85 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at vii.
- 86 At 53. See also *Royal Forest and Bird Protection Society v Minister of Fisheries* [2021] NZHC 1427.
- 87 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 48-50.
- 88 Dependent species must be considered in setting a TAC for a stock, but this process does not often take into account the complexity of marine ecosystems and trophic cascades.
- 89 See Wildlife Act 1953, s 7BA and sch 7A.
- 90 Wildlife Act 1953, s 68B.
- 91 Through providing a defence from prosecution if dolphins are killed "accidentally or incidentally" in the course of fishing and the event is reported and logged. See ss 26(4) and 16 of the Marine Mammals Protection Act 1978. Although it is an offence to fail to report incidental mortality under the Act, there is no penalty associated with the fishing-related mortality or injury itself.
- 92 OpenSeas NZ *Associated Species - Marine Mammals: Section Detail Report* (29 May 2019), at 4.
- 93 Fisheries Act 1996, ss 8 and 9.
- 94 Resource Management Act 1991, pt 2.
- 95 EEZ Act, s 10.
- 96 Resource Management Review Panel *New Directions for Resource Management in New Zealand* (June 2020) at 52.
- 97 At 52.
- 98 See New Zealand Coastal Policy Statement 2010.
- 99 *Environmental Defence Society Inc v The New Zealand King Salmon Co Ltd* [2014] NZSC 38, [2014] 1 NZLR 593 at [130].
- 100 See generally Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding Our Oceans* (Environmental Defence Society, Auckland, 2012) at Chapter 8.
- 101 *Environmental Defence Society Inc v Otago Regional Council* [2019] NZHC 2278 [*Port Otago*]. Now subject to appeal in the Supreme Court.
- 102 See Environmental Defence Society "Submission on the Natural and Built Environments Bill (Exposure Draft) 2021", <www.eds.org.nz/our-work/media/media-statements/media-statements-2021/changes-needed-to-strengthen-the-exposure/>.
- 103 See Marine Reserves Act 1971, s 5(6).
- 104 Resource Management Act 1991, s8.
- 105 EEZ Act, s 12. In practice, the Act has been read in a much wider light, despite this narrow Treaty clause. See *Trans-Tasman Resources Ltd v Taranaki-Whanganui Conservation Board* [2020] NZCA 86 at [133]-[180].
- 106 See Deidre Koolen-Bourke and Raewyn Peart *Conserving Nature: Conservation System Reform Issues Paper* (Environmental Defence Society, Auckland, 2021); *Ngāi Tai ki Tāmaki Tribal Trust v Minister of Conservation* [2018] NZSC 122; and Waitangi Tribunal *Ko Aotearoa Tenei: A Report into Claims Concerning New Zealand Law and Policy Affecting Māori Culture and Identity* (Wai 262, 2011).
- 107 *Ngāi Tai ki Tāmaki Tribal Trust v Minister of Conservation* [2018] NZSC 122.
- 108 Waitangi Tribunal *Ko Aotearoa Tenei: A Report into Claims Concerning New Zealand Law and Policy Affecting Māori Culture and Identity* (Wai 262, 2011); *Ngāi Tai ki Tāmaki Tribal Trust v Minister of Conservation* [2018] NZSC 122; and Ministry for the Environment *Natural and Built Environments Bill Exposure Draft* (June 2021).
- 109 See Ministry for the Environment *Natural and Built Environments Bill Exposure Draft* (June 2021), proposed s 6.
- 110 Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012, s 12. See Robert Joseph and others *Stemming the Colonial Tide: Shared Maori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand - Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2020) at 254-266.
- 111 See Robert Joseph and others *The Treaty, Tikanga Māori, Ecosystem-based Management, Mainstream Law and Power Sharing for Environmental Integrity in Aotearoa New Zealand - Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2019) at 3.
- 112 At 108.
- 113 See *Re Edwards (Te Whakatohea (No2))* [2021] NZHC 1025 on determining existence of customary marine title under the MACA Act.
- 114 Jo Moir "Govt no closer to landing Kermadec Sanctuary" *Newsroom* (online ed, 15 March 2021).
- 115 Waitangi Tribunal *The Marine and Coastal Area (Takutai Moana) Act 2011 Inquiry Stage 1 Report* (Wai 2660, 2020).
- 116 See Maxine Jacobs "Legal interpretations of wāhi tapu and rāhui sought in landmark rights hearing" *Stuff* (online ed, 17 February 2022) <<https://www.stuff.co.nz/pou-tiaki/300519200/legal-interpretations-of-whi-tapu-and-rhui-sought-in-landmark-rights-hearing>>
- 117 Marine Reserves Act 1971, long title.

- 118 Deidre Koolen-Bourke and Raewyn Peart *Conserving Nature: Conservation System Reform Issues Paper* (Environmental Defence Society, Auckland, 2021).
- 119 Parliamentary Commissioner for the Environment *Setting course for a sustainable future: The management of New Zealand's marine environment* (Office of the Parliamentary Commissioner for the Environment, Wellington, November 1999).
- 120 Parliamentary Commissioner for the Environment *Setting course for a sustainable future* (1999), preface.
- 121 That said, there have been other drivers for place-based legislation, such as the desire to implement packages of tools together (not just protected areas) and to be more focused in terms of implementation.
- 122 See for example Seachange Stakeholder Working Group *Sea Change Tai Timu Tai Pari Hauraki Gulf Marine Spatial Plan* (Hauraki Gulf Forum in partnership with others, April 2017), chapter 6; Minister for Oceans and Fisheries Revitalising the Hauraki Gulf – Government Sea Change Strategy (2 July 2021).
- 123 (Ngā Aho Incorporated Society, Auckland, October 2016). Compare TM Lenihan and J Bartley *Review of Māori planning futures: Review of the Productivity Commission's "Better urban planning" draft report* (Ngā Aho Incorporated Society, Auckland, October 2016) at 36.
- 124 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018).
- 125 MWD White *Australian Marine Pollution Laws* (2nd ed, The Federation Press, Annandale (NSW), 2007) at 186.
- 126 Bevan Marten *Maritime Law in New Zealand* (Thomson Reuters, 2016) at 66.
- 127 See *Laws of New Zealand* Maritime Law: Shipping and Navigation (online ed) at [2.11].
- 128 Under the Maritime Transport Act 1994 the Director of Maritime New Zealand and the Minister of Transport have broad powers to make delegated legislation by way of maritime rules to regulate maritime activity, and marine protection rules to implement international conventions and standards relating to the marine environment.
- 129 Bevan Marten *Maritime Law in New Zealand* (Thomson Reuters, 2016) at 66.
- 130 See for example *Attorney-General v The Trustees of the Motiti Rohe Moana Trust* [2019] NZCA 532.
- 131 The NES for Marine Aquaculture can hardly be said to have been designed to implement the policies of the NZCPS in the round – it is a much more targeted consenting tool.
- 132 *Attorney-General v The Trustees of the Motiti Rohe Moana Trust* [2019] NZCA 532.
- 133 See Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012, s 37A.
- 134 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 128.
- 135 At 14. See also Fisheries Act 1996, ss 25A and 25B.
- 136 At 14-15.
- 137 At vii, 38 and following.
- 138 Kate Mulcahy and Raewyn Peart *Wonders of the Sea: The protection on New Zealand's marine mammals* (Environmental Defence Society, Auckland, 2012) at 80.
- 139 Deidre Koolen-Bourke and Raewyn Peart *Conserving Nature: Conservation System Reform Issues Paper* (Environmental Defence Society, Auckland, 2021) at 72.
- 140 Kate Mulcahy, Raewyn Peart and Abbie Bull *Safeguarding Our Oceans* (Environmental Defence Society, Auckland, 2012) at 101.
- 141 See for example Seachange “Sea Change” <www.seachange.org.nz>.
- 142 See Department of Conservation *Te Mana o Te Taiao - Aotearoa New Zealand Biodiversity Strategy 2020* (Department of Conservation, Wellington, August 2020).
- 143 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018), ch 5.
- 144 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Synthesis Report* (Environmental Defence Society, Auckland, February 2019) at 114.
- 145 See for example, Waitangi Tribunal *The Final Report on the M V Rena and Motiti Island Claims* (Wai 2391 and 2393, 2015) at 39-43 and 57-58; Resource Management Review Panel *New Directions for Resource Management in New Zealand* (June 2020), ch 3.
- 146 See Robert Joseph and others *The Treaty, Tikanga Māori, Ecosystem-Based Management, Mainstream Law and Power Sharing for Environmental Integrity in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2019).
- 147 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 121.
- 148 Although protected areas have been/are being progressed on a considered basis in places like the Subantarctic Islands, the West Coast of the South Island, South-East Otago and the Hauraki Gulf, these processes are not framed by an overarching legislative framework or common/predictable process.
- 149 Parliamentary Commissioner for the Environment *A review of the funding and prioritisation of environmental research in New Zealand* (December 2020) at 35, 45 and following.
- 150 At 3.
- 151 At 72.
- 152 At 37.
- 153 At 57.
- 154 At 52.
- 155 Parliamentary Commissioner for the Environment *A review of the funding and prioritisation of environmental research in New Zealand* (December 2020) at 49.
- 156 At 41; see also Waitangi Tribunal *Ko Aotearoa Tenei: A Report into Claims Concerning New Zealand Law and Policy Affecting Māori Culture and Identity* (Wai 262, 2011) at 561 and following.
- 157 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 3. See also Fisheries New Zealand *Aquatic Environment and Biodiversity Annual Review 2019-20: A summary of environmental interactions between the seafood sector and the aquatic environment* (Fisheries Science Team, Ministry for Primary Industries, Wellington, June 2020).
- 158 See for example Office of the Prime Minister's Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021) at 108 and following.
- 159 Parliamentary Commissioner for the Environment *A review of the funding and prioritisation of environmental research in New Zealand* (December 2020) at 76.
- 160 Department of Conservation “New Zealand mangroves and seagrass database” (November 2018); Seasketch <<https://www.seasketch.org/#projecthomepage/5357cfa467a68a303e1bb87a>>; and Ministry for Primary Industries “National Aquatic Biodiversity Information System” <<https://maps.mpi.govt.nz/templates/MPViewer/?appid=96f54e1918554ebbf17f965f0d961e1>>.
- 161 Parliamentary Commissioner for the Environment *Setting course for a sustainable future: The management of New Zealand's marine environment* (Office of the Parliamentary Commissioner for the Environment, Wellington, November 1999)
- 162 For example, is our system of Crown Research Institutes and the way central government is configured to fund and directly deliver research adequate?
- 163 For example, are funding tools available to regional councils sufficient for them to discharge their responsibilities in managing marine habitats and biodiversity?
- 164 Parliamentary Commissioner for the Environment *A review of the funding and prioritisation of environmental research in New Zealand* (December 2020) at 3.
- 165 See Office of the Prime Minister's Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021).
- 166 The Convention on Biological Diversity 1760 UNTS 69 (opened for signature 5 June 1992, entered into force 29 December 1993).
- 167 See for example Eugenie Sage “International report card on New Zealand's indigenous nature - we must do better” (media release, 4 April 2019) <<https://www.beehive.govt.nz/release/international-report-card-new-zealand%E2%80%99s-indigenous-nature-%E2%80%93-we-must-do-better>>

4 The context of reform

Wellington harbour

4.1 Introduction

The description of the oceans management system in Chapter 3, and problems with it, shows a snapshot in time. The reality is that the system, like the sea itself, is dynamic and constantly evolving. It has an interesting history that informs both the present and the possibilities for future change. We also have a reform minded government that is taking a number of measures relevant to the marine space, and is contemplating deeper reform. In this chapter, we outline key aspects of the context that will influence systemic change to our oceans management system in the future.

4.2 Planned marine reforms

Some features of the current system are the product of recent change, and are described in Appendix 1. The implications of some of these are still playing out (including important case law under the EEZ Act about the interpretation of its principles,¹ under the RMA concerning the place of environmental bottom lines,² and under the Conservation Act relating to te Tiriti o Waitangi obligations).³ However, there are other features (including a number of legislative reforms or policy reviews) that are emerging at the time of writing or are planned for the future (see spotlight). They provide important context for future, deeper, oceans reforms. They also provide potential opportunities. In a few years' time, the "existing" system may look quite different.

A spotlight on emerging or planned reforms

There is a wide range of prospective reforms which could have implications for the future of an oceans management system. They include:

- Ongoing work on national direction under the RMA, including common wastewater standards, proposed NPS for Indigenous Biodiversity, and rollout of freshwater policy under the NPS for Freshwater Management.
- The development of emissions reduction plans under the Climate Change Response Act, which may have implications for marine activities, as well as for broader activities impacting on the marine environment.
- A review of the Biosecurity Act (including marine biosecurity).
- Ongoing reform of "three waters" – drinking water, stormwater and wastewater – where it is envisaged there will be a small number of publicly owned water service providers and a much smaller role for councils. This has implications for the funding, planning and provision of infrastructure that can cause or prevent harmful discharges to the marine environment.
- A review into the structure of local government, which has implications for the role of regional councils in marine management as well as territorial authorities in managing infrastructure and land use that can affect the ocean.
- Measures to tackle freshwater and sediment from land use practices being advanced by the Essential Freshwater package and the Productive and Sustainable Land Use package.⁴
- A review of waste legislation – the Waste Minimisation Act and the Litter Act, which will have implications for what kinds of waste end up in the marine environment, and the deployment of tools under the former to phase out various single use plastics.
- Continued efforts by regional councils to grapple with the implications of the *Motiti* decision⁵ (how they discharge their responsibilities in managing habitats impacted by fishing activities under the RMA).

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Flesh-footed shearwater, Hauraki Gulf

- Ongoing actions set out in *Responsibly delivering value - a minerals and petroleum resource strategy for Aotearoa New Zealand: 2019 – 2029*.⁶
- Changes to the Environmental Reporting Act to strengthen the environmental reporting system.
- Ongoing work by the Minister/Ministry of Transport to decarbonise the shipping sector.
- Investigations into legal frameworks for offshore wind energy.
- Ongoing development of case law, including in relation to how environmental limits under *King Salmon* are applied to marine infrastructure such as ports,⁷ relevant factors in determining how fish stocks recover under the Fisheries Act,⁸ and how the EEZ Act and its precautionary principle applies to deep sea mining activities.⁹
- International law developments, including “work with Pacific Islands Forum partners relating to sea level rise and a 2050 Blue Pacific Strategy, global engagement on marine plastic litter, negotiating a new High Seas Biodiversity Treaty, and a Post-2020 Global Biodiversity Framework”.¹⁰

Many iwi throughout the country have lodged applications with the Crown and in the High Court for recognition of customary title and protected customary rights under the MACA Act, and the extent of these applications is such that they cover the majority of the Aotearoa New Zealand coastline.¹¹ Few claims have been processed by Te Arawhiti (formerly the Office of Treaty Settlements) which on behalf of the government received over 380 applications by the statutory cut-off date.¹² While this involves a site by site assessment, cumulatively the ongoing process has potential to alter the way in which many parts of the foreshore and seabed are managed. Significant precedents on the the MACA Act and the nature of customary marine title are being determined through litigation in the High Court.¹³

Cutting across all these things are ongoing te Tiriti negotiations creating bespoke arrangements, not just for redress of grievances, but also for place-specific co-governance and other arrangements. These negotiations are, slowly, causing the system to evolve in an iterative way. And although it is not a reform measure itself, the release of *He puapua* (which means “a break” and usually refers to a break in the waves), which envisages how

Aotearoa New Zealand might implement the UNDRIP, has contributed to the deeper conversation about how the Māori-Crown relationship – and society – will work in the future, including at a constitutional level.¹⁴ This goes back to broad and evolving discussions about the nature of te Tiriti o Waitangi itself and what this means in the 21st century (see the discussion further below).

Careful consideration will be needed as to how the many different reform streams intersect (including over time), particularly in light of the capacity of the public service to support multiple reforms and investment priorities concurrently.

Two key packages of measures are particularly notable due to their breadth, depth and imminence. The first relates to resource management system reform, in which EDS has been intimately involved in recent years. The government’s independent resource management panel, chaired by Hon Tony Randerson QC, released a report in 2020 outlining a series of recommendations for sweeping reform.

The scope of the panel’s report, and the government’s programme of “resource management” reforms, is not as wide as the “system” we have defined in this report (in that the former is firmly focused on the replacement of the RMA, its connections with existing legislation, and the introduction of new legislation for spatial planning and climate change adaptation). This means that the oceans management system – at least as we have defined it – is not a “subset” of the government’s narrower conception of the resource management system, because oceans management has many components that lie beyond it (notably fisheries and conservation legislation). However, there is still considerable overlap, in that the RMA is intimately concerned with marine management in the coastal marine area (which roughly aligns with the territorial sea as defined under UNCLOS).



Nesting gannets, Mokohinau Islands

Raewyn Peart

The upshot is that a significant part of how our oceans are managed is set to be changed, and further reforms in this area need to be cognisant of that. The current resource management reform process represents both an opportunity to progress oceans reform (and to tailor current reform proposals to that context where needed) and a potential constraint (in that high level policy decisions, for example around legislative design, have already been taken and are unlikely to be revisited at a fundamental level).

A spotlight on the Randerson Panel's key recommendations on resource management reform¹⁵

The Randerson Panel produced an extensive report on resource management reform. Some of the key recommendations of significance for oceans reform are highlighted below.

- Repeal the RMA and replace it with a new NBA, with a revised purpose and principles. This signals a wider shift in the Act from managing environmental effects to achieving positive outcomes.
- Create a new Strategic Planning Act, requiring the preparation of regional spatial strategies encompassing both land and the coastal marine area. These strategies would align functions across other statutes, including the NBA, the Local Government Act, the Land Transport Management Act and the Climate Change Response Act.
- Enact a dedicated Managed Retreat and Climate Change Adaptation Act, which will provide for managed retreat and the establishment of a climate change adaptation fund.
- Require decision-makers to give effect to the principles of te Tiriti o Waitangi, and incorporate the overarching concept of te oranga o te taiao in the purpose statement of the NBA.
- Establish a National Māori Advisory Board to monitor the performance of central and local government in giving effect to te Tiriti and provide for an integrated partnership process between mana whenua and councils.
- Require national direction to be made on a range of core matters, and combine this into a coherent suite of instruments that clearly resolve conflicts and relationships between them.

- Require the establishment of environmental limits and targets.
- Reformulate existing RMA plans into combined regional plans, reducing the 100 or so plans we have now to just 14.
- Reform the planning process, including the establishment of joint planning committees comprising regional council, territorial authority and mana whenua representatives.
- Require an audit of plans by the Ministry for the Environment before they are notified.
- Alter how the notification framework operates, including removing the “no more than minor” threshold for notification of consents.
- Remove non-complying activity status.
- Provide an alternative dispute resolution pathway for minor matters.
- Strengthen the overall role of the Environment Court.
- Strengthen the framework for water conservation orders.
- Provide more flexibility to review existing resource consents to create a more responsive system.
- Provide for greater use of economic instruments to drive behaviour change.
- Establish a nationally coordinated environmental monitoring system led by the Ministry for the Environment.
- Expand the role of the Parliamentary Commissioner for the Environment to provide a stronger auditing and oversight role of the resource management system.
- Establish regional hubs for compliance, monitoring and enforcement.
- Strengthen offences and penalties for non-compliance.

The above is a selective summary. The recommendations of the Randerson Panel are wide ranging and complex, and they arise at a number of points in

this report. There are two overarching points to note for the marine context, though. First, most that is changing about the RMA will affect management in the coastal marine area. This ranges from the mandatory establishment of limits for the area, to a more comprehensive and integrated approach to national direction (and its implications for the NZCPS), to how plans are made, consents dealt with and enforcement undertaken.

Secondly, it remains uncertain as to how proposed new legislation for regional spatial planning, the Strategic Planning Act, will (or should) apply to the marine context, despite the fact that it is intended to extend out to the seaward edges of the coastal marine area. We look at this in the context of legislative design in Chapter 11. There are also questions about how a new statute on climate change adaptation will apply offshore – whether it will be primarily about issues on land like managed retreat (including the impact of things like hard seawalls on marine life), or whether it will also seek to manage climate induced changes *in* the marine area (eg how sectors like aquaculture can adapt).

The process of implementing reform is now well underway. In particular, the government has recently released an “exposure draft” of some of the key provisions (including a purpose and principles part) of a new NBA, which has been considered through a novel select committee process.¹⁶ For the most part it confirms the spirit of the Randerson Panel’s proposals.¹⁷ We understand key policy decisions will be taken during 2022.

We emphasise, however, that resource management reform does not obviate the need for wider ranging oceans reform. For instance, there are the spatial limitations to the current process – it does not include anything beyond the territorial sea. There are also important sectoral exclusions (especially fishing and marine conservation). And its normative foundations are focused on land rather than on the sea. Indeed, the spirit of reform is very much about improving the RMA and its relationships with infrastructure provision and climate change legislation, not grappling with the much wider issue of fragmented management of te moana.

The other key element of context worth noting is the government’s interest in pursuing holistic oceans reform directly. This was signalled by the establishment of a new ministerial portfolio for oceans and fisheries (previously just fisheries). While the extent of appetite for reform remains unclear, there have been strong indications in a series of recent Cabinet papers that significant fisheries-focused reform is planned.¹⁸ That goes as far as saying that “significant reform of the fisheries system is required”,¹⁹ including commercial fishing (which was the subject of a report by the Prime Minister’s Chief Science Advisor earlier in 2021).²⁰

Among the reforms signalled are targeted measures like the rollout of cameras on boats,²¹ potential changes to the QMS (although retention of the basic tool), revisiting the National Plan of Action on Sharks, significant changes to rules around discarding and landing fish,²² a more graduated and nuanced system of penalties for non-compliance, the ability to establish pre-set changes to catch limits and other sustainability measures in advance, more responsive tools to set recreational fishing (eg bag) limits, and an industry transformation plan (with possible support for innovation, including transitioning away from harmful fishing methods like bottom trawling).²³

Proposals also include targeted initiatives, such as implementing some of the recommendations of the Sea Change Tai Timu Tai Pari marine spatial plan developed for the Hauraki Gulf. That includes using available tools to do things like reducing the trawling footprint²⁴ and creating a spatially based fishing plan (the first of its kind);²⁵ a trial and eventual framework for novel locally-controlled Ahu Moana areas; and likely bespoke legislation to establish novel high protection areas that allow some customary activities.²⁶ The further development of legislation for Rangitāhua/the Kermadec Islands is also signalled, as is support for new marine development opportunities (eg offshore aquaculture). The South-East Marine Protection initiative is also continuing (to implement a network of MPAs in the south-eastern South Island coastal marine area). And despite a chequered history, there is still a commitment to progress new MPA legislation (although that may come after more targeted protections through Sea Change and for Rangitāhua/Kermadec Islands).



Cape Rodney-Okakari Point marine reserve

Craig Potton

Overall, this package is significant. However, equally significant is the signal that this will take place within a broad “vision” for the oceans founded in ecosystem-based management, described as “ensuring the long-term health and resilience of ocean and coastal ecosystems, including the role of fisheries”. Cabinet papers refer to a suite of objectives and principles that are intended to support this vision, and the establishment of an Oceans and Marine Ministers Group²⁷ as well as an inter-agency Oceans Secretariat²⁸ to progress an integrated work programme. Although what it will comprise and how far it will go is still up in the air, longer-term and deeper reform measures are also mentioned as a possibility,²⁹ with EDS’s project specifically mentioned as an input. There is to be “an assessment of how far the initial work programme will go in realising the vision and objectives, and what future longer-term work may be necessary”.³⁰ It is heartening to see that the horizons

for fundamental reform are broadening – including the possibility of institutional change and legislative rearrangement.

The articulation of a “vision” for Aotearoa New Zealand’s oceans does not really form part of the formal “system”, and instead should be regarded as a manifesto and mandate for change. But combined with an integrated oceans portfolio, and the coordinated way in which surgical measures are being progressed, it indicates that the winds of change are blowing across our seas.

The concept of developing a common vision and set of principles and objectives for our oceans is not new. Twenty years ago there were similar efforts to progress a conversation about marine reform, but these foundered on the rocky reefs of the controversial foreshore and seabed debate (see further below). Those events provide valuable historical context for efforts to undertake reform today.

Biodiversity is also on the agenda. The proposed NPS for Indigenous Biodiversity (which although largely excluding biodiversity in freshwater and the coastal marine area still has some relevance to the marine space)³¹ remains on the table, although it appears to have stalled somewhat. Yet a non-statutory biodiversity implementation plan, and biodiversity targets to (among other things) develop a network of representative MPAs, is expected in 2022.³² *Te mana o te taiao – the Aotearoa New Zealand biodiversity strategy 2020* states that implementation plans will include further goals, actions to progress towards goals, responsibility for the delivery of goals, and indicators to measure progress.³³

Reforms to the oceans management system will take place against a constantly shifting background of policy, legal and institutional change. The current government has an active reform agenda, and this presents opportunities and potentially constraints for future reform. Most significant will be the implementation of the Randerson Panel’s recommendations on resource management reform and targeted changes to fisheries law.

4.3 An evolving constitutional conversation

Although it extends well beyond the marine context, *te Tiriti o Waitangi* forms a critical part of the backdrop for both resource management and oceans reform. The legal place of *te Tiriti* itself, the principles of *te Tiriti*, and statutory references to them, have evolved over time.³⁴

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Rocky coast, Northland

However, it is worth considering the origins of the document and for this to be held in the minds of policy makers. This is in the context of growing calls for the relationship between Māori and the Crown to be reimagined or reinvigorated beyond just the settlement of historical grievances and generic references to paternalistic principles about active protection and partnership. A focus is being placed, not just on te Tiriti, but also on He Whakaputanga – the Declaration of Independence of the United Tribes of New Zealand (1835) and UNDRIP (see Chapter 3). It has been suggested that the “full and final” settlement of some matters could be revisited, which may have implications for fisheries and aquaculture.³⁵

In short, the Crown-Māori relationship is no longer just about righting the wrongs of the past and moving on. It is about the nature of sovereignty and constitutional arrangements in post-colonial Aotearoa New Zealand.³⁶ The settlement process has taken on quite a different character from 20 years ago. *He Puapua* and the report of Matike Mai Aotearoa – The Independent Working Group on Constitutional Transformation have contributed to a deeper conversation about how the Māori-Crown relationship – and society – will work in the future, including at a constitutional level.³⁷

Core to all this is whether sovereignty was ceded to the Crown, and the nature of rights to manage resources. Parliamentary sovereignty is a practical legal reality in Aotearoa New Zealand. However, the Waitangi Tribunal when hearing the Wai 1040 Treaty Claim has said (among other things) that:

- The rangatira who signed te Tiriti o Waitangi in February 1840 did not cede their sovereignty to Britain. That is, they did not cede authority to make and enforce law over their people or their territories.
- The rangatira agreed to share power and authority with Britain. They agreed that the Governor would have authority to control British subjects in Aotearoa New Zealand, and thereby keep the peace and protect Māori interests.
- The rangatira consented to te Tiriti on the basis that they and the Governor were to be equals, though they were to have different roles and different spheres of influence.
- The rangatira agreed to enter into land transactions with the Crown, and the Crown promised to investigate pre-Treaty land transactions and to return any land that had not been properly acquired from Māori.

- Though Britain went into the Treaty negotiation intending to acquire sovereignty, and therefore the power to make and enforce law over both Māori and Pākehā (non-Māori), it did not explain this to the rangatira. Rather, the explanations of the texts and the verbal assurances given by Hobson and his agents, said that Britain sought the power to control British subjects and thereby to protect Māori.

There has been considerable dispute due to varying interpretations of the te reo (Māori language) and English language versions of te Tiriti. Many are of the view that there was not a proper meeting of the minds. Significant questions will be grappled with over the coming years in the political sphere, and any oceans reforms will need to provide space for that to be worked out, without implementing changes that will require further overhaul. In short, reforms will need to respect te Tiriti as a living document, give effect to its principles, and defend existing settlements from erosion. As Dame Anne Salmond has pointed out in a series of articles,³⁸ the evolving context of reform might be one in which there is a strong sense of partnership, reciprocity and above all a weaving of different outlooks in a non-binary way. It is about our identity as a nation.



Waharoa (customary gateway), Rangitoto Island

Raewyn Peart

A spotlight on recent recommendations for constitutional transformation

The government's endorsement of UNDRIP in 2010 encouraged deeper conversations about how the rights of Māori could (and should) be realised within Aotearoa New Zealand's constitutional framework. In 2010, the National Iwi Chairs forum founded Matike Mai Aotearoa, an independent working group tasked with setting out what an inclusive constitution might look like for Aotearoa New Zealand.³⁹ The specific terms of reference for Matike Mai were to develop and implement a model of governance based on tikanga and kawa (Māori philosophy and law), existing agreements between Māori and the Crown (He Whakaputanga and te Tiriti),⁴⁰ and indigenous human rights instruments.⁴¹

In pursuit of this objective, between 2012 and 2015, the working group engaged extensively with Māori on potential models of governance by facilitating 252 hui and working groups, inviting written submissions, and conducting interviews.⁴² Drawing on the responses received, the working group published a collective Māori vision of a constitutional framework that was consistent with te Tiriti and the rights and duties affirmed under UNDRIP (refer to spotlight in Chapter 3).

The Matike Mai Report emphasises the need to establish a values-based constitutional framework.⁴³ It identifies “political and social inclusiveness” as a central theme deriving from te Tiriti, in addition to seven specific interrelated values: tikanga, community, belonging, place, balance, conciliation and structure.⁴⁴ The value of place expressly recognises the need to “promote relationships with, and ensure the protection of Papatūānuku”.⁴⁵

The working group set out a series of six indicative models for constitutional transformation. All the options require shared authority between Māori and the state, but they provide for this constitutional arrangement in different ways. The models provide for different spheres of influence described as “the rangatiratanga sphere” (where Māori make decisions for Māori); “the kāwanatanga sphere” (where the Crown will make decisions for its people); and “the relational sphere” (where Māori and the Crown will work together as equals).⁴⁶

In 2019, in response to ongoing pressure at the domestic and international level, Te Puni Kōkiri (the Ministry of Māori Development) established a working group to advise the government on a plan for implementing UNDRIP in Aotearoa New Zealand.⁴⁷ The report of the working group (“He Puapua”) was released in 2019 as the result of an official information request.

He Puapua sets out a roadmap for realising the rights of Māori under UNDRIP by 2040, which is described as “Vision 2040”. The overarching vision expressly recognises the central importance of the environment:⁴⁸

Our vision is that in 2040 rangatiratanga Māori is realised, Māori and the Crown enjoy a harmonious and constructive relationship and work together to restore and uphold the wellbeing of ngā tangata, Papatūānuku and the natural environment.

The roadmap for Vision 2040 comprises five core elements, which draw on the recommendations of the Matike Mai Report: rangatiratanga (in recognition of Māori authority over Māori matters); Māori participation in kawanatanga (“a bicultural sphere”); lands, territories and resources; culture; and equity.⁴⁹ The vision expressly recognises that effective participation requires that “Māori will have a meaningful and sometimes dominant voice in resource management decisions”.⁵⁰

The roadmap envisages an enlarged Māori estate supported by significantly increased return of lands and waters, including takutai moana, to Māori ownership.⁵¹ In addition, it seeks to ensure iwi can contribute towards the control of, access to, and management of lands and resources within their rohe in accordance with tikanga and mātauranga Māori.⁵² A central overarching theme of He Puapua is the need to provide for equity between Māori and non-Māori in constitutional arrangements.

This context has a heavy bearing on what the principles and objectives of a future system should be (see Chapter 7). That is not just in terms of incorporating Māori values, information and worldviews into the system, but also the more direct issue of whether the system should actively pursue specific things like co-governance, the transfer of powers, and parallel Māori institutions, and the extent to which reforms impacting settlement rights are off the table. Some options for doing these things in the marine context appear throughout the report, although there will be many others.

Oceans reform will take place in the context of an ongoing and evolving conversation about Māori sovereignty under te Tiriti o Waitangi and the nature of sovereignty and constitutional arrangements in post-colonial Aotearoa New Zealand.

4.4 The importance of historical context

As mentioned above, the idea of oceans system reform is not new. A concerted effort was made two decades ago to do a similar thing, and although many contextual factors have moved on (including te Tiriti and biophysical change), lessons can still be learned from this attempt. We summarise the oceans policy process of the 2000s in Appendix 4. But it is also interesting to consider where the current system – which failed to be fundamentally transformed in the 2000s – has come from. This is intimately linked to the development of the broader resource management

system. A summary of this can be found in our previous work on resource management reform, as well as other publications.⁵³ In short:

- With the advent of a largely Western system of laws following British colonisation of Aotearoa New Zealand, the management of resources was initially considered to be one of property protection and minimising impacts of resource use on other people's health and property.
- It was not until the 1950s and 1960s, in the wake of the international environmental movement, that Aotearoa New Zealand started to implement laws specifically targeting environmental health. This was done in a fairly ad hoc way, and by the start of the 1980s a large number of resource and issue-specific statutes were in existence. Some remain today, such as the Wildlife Act, Marine Reserves Act and Marine Mammals Protection Act. This reflected an evolving international recognition of environmental issues.



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Russell waterfront

- The domestic politics of the late 1970s and 1980s played a central role in producing the resource management system we have today. Under the oversight of Prime Minister Robert Muldoon, development became centralised and overrode many environmental considerations, epitomised in the National Development Act and bespoke legislation for the Clyde Dam.
- David Lange's Labour-led government came to power in 1984 on the back of political and public pressure for fundamental economic and social change across the board. There was a reaction against the centralised, untransparent and economically interventionist approach of previous times. The country also rode the wave of the broader international zeitgeist of economic rationalism and free market thought, and sought to reconcile this with a budding environmental movement.
- In the reforms devised in the later 1980s, central government took a back seat in both development and environmental management (most powers were devolved), many things were rationalised into a single effects-based RMA (that sought integrated management but rejected economic planning), institutional checks and balances were put in place (eg in the roles of the Environment Court and Parliamentary Commissioner for the Environment) and fragmented councils were amalgamated. Small government was the watchword of the day. Subject to limits (which were, admittedly, often not put in place), markets would provide.
- Fisheries management was transformed in this era with the advent of the QMS, where commercial fishing rights were privatised (reflecting the spirit of the times). This provided the currency for settling te Tiriti claims for fisheries, and aquaculture claims were also settled (although not through perpetual occupation rights).
- Conservation legislation was not rationalised or integrated, but a unifying force was provided through the creation of a statutory Department of Conservation that was to administer other statutes under the umbrella of the Conservation Act.
- The core frameworks born in this period remain largely intact today, even though there has been no shortage of reform around them (eg many amendments to the RMA including for aquaculture, the creation of an EPA, new climate change legislation, two iterations of legislation for the foreshore and seabed, numerous te Tiriti settlement acts,

special legislation for Auckland Council, and a new EEZ Act that roughly resembles the RMA).

Within this broader picture one can observe that the arc of marine legislation has been slightly different. Initially, conservation and pollution control efforts were concentrated on terrestrial landscapes – that is what people could see and understand – and to some extent this mentality has persisted until the modern day. The ocean and its resources were for a long time seen as so bountiful – including as a sink for waste – that human activity could have no impact. Until relatively recently, estuaries were commonly used as municipal rubbish tips, and aside from the most egregious forms of dumping, control of marine pollution and litter still does not have the same rigour about it as freshwater matters have seen in recent times.⁵⁴

Assumptions about the inexhaustibility of the oceans changed with a realisation that fish stocks were vulnerable things. As far back as 1866, legislation was passed to control the taking of oysters and in 1877 the Fish Protection Act introduced closed seasons and limits on the mesh size of nets for finfish.⁵⁵ This was combined with other measures in the Fisheries Act 1908. The ethos of the protective measures was to ensure fish and shellfish would be available for future catches – the collapse of stocks was bad for industry and people – rather than for the intrinsic value of the creatures or the health of surrounding ecosystems.⁵⁶ Indeed, in 1982 UNCLOS introduced and embedded the goal of “maximum sustainable yield” for fisheries at an international level,⁵⁷ which essentially remains the management aim in domestic fisheries management to this day.

The 1950s and 60s was a time when more New Zealanders ventured under the surface of the sea with the growing availability of SCUBA equipment. This revealed a vibrant and diverse underwater world that had been largely hidden from public view. This was also a time when marine science became more firmly established. The opening of a marine laboratory near Leigh, in 1962, directly led to the passing of the Marine Reserves Act in 1971 and the establishment of the first marine reserve in 1975, often said to be the first no-take protected area in the world. But the marine reserve legislation was modest in ambition, reserves were only to be set aside for scientific purposes, and those that slowly appeared around the coast were hard fought for, and generally very small in order to minimise impacts on fishers.

The 1970s was also a time when the anti-whaling movement lit up the popular imagination, due to the ongoing slaughter of diminishing whale populations. This prompted special legislation in Aotearoa New Zealand

to protect marine mammal species, although a carve out was included for fisheries bycatch.⁵⁸ There was also recognition that other marine wildlife (albeit only a few species) deserved protection that went beyond the achievement of sustainable fish stocks.⁵⁹

Shipping, and the threat of catastrophic oil spills from ships and offshore installations, are now managed.⁶⁰ The most egregious point source discharges into the oceans have been cleaned up, albeit mainly because of human health imperatives. Broad responsibilities to manage pollution and marine biodiversity were created in the RMA. Some glaring gaps in the management of the EEZ were finally filled in 2012 with the passage of the EEZ Act. Biosecurity concerns with shipping and aquaculture are now being tackled. And debates about the rights and interests of mana whenua have been addressed – albeit with many questions and issues still to be resolved – through the MACA Act.

However, technological advances over the last 50 years have not only brought better scientific understanding of the oceans and human impacts on life within them, but also the ability to exploit it at a larger scale (and in new ways). Overall, the story of the oceans management system has been one of piecemeal adjustment based on the gradual realisation of new threats. Things have been tacked on. Gaps have been filled slowly and, arguably, reluctantly. Many still remain, not least with respect to MPAs, climate change and systemic land-based pressures. On land, greater integration has been achieved through the RMA.

4.5 Other contextual features

A few other contextual points should be noted, that are important for where policy makers find themselves in the present day, and what they are likely to face in the future. These are not always obvious from reading a statute or government policy, but can be very powerful drivers behind the scenes.

- The presence of many existing rights and interests in the marine space, including property rights under the QMS.
- A sometimes fraught and unresolved relationship between commercial and recreational fishing sectors.
- Growing competition for different uses in the marine environment, including novel ones like offshore aquaculture, renewable energy, deep-sea mining and carbon capture and storage.

- Increasing environmental awareness amongst the public, particularly in younger generations.
- A volatile international context, including a broader fracturing of consensus within liberal democracies, supply chain disruptions and economic volatility.

A future system will need to pre-empt and address future challenges that could emerge, including the following.

- Human population change, and its implications for the pressures humans will put on oceans (especially in urban areas where population growth will be fastest), food security (including the role of protein from fish and fish farming), energy needs (including renewables in the marine space) and basic services (eg the need for drinking water and potential for desalination).
- Technological change, including the risks this poses for the marine environment (eg novel chemicals, more intensive forms of resource exploitation such as deep sea mining) but also the opportunities it affords in terms of cleaner industry and enhanced monitoring and information gathering.



Mussel harvesting, Pelorus Sound

Raewyn Peart

- Increasing political and economic expectations of Māori.⁶¹
- The unpredictability of climate change and its impacts, not just directly on the oceans, but also on human society and its use of the marine space.
- Ongoing and potentially escalating risks from marine biosecurity, particularly for the aquaculture industry which is facing new threats associated with competition from invasive species (which can foul up lines) and diseases (which can wipe out stock). Biosecurity risks also exist for wild fisheries to the extent that habitats are lost through invasive species.
- Social change and the evolving expectations of society.
- International developments, including in international marine law, but also in macro-level socio-political trends.
- The ongoing, unpredictable and cross-cutting implications of Covid-19 and future social and health emergencies (including implications for funding other things).

4.6 Concluding comments

Placing the oceans management system in its broader historical, social and reform context highlights that it is in a constant state of evolution. The system is never truly created or dismantled; it is an organic thing that is moulded to the changing shape of human society. However, in a first principles rethink of our oceans management system, it is legitimate to go further than just putting out fires and anticipating problems.

With the possible exception of the late 1980s, the 2020s can arguably be seen as an unprecedented hive of policy activity and creative thinking. Environmental awareness is growing, and Covid-19 has caused an important moment of reflection about our society and what it stands for. Thus, we need to think positively – the system is a tool for building our relationship with te moana, not just for reigning in the more destructive tendencies of humanity. What opportunities are out there? What do we want our future to be, keeping in mind it is likely to be quite different to the present?⁶² We have an opportunity to create something new, meaningful and reflective of who we are as a people and a country. What values do we want the oceans management system to reflect, in our unique cultural context?⁶³

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Endnotes

- 1 For example, see *Trans-Tasman Resources Ltd v Taranaki-Whanganui Conservation Board* [2020] NZCA 86.
- 2 For example, see *Environmental Defence Society Inc v Otago Regional Council* [2019] NZHC 2278, which is subject to appeal in the Supreme Court at the time of writing.
- 3 For example, see *Ngāi Tai ki Tāmaki Tribal Trust v Minister of Conservation* [2018] NZSC 122.
- 4 See Ministry for the Environment “Essential freshwater new rules and regulations” (April 2021) <www.environment.govt.nz>; and Ministry for Primary Industries “Productive and Sustainable Land Use” (16 November 2020) <www.mpi.govt.nz>.
- 5 *Attorney-General v The Trustees of the Motiti Rohe Moana Trust* [2019] NZCA 532.
- 6 Ministry of Business, Innovation and Employment *Responsibly Delivering Value: A Minerals and Petroleum Resource Strategy for Aotearoa New Zealand: 2019-2029* (November 2019).
- 7 *Environmental Defence Society Inc v Otago Regional Council* [2019] NZHC 2278 [Port Otago], which is subject to appeal in the Supreme Court.
- 8 *Royal Forest and Bird Protection Society v Minister of Fisheries* [2021] NZHC 1427 which is now subject to appeal.
- 9 *Trans-Tasman Resources Ltd v Taranaki-Whanganui Conservation Board* [2020] NZCA 86.
- 10 Minister for Oceans and Fisheries “Oceans and Fisheries portfolio – ensuring healthy ecosystems” (2 July 2021).
- 11 Paul Majurey and Christian Whata “Māori and Environmental Law” in Derek Nolan (ed) *Environmental and Resource Management Law* (LexisNexis, online ed, 2021) at [14.69].
- 12 Robert Joseph and others *Stemming the Colonial Tide: Shared Māori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2020)
- 13 *Re Edwards (Te Whakatohea (No2))* [2021] NZHC 1025.
- 14 Claire Charters and others *He Puapua: Report of the Working Group on a Plan to Realise the UN Declaration on the Rights of Indigenous Peoples in Aotearoa New Zealand* (Te Puni Kōkiri, 1 November 2019).
- 15 Resource Management Review Panel *New Directions for Resource Management in New Zealand*, (June 2020).
- 16 Ministry for the Environment *Natural and Built Environments Bill Exposure Draft* (June 2021).
- 17 With some exceptions, notably the absence of mandatory targets. See also Environmental Defence Society “Submission on the Natural and Built Environments Bill (Exposure Draft) 2021”, accessible at <https://www.eds.org.nz/our-work/media/media-statements/media-statements-2021/changes-needed-to-strengthen-the-exposure/>.
- 18 See Minister for Oceans and Fisheries “Oceans and Fisheries portfolio – ensuring healthy ecosystems” (2 July 2021); Minister for Oceans and Fisheries “Fisheries system reform agenda” (2 July 2021); Minister for Oceans and Fisheries “Fisheries Amendment Bill: Strengthening fishing rules and policies: landings and discards” (2 July 2021); Minister for Oceans and Fisheries “Fisheries Amendment Bill: Strengthening fishing rules and policies: offences and penalties and agile decision-making” (2 July 2021); Minister for Oceans and Fisheries “Revitalising the Hauraki Gulf – Government Sea Change Strategy” (2 July 2021); Minister for Oceans and Fisheries “Initial response to Prime Minister’s Chief Science Advisor’s report on commercial fishing” (2 July 2021); and Minister for Oceans and Fisheries “On-board cameras across the inshore fishing fleet” (2 July 2021).
- 19 Minister for Oceans and Fisheries “Fisheries system reform agenda” (2 July 2021).
- 20 Office of the Prime Minister’s Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021).
- 21 Around 300 inshore fishing boats are intended to have cameras by 2024.
- 22 Which is to be rolled out over several years to maintain the value of quota as fishers transition to more selective methods of fishing to avoid non-target species. QMS species, whether alive or dead, must be landed unless an exemption is issued by the Minister for Oceans and Fisheries. In other words, fishers will have to account for all fishing mortality against ACE or deemed values, creating an incentive to avoid fish species that have low or no value to them – in simple terms; if you kill it you will need to pay for it.
- 23 Some of these measures are reflected in an amendment bill that has just been introduced to the House at the time of writing, including tightening commercial fishing rules for landings and discards; introducing new mechanisms for commercial and recreational management decision-making (including pre-set decision-rules); enabling the further use of on-board cameras on vessels; and creating a new defence to help save marine mammals and protected sharks and rays.
- 24 Methods such as bottom trawling and Danish seining will be excluded from the Hauraki Gulf except within limited trawl corridors. Recreational scallop dredging will be banned and commercial scallop dredging limited to its current footprint.
- 25 And establishment of a multi-stakeholder Hauraki Gulf Fisheries Plan Advisory Group.
- 26 Expanding the Leigh and Hahei marine reserves as well as establishing 11 new High Protection Areas and five new Seafloor Protection Areas. There is an intention to use new legislation to achieve this, to avoid “the antiquated procedures in the Marine Reserves Act”.
- 27 Comprised of the Minister of Oceans and Fisheries, Minister for the Environment, Minister of Conservation and the Under-Secretary for Oceans and Fisheries.
- 28 Hosted by the Department of Conservation. This comprises officials from the Ministry for Primary Industries and Ministry for the Environment, and is supported by other agencies as required.
- 29 One Cabinet paper suggests this “could consider issues such as: Marine spatial planning across the territorial sea and the EEZ... comparisons with oceans governance and ecosystem-based management in other jurisdictions, and their potential suitability in the New Zealand context...i.e. legislative, institutional and funding arrangements, including the incorporation of Māori world views and interests...” (Minister for Oceans and Fisheries “Oceans and Fisheries portfolio – ensuring healthy” ecosystems (2 July 2021) at 8-9).
- 30 At 8.
- 31 See Ministry for the Environment “Proposed national policy statement for indigenous biodiversity” (25 June 2021) <<https://environment.govt.nz/acts-and-regulations/national-policy-statements/proposed-nps-indigenous-biodiversity/>>
- 32 Department of Conservation *Te Mana o te Taiao - Aotearoa New Zealand Biodiversity Strategy* (Department of Conservation, Wellington, August 2020) at 57 and 53.
- 33 Department of Conservation *Te Mana o te Taiao - Aotearoa New Zealand Biodiversity Strategy* (Department of Conservation, Wellington, August 2020) at 53.
- 34 Robert Joseph and others *Stemming the Colonial Tide: Shared Māori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2020)
- 35 Michael Neilson “Land Back’: Green Party calls for revamp of Treaty settlements” *NZ Herald* (online ed, 7 February 2022) <www.nzherald.co.nz>
- 36 See generally Jacinta Ruru “Indigenous Ancestors” in Sumundu Atapattu and others (eds) *The Cambridge Handbook of Environmental Justice and Sustainable Development* (Cambridge University Press, Cambridge, 2021) 183.
- 37 Claire Charters and others *He Puapua: Report of the Working Group on a Plan to Realise the UN Declaration on the Rights of Indigenous Peoples in Aotearoa New Zealand* (Te Puni Kōkiri, 1 November 2019).
- 38 Dame Anne Salmond “He Puapua and a Forgotten Promise” *Newsroom* (online ed, 12 July 2021) <www.newsroom.co.nz>
- 39 See Margaret Mutu and Moana Jackson *Whakaaro Here Whakaumu mō Aotearoa* (Matike Mai Aotearoa, Independent Iwi Working Group on Constitutional Transformation, January 2016).
- 40 He Whakaputanga o te Rangatiratanga o Nu Tirenē (the Declaration of Independence of the United Tribes of Aotearoa New Zealand) was initially signed by 34 rangatira on 28 October 1835 and preceded the signing of te Tiriti. It was formally acknowledged by the Crown in 1836, and subsequently signed by additional rangatira. He Whakaputanga expressly recognises, inter alia, Māori authority and independence and informs the constitutional framework of Aotearoa New Zealand.
- 41 Margaret Mutu and Moana Jackson *Whakaaro Here Whakaumu mō Aotearoa* (Matike Mai Aotearoa, Independent Iwi Working Group on Constitutional Transformation, January 2016) at 7.
- 42 Margaret Mutu and Moana Jackson *Whakaaro Here Whakaumu mō Aotearoa* (Matike Mai Aotearoa, Independent Iwi Working Group on Constitutional Transformation, January 2016) at 7.
- 43 Margaret Mutu and Moana Jackson *Whakaaro Here Whakaumu mō Aotearoa* (Matike Mai Aotearoa, Independent Iwi Working Group on Constitutional Transformation, January 2016) at 68.
- 44 Margaret Mutu and Moana Jackson *Whakaaro Here Whakaumu mō Aotearoa* (Matike Mai Aotearoa, Independent Iwi Working Group on Constitutional Transformation, January 2016) at 69.
- 45 Margaret Mutu and Moana Jackson *Whakaaro Here Whakaumu mō Aotearoa* (Matike Mai Aotearoa, Independent Iwi Working Group on Constitutional Transformation, January 2016) at 69.

- 46 Margaret Mutu and Moana Jackson *Whakaaro Here Whakaumu mā Aotearoa* (Matike Mai Aotearoa, Independent Iwi Working Group on Constitutional Transformation, January 2016) at 9.
- 47 Claire Charters and others *He Puapua: Report of the Working Group on a Plan to Realise the UN Declaration on the Rights of Indigenous Peoples in Aotearoa New Zealand* (Te Puni Kōkiri, 1 November 2019).
- 48 Claire Charters and others *He Puapua: Report of the Working Group on a Plan to Realise the UN Declaration on the Rights of Indigenous Peoples in Aotearoa New Zealand* (Te Puni Kōkiri, 1 November 2019) at iv.
- 49 Claire Charters and others *He Puapua: Report of the Working Group on a Plan to Realise the UN Declaration on the Rights of Indigenous Peoples in Aotearoa New Zealand* (Te Puni Kōkiri, 1 November 2019) at v.
- 50 Claire Charters and others *He Puapua: Report of the Working Group on a Plan to Realise the UN Declaration on the Rights of Indigenous Peoples in Aotearoa New Zealand* (Te Puni Kōkiri, 1 November 2019).
- 51 Claire Charters and others *He Puapua: Report of the Working Group on a Plan to Realise the UN Declaration on the Rights of Indigenous Peoples in Aotearoa New Zealand* (Te Puni Kōkiri, 1 November 2019).
- 52 Claire Charters and others *He Puapua: Report of the Working Group on a Plan to Realise the UN Declaration on the Rights of Indigenous Peoples in Aotearoa New Zealand* (Te Puni Kōkiri, 1 November 2019).
- 53 See generally C Knight *Beyond Manapouri: 50 years of environmental politics in New Zealand* (Canterbury University Press, 2018) and D Young *Values as law: The history and efficacy of the Resource Management Act* (Victoria University of Wellington Institute of Policy Studies, 2001).
- 54 For example, wastewater overflows, fishing nets as opposed to litter in urban streets, nutrient and sediment inputs and so forth.
- 55 Gerard Hutching and Carl Walrond "Marine conservation – Fisheries" (12 June 2006, updated 1 September 2015) *Te Ara - the Encyclopedia of New Zealand* <www.TeAra.govt.nz/>
- 56 Gerard Hutching and Carl Walrond "Marine conservation – Fisheries" (12 June 2006, updated 1 September 2015) *Te Ara - the Encyclopedia of New Zealand* <www.TeAra.govt.nz/>.
- 57 Arts 61 and 119.
- 58 Marine Mammals Protection Act 1978.
- 59 Through scheduling in the Wildlife Act 1953.
- 60 Through the Maritime Transport Act 1994.
- 61 A specifically Māori economy is thought to currently be around \$50 billion. See Chapman Tripp *Te Ao Māori: Trends and insights* (Chapman Tripp, online ed, June 2017); The Treasury *He Tirohanga Mokopuna: 2016 Statement on New Zealand's Long-Term Fiscal Position, New Zealand* (22 November 2016).
- 62 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation* (Environmental Defence Society, Auckland, December 2018) at ch 3.
- 63 At 38; Waitangi Tribunal *Te Paparahi o Te Raki (Northland)* (Wai 1040, 2014); Waitangi Tribunal *Ko Aotearoa Tenei: A Report into Claims Concerning New Zealand Law and Policy Affecting Māori Culture and Identity* (Wai 262, 2011); and Waitangi Tribunal *The stage 2 report on the national freshwater and geothermal resources claims* (Wai 2358, 2019).